

N-Substituted Ephedrine as Chiral Auxiliaries in Enantioselective Alkylation Reactions of Carbonyl Compounds

Alejandro Cruz,^{1*} Itzia Irene Padilla-Martínez¹ and María Esther Bautista-Ramírez²

¹Departamento de Ciencias Básicas; ²Departamento de Ciencia y Tecnología Farmacéutica del Instituto Politécnico Nacional-UPIBI, Av. Acueducto de Guadalupe S/N Col. Barrio la Laguna Ticomán, C.P. 07340, México D.F.



Alejandro Cruz

Abstract: The synthesis of high optical purity chiral compounds is of current importance in organic and organometallic areas of the chemistry with further boost. Chiral compounds are important because of wide use in many areas such as pharmacy and food industry, etc. Reagents, catalysts or chiral auxiliaries that can control the stereochemistry of the products are required for the synthesis of chiral compounds. Furthermore, the reagents used must be able to be recycled. In recent decades, Ephedra derivative compounds have been widely used as catalyst chiral ligands or chiral inductors in asymmetric synthesis. To our knowledge, no review about Ephedra derivative compounds as chiral auxiliaries used with organometallics in alkylation reactions to carbonyl compounds, has been published. In this paper, the use of N-substituted and N-disubstituted ephedra compounds as chiral auxiliary ligands in alkylation, alkenylation and alquination reactions of carbonyl and imine compounds is summarized.

Keywords: Chiral auxiliaries, Asymmetric synthesis, Enantioselectivity, Ephedrine, Organometallics.

INTRODUCTION

In nature, chirality is an important phenomenon. Several key biological functions are possible when molecular recognition is carried out, which requires matching of chirality. On the other hand, chiral molecules also contribute to the formation of advanced materials to obtain different physical properties. Therefore, the enantioselective synthesis of chiral organic compounds is an important labor for synthetic chemists. Chiral metal complexes in asymmetric catalysis, is a powerful tool in this context [1].

The reaction to form a carbon-carbon bond is one of the key operations for the construction of organic molecules. In organic synthesis, the nucleophilic addition of organometallic reagents to carbonyl substrates is a basic reaction, this reaction and the reduction of aldehydes or ketones, provide a reliable way to give alcohols. Organometallic chemistry begins with the discovery of organozinc compounds by Frankland in 1849 [2]. In the alkylation to carbonyl compounds, Reformatsky [3] and Simmons-Smith [4] pioneering the use of Zn metal. After, organozinc compounds were replaced by Grignard reagents [5] and organolithium compounds [6], owing to their higher generality, reactivity and ease of manipulation. Recently, the development of organo-boron, -aluminum, -titanium and -lanthanide compounds, has been carried out with the purpose of obtaining high selectivities. On these bases, the enantioselective addition reactions of organometallics to carbonyl compounds to give optically active alcohols are of general synthetic importance. The modification of organometallic compounds by chiral auxiliaries provides a synthetic pathway that leads to optically active alcohols through catalytic reactions of high efficiency.

One way to introduce a chiral environment into an organometallic compound is the coordination of aprotic chiral solvents or a chiral ligand to the metallic center. Other way is the modification of

the organometallic reagent by a protic chiral auxiliary, such as chiral β -aminoalcohols, to give the corresponding alkoxides or amides $MR^3_nL^*$, Fig. 1. The structural X-ray diffraction analysis and spectroscopic methods of several main-group organometallic compounds revealed that the steric and electronic properties of the chiral ligand modify the structure of the complexes [7-9]. Several organometallic compounds form labile molecular associates, so the stoichiometry of the associated complex, responsible for the reaction, can be different to that of the ground-state structure. Due to the amphiphilic nature of organometallic compounds, the nucleophilicity of the carbanion and Lewis acidity of the metal, cooperate together in the transfer of the alkyl group in the reaction. Thus, the chiral ligand (L^*) modifies the structure and stability of the transition state.

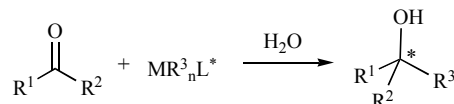


Fig. (1). Chiral organometallics in the enantioselective alkylation of carbonyl compounds.

The well-designed chiral auxiliaries and the appropriate combination of organometallic reagents have been used in the last decades to build up a wide collection of enantioselective alkylation reactions of carbonyl compounds. Chiral auxiliaries (L^*), in stoichiometric or excess amounts, have been used so far in more selective reactions. In this context, some unexpected aspects have been observed with the typical organozinc chemistry. Among these approaches, the enantioselective dialkylzinc addition to aldehydes is the most commonly used methodology [10-13].

One of the benefits of using ephedrine, norephedrine and their pseudostereoisomers as chiral auxiliaries (L^*) in alkylation reactions of carbonyl compounds, is that the two enantiomers of each compound, in total, eight stereoisomers of these *ephedra* compounds, are readily available, as shown in Figs. (2 and 3). By using the appropriate enantiomer of the chiral aminoalcohol as chiral auxiliary in this reaction, both enantiomers of the chiral alcohol are synthesized in the same synthetic yield and enantioselectivity. The

*Address correspondence to this author at the Departamento de Ciencias Básicas; Departamento de Ciencia y Tecnología Farmacéutica del Instituto Politécnico Nacional-UPIBI, Av. Acueducto de Guadalupe S/N Col. Barrio la Laguna Ticomán, C. P. 07340, México D.F.; Tel/Fax: 57296000-56325, Proyecto SIP-20150392; E-mails: alecruz@ipn.mx, ipadillamar@ipn.mx, maestherbr@gmail.com

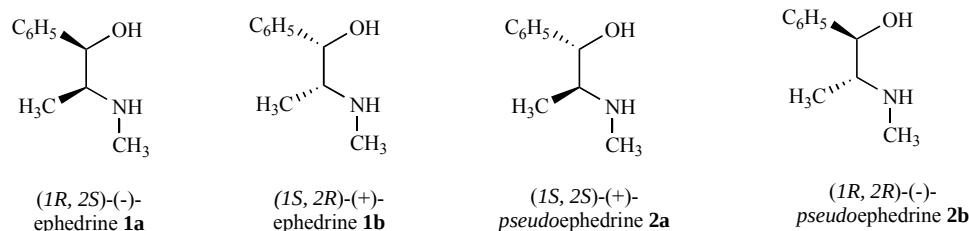


Fig. (2). Ephedrine stereoisomers.

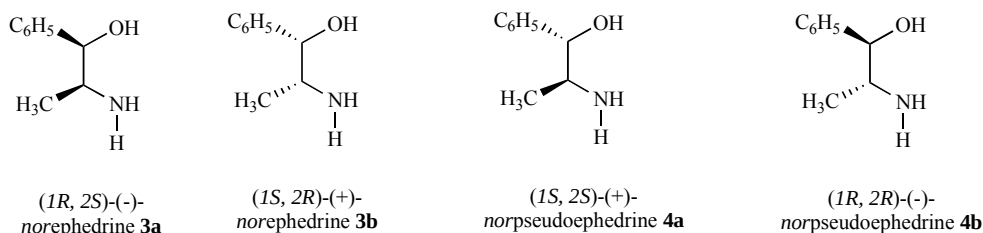


Fig. (3). Norephedrine stereoisomers.

chemical access to both enantiomers of the alcohol is predictable and superior to the biochemical and biological methods.

Ephedrine **1** and **2** have the β -aminoalcohol skeleton with a methyl group on the nitrogen atom. The phenyl and methyl groups on each carbon atom of the ethyl chain originate two chiral centers and generate four optically active stereoisomers: *l*-(1*R*, 2*S*)-ephedrine **1a** and *d*-(1*S*, 2*R*)-ephedrine **1b**; *d*-(1*S*, 2*S*)-pseudoephedrine **2a** and *l*-(1*R*, 2*R*)-pseudoephedrine **2b**, (Fig. 2).

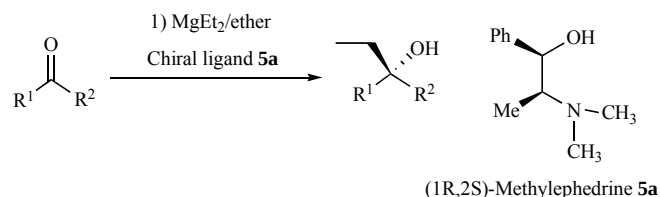
The stereoisomers with lack of methyl group on nitrogen atom of ephedrine are called nor-ephedrine: *l*-(1*R*, 2*S*)-norephedrine **3a** and *d*-(1*S*, 2*R*)-norephedrine **3b**; *d*-(1*S*, 2*S*)-norpseudoephedrine **4a** and *l*-(1*R*, 2*R*)-norpseudoephedrine **4b**, (Fig. 3).

Due to the relevant importance of *ephedra* compounds, we have published some reviews about the methodologies of synthesis of cyclic [14] and acyclic [15] derivatives. Now, in this review, we describe the historical background of the use of *N*-substituted-ephedrine, -pseudoephedrine, -norephedrine, -norpseudoephedrine and *N*-disubstituted-norephedrine and -norpseudoephedrine as chiral auxiliaries in the addition reaction of organo-metallics to carbonyl or imine compounds in order to carry out enantioselective alkylations. The use of some thiols, sulfides and disulfides derived from *ephedra* compounds are compiled as well. Even the use of polymeric supports anchored with ephedrine derived ligands as chiral auxiliaries have been found to promote asymmetric reactions. The review includes alkylation, alkenylation, alkinylation, furylation and arylation reactions. More elaborate nucleophiles, such as metal enolates, alkynyl or allylic metal compounds, are also included.

1. ENANTIOSELECTIVE 1, 2-ADDITIONS TO CARBONYL COMPOUNDS

1. 1. Organometallic Chiral Catalysts

In the seventies, ephedrine **1a** and methylephedrine **5a** were reported as chiral auxiliaries in the asymmetric addition of organometallics to carbonyl compounds. In 1972, Battioni *et al.* examined the influence of ephedrine **1a** and methylephedrine **5a** in alkylation reactions of aldehydes and ketones using Grignard reagents [16], in table 1. It was found that chiral aminoalcohols induced certain asymmetric alkylation, leading to secondary and tertiary alcohols, respectively. In the ethylmagnesium addition reaction to benzaldehyde by using **1a**, a racemate was obtained in 50% yield of benzyl alcohol. However, when **5a** was used, the yield increased (up to 80%) but poor enantioselectivity (up to 15.3% ee) was obtained.

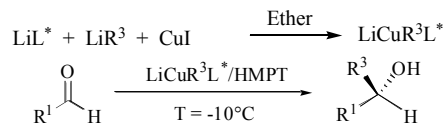
Table 1. Asymmetric alkylation of aldehydes and acetophenone when diethylmagnesium is used with **5a** as chiral auxiliary.

Entry	R ¹	R ²	Auxiliary	Yield (%)	Ee (%) ^b	Config.
1	Ph	H	5a	80	15.3	<i>S</i>
2	Ph	H	1a	50	00.0	-
3	<i>t</i> -Bu	H	5a	50	20.0	<i>R</i>
4	C ₆ H ₁₁	H	5a	25	20.0	<i>R</i>
5	Ph	CH ₃	5a	50	14.5	<i>R</i>

^aProducts were identified by comparison with authentic samples.

^bWithout solvent.

The induction of asymmetry of prochiral aldehydes with chiral organocuprates was studied. Chiral organocuprates were prepared with equimolar quantities of lithium alcoholate of **5a** (LiL*), alkyl-lithium and cuprous iodide [17]. The best result was obtained for benzaldehyde when methyl lithium was used to get the (*R*)-1-phenylethanol in only 16.5% ee (Table 2, entry 1).

Table 2. Enantioselective 1,2-additions to aldehydes when chiral organocuprates (LiCuR³L*) derived from methylephedrine **5a** was used as chiral auxiliary (L*).^a

Entry	R ¹	R ³ Li	[α] _D (°) ^b	Ee (%) ^b	Config.
1	Ph	Me	+6.6	16.5	<i>R</i>
2	Ph	<i>n</i> -Bu	+0.6	1.5	<i>R</i>
3	<i>n</i> -C ₇ H ₁₅	Me	+0.27	3.0	<i>R</i>

^a The reactions were carried out by using ether-HMPT.

^b The optical rotations were taken at room temperature in chloroform, C = 1M.

The chiral organometallic reagent $[\text{LiAl}(n\text{-Bu})_3\text{L}^*]$ was prepared *in situ* in good yields when equimolar amounts of lithium tetrabutyl aluminate and **5a** were combined [18], as shown in table 3. The chiral reagent was used for the asymmetric alkylation of aldehydes and ketones, leading to secondary (70% yield, 24% ee) and tertiary alcohols (up to 56% yield, 31% ee), respectively.

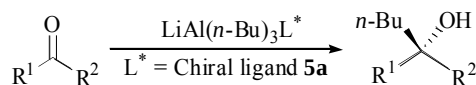
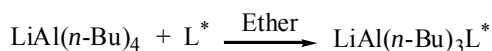


Table 3. Enantioselective 1,2-addition to carbonyl compounds of organoaluminate chiral reagent $[\text{LiAl}(n\text{-Bu})_3\text{L}^*]$ derived from methylephedrine **5a**.

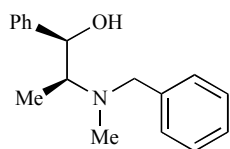
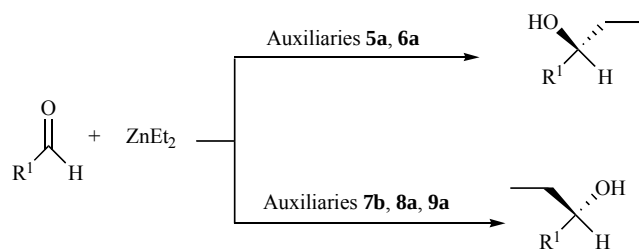
Entry	R ¹	R ²	Time (h)	Yield (%) ^a	Ee (%) ^b	Config.
1	C ₂ H ₅	H	24	70	24	S
2	C ₆ H ₅	H	4	79	8	S
3	C ₆ H ₅	Me	24	56	31 ^b	S

^aEvaluated by GC, using the internal standard method.

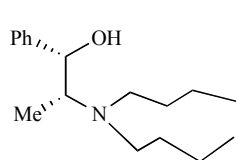
^bEvaluated by NMR with chiral shift reagent Eu(t. fac. cam.)₃

I. 2 Dialkylzincs Addition to Carbonyl Compounds

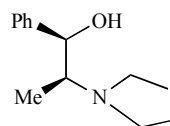
In the 80's decade, it was found that chiral *N*-alkylephedrine [19-24] such as **5a** and (1*R*, 2*S*)-*N*-benzylephedrine **6a** as well as *N*-dialkylnorephedrine: (1*S*, 2*R*)-*N*-dibuthylnorephedrine **7b**, (1*R*, 2*S*)-*N*-diethylnorephedrine **8a** and (1*S*, 2*S*)-*N*-diethylnorpseudoephedrine **9a** showed to be efficient catalysts for the enantioselective alkylation of aldehydes by using ZnEt₂, Table 4.



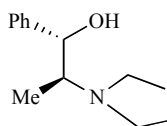
(1*R*, 2*S*)-**6a**



(1*S*, 2*R*)-**7b**



(1*R*, 2*S*)-**8a**

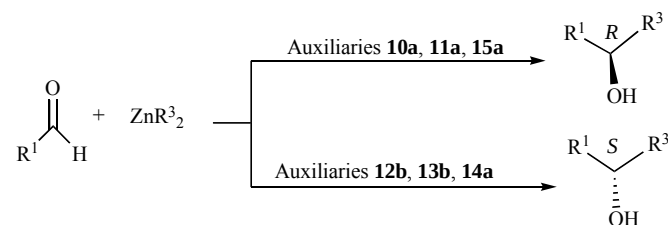


(1*S*, 2*S*)-**9a**

Table 4. Alkylephedrine and dialkylnorephedrine as chiral auxiliaries in the enantioselective addition of diethylzinc to aldehydes.

Auxiliary	R ¹	Yield (%)	Ee (%)	Config.
5a	C ₆ H ₅	81	81	<i>R</i>
6a	C ₆ H ₅	91	86	<i>R</i>
7b	C ₆ H ₅	100	90	<i>S</i>
7b	<i>n</i> -C ₆ H ₁₃	95	88	<i>S</i>
7b	C ₆ H ₅ (CH ₂) ₂	94	95	<i>S</i>
7b	(CH ₃) ₂ CHCH ₂	92	93	<i>S</i>
7b	C ₆ H ₅ COC ₆ H ₄	84	91	<i>S</i>
7b	C ₂ H ₅ OCO(CH ₂) ₂	90	92	<i>S</i>
7b	C ₂ H ₅ OCO(CH ₂) ₃	87	95	<i>S</i>
8a	C ₆ H ₅	93	94	<i>S</i>
9a	C ₆ H ₅	97	81	<i>S</i>

(1*R*, 2*S*)-*N*-isopropylephedrine **10a** promoted the enantioselective addition of ZnEt₂ reagent to benzaldehydes and some aliphatic aldehydes, to afford (*R*)-1-phenylpropanol in 72% yield with 80% ee (Table 5, entry 1) [25]. In contrast, a racemic mixture of the alcohol was obtained in 40% yield (entry 4) in the case of the organometallic addition to aliphatic aldehyde. However, the use of high molar ratios (4.5) of ZnEt₂ reagent to alkyl aldehydes, increases the ee values to reach up to 97% (entry 5) [26].



The lithium salt of (1*R*, 2*S*)-*N*-[2-(dimethylamino)ethyl]ephedrine **11a**, catalyzes the ZnEt₂ reagent addition to benzaldehyde to afford (*R*)-1-phenylpropanol in 90% ee (table 5, entry 6) [27]. This chiral lithium salt was used in the conjugate addition of organocopper reagents [28]. When (1*S*, 2*S*)-pseudoephedrine-derivative **12a** was used as chiral auxiliary, (*S*)-1-phenylpropanol in 91% ee was produced (entry 7).

When the lithium salt of (1*S*, 2*R*)-*N*-[1-methylpyrrolidin-2-yl)methyl]ephedrine **13b** was employed, (*S*)-1-phenylpropanol was obtained with 95% ee (entry 8). The chiral phenol **14a** derived from (1*S*, 2*S*)-pseudoephedrine, promoted the enantioselective addition of ZnEt₂ reagent to benzaldehyde to give (*S*)-1-phenylpropanol with 86% ee (entry 10) [29]. The enantiomer (*R*)-1-phenylpropanol with 85% ee (entry 11) is obtained when the dilithium salt of **15a** was used [30].

The structure of the *N*-alkyl substituent groups on the (1*S*, 2*R*)-*N*, *N*-di-*n*-alkylnorephedrine (table 6) showed an important effect on the enantioselective ZnEt₂ addition to 3-methylbutanal and nonanal (entries 5, 8-10) [21, 31]. When *N*-alkyl substituent length chain is of four carbon atoms, the optical purity of the product has the maximum value (93% ee, entry 5) [31]. Thus, the chiral auxiliary **7b** acts with high enantioselectivity for the ZnEt₂ reagent addition

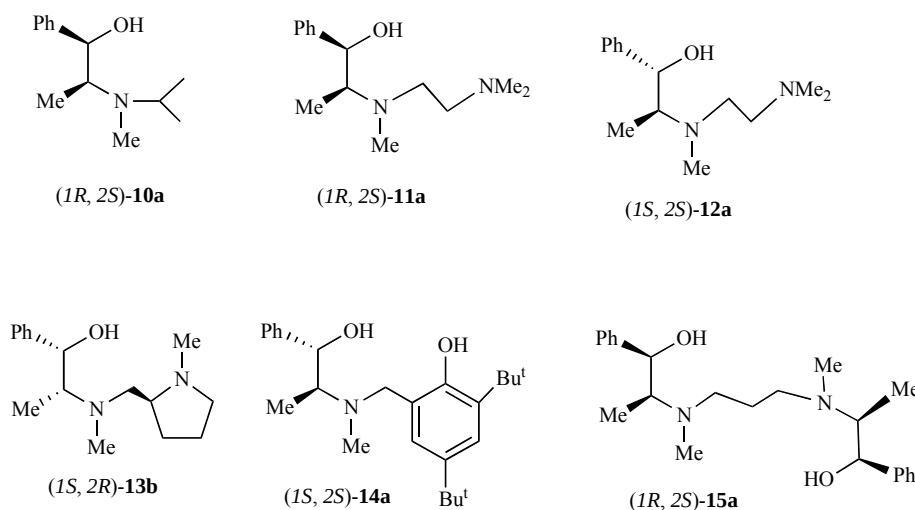
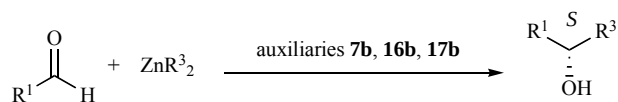
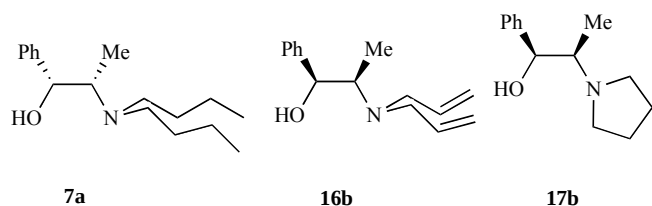


Table 5. Enantioselective dialkylzinc additions to aldehydes with ephedrine- and pseudoephedrine- derived amino alcohols as chiral auxiliaries.

Entry	R ¹	R ³	Auxiliary (%mol)	Solvent	T (°C)	Time (h)	Yield (%)	Ee (%)	Config.
1	Ph	Et	10a (10)	hexane	25	48	72	80	<i>R</i>
2	3-ClC ₆ H ₄	Et	10a (10)	hexane	25	48	60	78	<i>R</i>
3	(CH ₂) ₅ CH ₃	Et	10a (10)	hexane	25	48	88	79	<i>R</i>
4	cyclohexyl	Et (1 eq.)	10a (10)	hexane	25	48	40	0	-
5	cyclohexyl	Et (4.5 eq.)	10a (10)	hexane	25	48	90	97	-
6	Ph	Et	Li- 11a (10)	toluene	23	14	81	90	<i>R</i>
7	Ph	Et	Li- 12a (10)	toluene	23	14	69	91	<i>S</i>
8	Ph	Et	Li- 13b (10)	toluene	0	7	68	95	<i>S</i>
9	Ph	<i>n</i> -Bu	Li- 13b (10)	toluene	0	7	8	84	<i>S</i>
10	Ph	Et	14a (13)	toluene	3	8	93	86	<i>S</i>
11	Ph	Et	Li- 15a (4)	toluene/ hexane	0	48	59	85	<i>R</i>
12	2-MeOC ₆ H ₄	Et	Li- 15a (4)	toluene hexane	0	48	93	73	-

to both aliphatic (entries 5-8) and aromatic aldehydes (entries 1, 2, and 4). Optically active fluorinated (entries 11 and 12) [32] and deuterated alcohols (entries 13 and 14) [33] synthesized from **7b**, (*1S,2R*)-*N,N*-diallylnorephedrine **16b** and (*1S,2R*)-1-phenyl-2-(1-pyrrolidinyl)propan-1-ol **17b** were as well effective [31]. Diisopropylzinc was used with success (entry 4). Also, both enantiomers of each secondary alcohol were synthesized by using the appropriate enantiomer **7a** or **7b**.



On the other hand, the amino alcohol **17b** has been found to be an efficient chiral auxiliary in the reaction of dialkylzincs with aromatic aldehydes [34]. Very good enantioselectivities of the corresponding (*S*)-alcohol (87-92% ee) were obtained, Table 7.

Table 6. Enantioselective dialkylzincs addition to aldehydes, promoted by norephedrine-derived amino alcohols as chiral auxiliaries (6% mol, hexane; T, 0°C).

Entry	R ¹	R ²	Auxiliary	T (h)	Yield (%)	Ee (%)	Config.
1	Ph	Et	7b	16	100	90	S
2	2-MeOC ₆ H ₄	Et	7b	16	100	94	S
3	C ₆ H ₅ (CH ₂) ₂	Et	7b	16	94	95	S
4	Ph	<i>i</i> -Pr	7b	16	53	75	S
5	(CH ₃) ₂ CHCH ₂	Et	7b	16	92	93	-
6	CH ₃ (CH ₂) ₅	Me	7b	16	70	90	S
7	CH ₃ (CH ₂) ₅	<i>n</i> -Pr	7b	16	100	90	-
8	CH ₃ (CH ₂) ₇	Et	7b	16	95	87	S
9	CH ₃ (CH ₂) ₇	Et	17b	16	87	95	S
10	CH ₃ (CH ₂) ₇	Et	16b	16	61	88	S
11	4-CF ₃ C ₆ H ₄	Et	7b	22	92	91	S
12	4-FC ₆ H ₄	Et	7b	24	83	93	-
13	PhCDO	Et	7b	22	92	94	S
14	CH ₃ (CH ₂) ₅ CDO	Et	7b	15	79	84	S

Table 7. Dialkylzinc additions to aromatic aldehydes using **17b** as chiral auxiliary.

Entry	R ¹	R ²	Yield (%)	ee (%)
1	C ₆ H ₅	Et	91	88
2	C ₆ H ₅	<i>n</i> -Bu	57	89
3	<i>p</i> -MeC ₆ H ₄	Et ^b	93	88
4	<i>p</i> -MeC ₆ H ₄	Et	87	92
5	<i>p</i> -MeC ₆ H ₄	<i>n</i> -Bu	64	91
6	<i>p</i> -MeC ₆ H ₄	<i>i</i> -Pr ₂	71	92
7	<i>o</i> -MeC ₆ H ₄	Et	90	89
8	<i>p</i> -ClC ₆ H ₄	Et	92	87
9	1-naphthyl	Et	86	92
10	2-naphthyl	Et	94	87
11	<i>p</i> -PhC ₆ H ₄	Et	97	88
12	3,4-(CH ₂ O) ₂ C ₆ H ₃	Et	91	91
13	3,4-(CH ₃ O) ₂ C ₆ H ₃	Et	96	88

A tentative mechanism has been proposed by Soai [31]. It is illustrated with modifications in Fig. (4). Complexes **19** and **20** are formed from Zinc monoalkoxide-chiral auxiliary **18**. The nucleophilicity of the alkylzinc group is increased by coordination of the heteroatom (O, N) of the chiral aminoalcohol to the zinc atom [35], as well as by the tetrahedral geometry of the metal [36, 37]. The *si* face of the aldehyde is attacked to form a six-membered ring dinuclear complex in the transition state (**21** or **22**, respectively) [28], to give the chiral alkylzinc alkoxide (S)-**23**. Steric and electronic effects of the *N*-substituents of the chiral aminoalcohol may influence both the formation and stability of the transition states. Some structural features such as one β-carbon atom in the *N*-Alkyl substituents and longer chains than *n*-butyl are required to obtain

high enantioselectivity. On the other hand, the six-membered ring transition states (**21** or **22**) are inhibited, if the *N*-alkyl substituents are bulkier than *n*-butyl, resulting in low enantioselection.

Chiral alcohols in good ee's (up to 74%) are obtained from the reaction of ZnEt₂ and aldehydes promoted by chiral (1*S*, 2*R*)-*N*-benzyl-*N*-methylephedrinium halides (**24b**) [38] (Table 8). The effect of the solvents used is significant in the enantioselectivities. Higher enantioselectivities are observed in suspension (hexane or benzene) than in solution when the chiral ammonium auxiliary ligands are used. The reaction of benzaldehyde with ZnEt₂ in a suspension of **24b** in hexane, afforded the corresponding chiral alcohol in 74% ee (entry 1). However, in dimethylformamide-

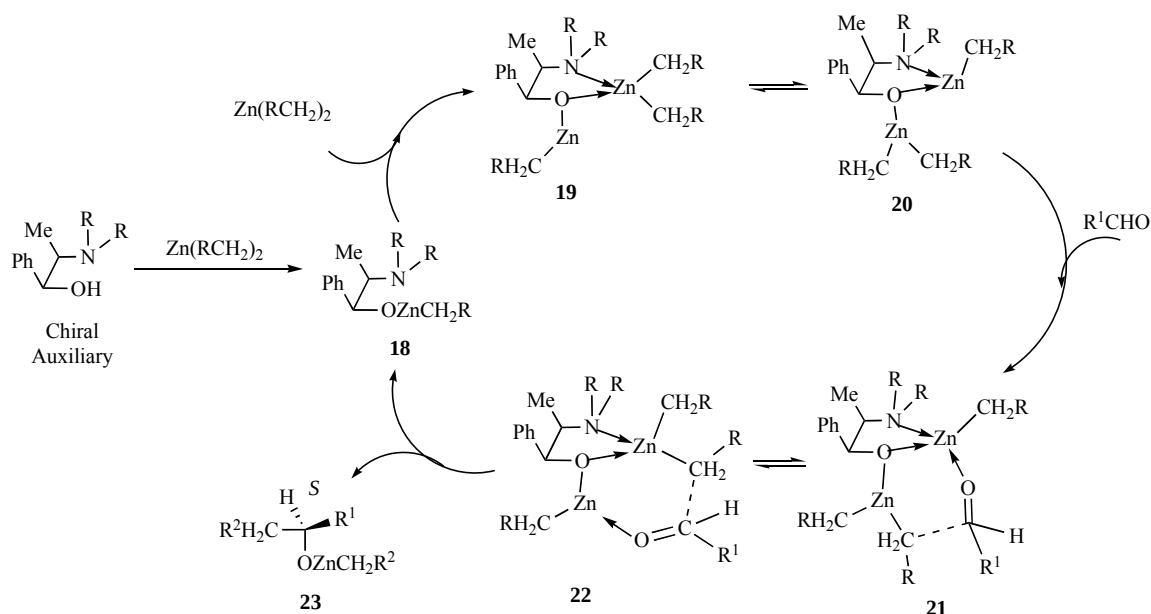


Fig. (4). Proposed mechanistic path way for the enantioselective dialkylzinc additions to aldehydes using *N*-dialkyl-norephedrine as chiral auxiliary.

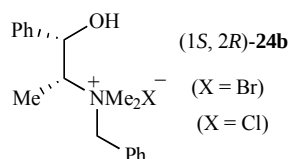
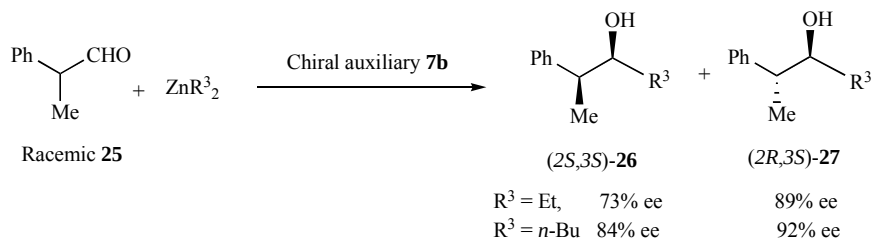


Table 8. Enantioselective diethylzinc addition to aromatic aldehydes ($R^1\text{CHO}$) promoted by *N*-benzyl-*N*-methylephedrinium halides 24b as chiral auxiliaries.

Entry	R^1	Auxiliary	Solvent	Yield (%)	Ee (%)	Config.
1	Ph	X = Br	hexane	90	74	S
2	Ph	X = Br	Benzene/hexane	76	73	S
3	Ph	X = Br	DMF/hexane	71	0	-
4	Ph	X = Cl	hexane	72	62	S
5	4-MeC ₆ H ₄	X = Br	hexane	81	61	S



hexane solution (entry 3), the same reaction proceeded without enantioselectivity. This is a rare example because in general, the catalyst in solution is more enantioselective than in the solid state. It was found that very little solvation of the ammonium cation in hexane is needed for the asymmetric induction.

Reactions were run at room temperature, using 6% mol of the chiral auxiliary

The diastereo and enantioselective addition of dialkylzinc to racemic 2-phenylpropanal **25**, using **7b** as chiral auxiliary, yields chiral alcohols with two stereocenters in very good ee's [39].

The racemic mixture of **25** with ZnEt_2 reagent using **7b** as a chiral auxiliary (mol ratio aldehyde: ZnEt_2 :**7b** = 1:1:0.1) in hexane at room temperature, produces 2-phenylpentan-3-ol in 58% yield (erythro/threo = 81:19). The predominant enantiomers were (2*S*,3*S*)-**26** in 73% ee for the erythro isomer and (2*R*,3*S*)-**27** in 89% ee for the threo isomer. The stereochemistry, yields and ee's are almost independent of the nature and quantity of the dialkylzinc employed, since when aldehyde: $\text{Zn}(n\text{-Bu})_2$ mol ratio 2:1 is used, the ee's of (2*S*,3*S*)-**26** and (2*R*,3*S*)-**27** were 84% and 92%, respectively (erythro/threo = 83:17) [40]. This reaction proceeds through the selective addition of dialkylzinc to racemic **25** on the aldehyde *si*

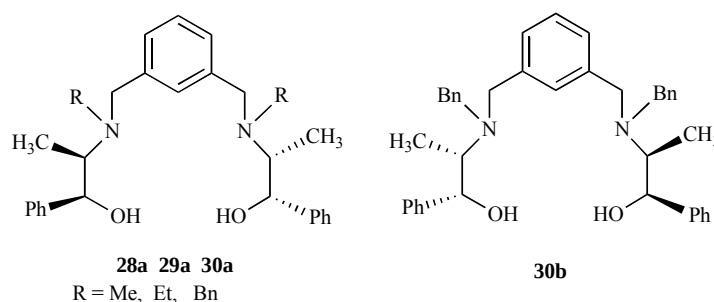
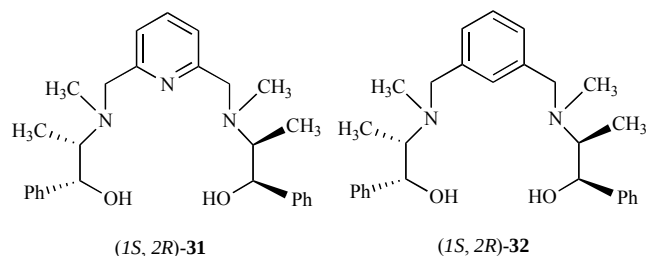


Table 9. Dialkylzinc additions to aromatic aldehydes promoted by C₂-symmetric *N*-benzylphedrine **28a** and *N*-alkylbenzylnorephedrine **29a**, **30a** and **30b** as chiral auxiliaries.

Entry	Auxiliary	R ¹	Yield (%)	Ee (%)	Config.
1	28a	<i>p</i> -ClC ₆ H ₄	95	78	<i>S</i>
2	29a	<i>p</i> -ClC ₆ H ₄	82	95	<i>S</i>
3	30a	<i>p</i> -ClC ₆ H ₄	80	78	<i>S</i>
4	30b	<i>p</i> -ClC ₆ H ₄	82	78	<i>R</i>
5	28a	C ₆ H ₅	72	80	<i>S</i>
6	29a	C ₆ H ₅	85	93	<i>S</i>
7	28a	2-naphthyl	84	87	<i>S</i>
8	29a	2-naphthyl	85	94	<i>S</i>



face, no matter of the starting configuration of **25**, to form the new chiral center with *S* configuration. Hence, the selected face of the dialkylzinc addition to **25** was determined by the chiral aminoalcohol **7b** rather than by the configuration of **25**. The yields of (2*S*,3*S*)-**26** and (2*S*,3*R*)-**27** are higher than those of the enantiomers (2*R*,3*R*)-**26** and (2*R*,3*S*)-**27**. This result indicated that, in presence of **7b**, (*S*)-**25** reacted with ZnEt₂ reagent faster than its (*R*)-enantiomer.

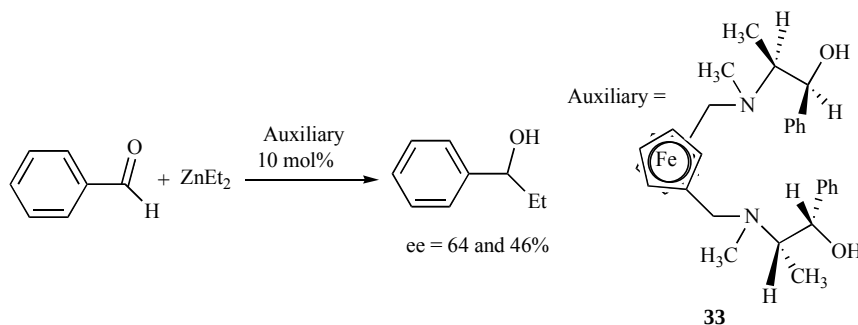
The results obtained when compounds (1*S*,2*R*)-(28-30)**a** and (1*R*,2*S*)-**30b** containing two ephedrine units were used as chiral auxiliaries in the addition of ZnEt₂ to aromatic aldehydes [41], are summarized in table 9. From diaminoalcohols **28a-30a**, with different

groups on the nitrogen atom, the ethyl-substituted ephedrine **29a** induces the highest enantioselectivity (95% ee) (entries 1-3) whereas diaminoalcohol **30b**, induces the expected opposite enantioselectivity (entry 4).

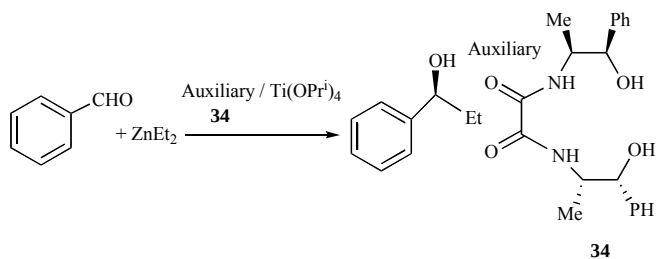
The reactions were carried out at 0 °C in toluene:hexane (2:1) at room temperature and 10% mol of the chiral auxiliary was used

A comparative study of the dialkylzinc addition to aldehydes using the C₂-symmetric pyridine-based dimeric ephedrine ligand **31** in comparison with the chiral diaminoalcohol **32** contained no pyridine ring, was performed [42]. Diethylzinc was added to aromatic aldehydes in the presence of **31** (20% mol) to yield (*S*)-alcohols, and in the same conditions, **32** affords the (*R*)-alcohols. It was found that using either high (50% mol) or small (<10% mol) quantities of **32**, both enantioselectivity and yield were decreased. In addition, compound **31** gives higher enantioselectivity (up to 90% ee) than compound **32**.

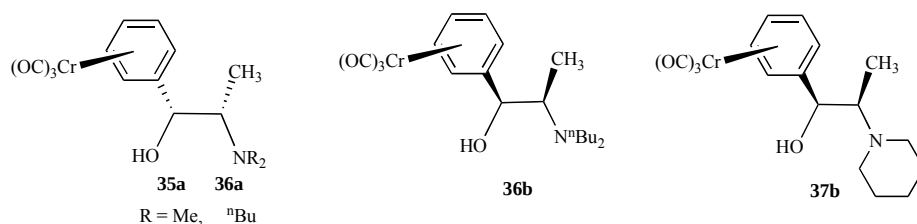
Optically active 1,1'- and 1,2-disubstituted ferrocenylpseudoephedrine ligands **33** were used in 10% mol as chiral auxiliaries in the asymmetric ZnEt₂ addition to benzaldehyde [43]. The corresponding (*S*)-1-phenylpropanol was obtained in 64 and 46% ee, respectively.



Poor yield (25%) and low ee (4%) of the corresponding (S)-alcohol are obtained when the bis(norephedrine)oxalamide **34** ligand as chiral auxiliary in the presence of titanium isopropoxide was used in the enantioselective ZnEt_2 addition to benzaldehyde [44].



Chiral η^6 -arene-chromium complexes **35a**, **36a**, **36b** and **37b** derived from **5a**, **7a**, **7b** and (1*S*,2*R*)-pentylennorephedrine **200b**, respectively, have been used as chiral auxiliaries in the enantio-



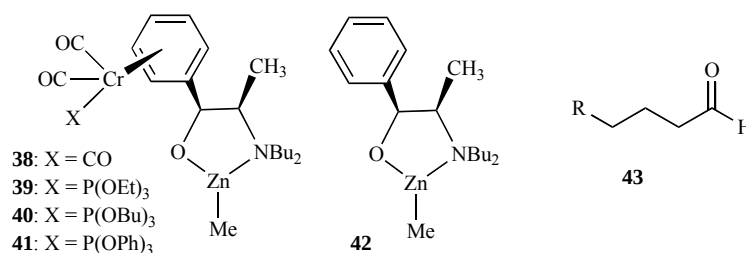
selective dialkylzinc addition to aldehydes (table **10**) [45]. Aminoalcohol **36a** showed higher enantioselectivity (99% ee) than **7a** (94% ee) in the addition reaction of ZnEt_2 reagent to benzaldehyde (entries 5 and 6) [45a]. The high steric bulkiness of **36a** and the electroattractive effect of the $\text{Cr}(\text{CO})_3$ group were proposed to be responsible of the increased enantioselectivity. Aminoalcohol **36a** and its enantiomer **36b** showed very good enantioselectivities for the addition of dialkylzinc ($\text{R} = \text{Me}, \text{Et}, n\text{-Bu}$) to aromatic aldehydes (entries 9-24) [45b]. For aliphatic aldehydes, the enantioselectivities were lower but still higher than the ligand without the coordinated chromium group. Some application examples are described elsewhere [45b, c].

Complexes containing various η^6 -arene-chromium organometallic auxiliaries **38-41** derived from **7b** were used to catalyze the dimethylzinc addition to **43** to get the (S)-alcohols. The enantioselectivities were increased with the size of ligand (X) (**38-41**) [46] (Table **11**). Complex **41** having the bulkiest phosphite group exhibited the highest enantioselectivity (98% ee, entry 8) in contrast, compound **42**, with no η^6 -arene-chromium coordination, gave low

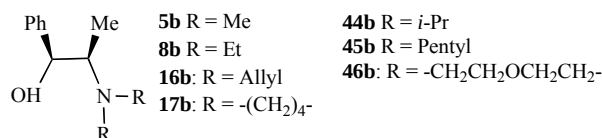
Table 10. Dialkylzinc additions to aldehydes promoted by the chromium-complexed and nocomplexed ephedra amino alcohols as chiral auxiliaries^a.

Entry	R ¹ CHO	ZnR ³ ₂	Auxiliary	Yield (%)	Ee (%)	Config.
1	C ₆ H ₅ CHO	ZnEt ₂	35a	99	86	<i>R</i>
2	C ₆ H ₅ CHO	ZnEt ₂	5a	99	67	<i>R</i>
3	C ₆ H ₅ CHO	ZnEt ₂	37b	99	96	<i>S</i>
4	C ₆ H ₅ CHO	ZnEt ₂	7b	99	90	<i>S</i>
5	C ₆ H ₅ CHO	ZnEt ₂	36a	99	99	<i>R</i>
6	C ₆ H ₅ CHO	ZnEt ₂	7a	99	94	<i>R</i>
7	C ₆ H ₅ CHO	ZnEt ₂	36b	99	99	<i>S</i>
8	C ₆ H ₅ CHO	ZnEt ₂	7b	99	94	<i>S</i>
9	1-naphthaldehyde	ZnEt ₂	36b	98	99	<i>S</i>
10	2-naphthaldehyde	ZnEt ₂	36a	99	99	<i>R</i>
11	<i>p</i> -MeOC ₆ H ₄ CHO	ZnEt ₂	36b	99	94	<i>S</i>
12	Phenanthrene-9-carboxaldehyde	ZnEt ₂	36a	97	93	<i>R</i>
13	3-Phenylpropionaldehyde	ZnEt ₂	36b	89	92	<i>R</i>
14	3-Phenylpropionaldehyde	ZnEt ₂	7b	90	38	<i>R</i>
15	3-(4- <i>tert</i> -butyldimethyl-silylphenyl)propionaldehyde	ZnEt ₂	36b	88	85	<i>R</i>
16	C ₆ H ₅ CHO	ZnMe ₂	36b	85	92	<i>S</i>
17	1-naphthaldehyde	ZnMe ₂	36b	95	92	<i>S</i>
18	2-naphthaldehyde	ZnMe ₂	36b	90	94	<i>S</i>
19	3-Phenylpropionaldehyde	ZnMe ₂	36b	87	81	<i>S</i>
20	7-(<i>tert</i> -butyldimethyl-sililoxy)heptanal	ZnMe ₂	36a	82	85	<i>R</i>
21	4-benzyloxybutiraldehyde	ZnMe ₂	36b	86	72	<i>S</i>
22	5-hexyn-1-al	ZnMe ₂	36a	95	64	<i>R</i>
23	C ₆ H ₅ CHO	ZnBu ₂	36b	75	91	<i>S</i>
24	2-naphthaldehyde	ZnBu ₂	36b	95	93	<i>R</i>

^aThe reactions were carried out in toluene at 0 °C for 12 h in the presence of 10% mol of the chiral auxiliary.

**Table 11.** Dimethylzinc additions to aldehyde **43** catalyzed by zinc complexes **38-42**.

Entry	R (43)	Auxiliary (mol %)	Yield (%)	Ee (%)
1	OTPS	38 (10)	88	86 ^a
2	Cl	38 (10)	91	75 ^a
3	OBn	38 (10)	92	87 ^a
4	OTr	38 (10)	94	89
5	OTr	39 (10)	90	94
6	OTr	40 (10)	91	95
7	OTr	41 (10)	82	96
8	OTr	41 (20)	91	98
9	OTr	42 (10)	91	59

^a The reactions were carried out in toluene-hexanes at 0 °C for 12 h. and 18 h.**Table 12.** Diisopropylzinc additions to aldehydes promoted by (*1S*, *2R*)-*N*-dialkylnorephedrine as chiral auxiliaries.

Entry	Auxiliary	NR ₂	Aldehyde	Yield (%)	Ee (%)	Config.
1	7b	<i>n</i> -Bu	C ₆ H ₅ CHO	73	92	<i>S</i>
2	44b	<i>i</i> -Pr	C ₆ H ₅ CHO	73	91	<i>S</i>
3	5b	Me	C ₆ H ₅ CHO	69	36	<i>S</i>
4	8b	Et	C ₆ H ₅ CHO	72	88	<i>S</i>
5	45b	Pentyl	C ₆ H ₅ CHO	61	90	<i>S</i>
6	16b	Allyl	C ₆ H ₅ CHO	63	87	<i>S</i>
7	17b	-(CH ₂) ₄ -	C ₆ H ₅ CHO	68	86	<i>S</i>
8	46b	-(CH ₂) ₂ O(CH ₂) ₂ -	C ₆ H ₅ CHO	72	88	<i>S</i>
9	7b	<i>n</i> -Bu	cinnamaldehyde	73	93	<i>S</i>
10	44b	<i>i</i> -Pr	cinnamaldehyde	63	91	<i>S</i>
11	7b	<i>n</i> -Bu	3-phenylpropionaldehyde	62	96	<i>S</i>
12	44b	<i>i</i> -Pr	ferrocenecarboxaldehyde	97	98	<i>S</i>
13	7b	<i>n</i> -Bu	ferrocenecarboxaldehyde	92	97	<i>S</i>
14	44b	<i>i</i> -Pr	ferrocene-1,1'-dicarboxaldehyde	39	100	SS
15	7b	<i>n</i> -Bu	ferrocene-1,1'-dicarboxaldehyde	47	100	SS

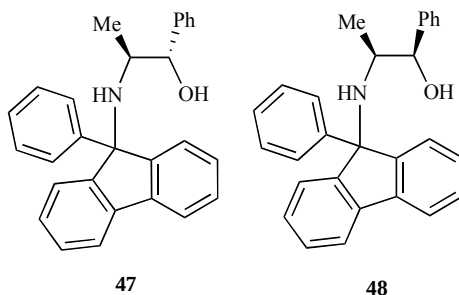
The reactions were carried out at 0 °C in hexane in the presence of 10% mol of chiral auxiliary.

enantioselectivity (59% ee, entry 9). The complexes **38-41** were used in the enantioselective addition of ZnMe₂ (4 eq) to benzaldehyde, **41** gave the highest selectivity (98% ee) of the (*S*)-2-phenetyl alcohol, even using 5, 10 or 20% mol in toluene-hexanes at 0 °C for 16 h.

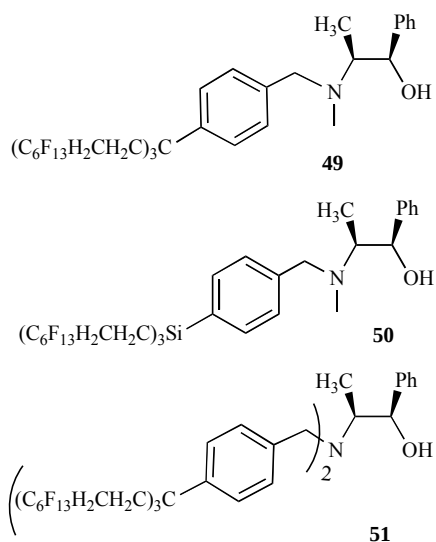
The (*1S*,*2R*)-*N*-dialkylnorephedrine **5b**, **8b**, **16b**, **17b**, (**44-46b**), were tested in the diisopropylzinc addition to aldehydes from which **7b** and **45b** gave the highest enantioselectivities with over 90% ee (Table 12) [47a]. Ligands **7b** and **44a** were used in a solvent-free system to promote the enantioselective ZnEt₂ addition to aldehydes

with 84-91% ee [47b]. In these conditions, the reaction rates were faster than in solution.

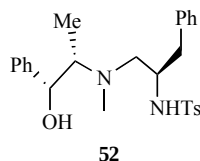
The effect of substituents on the nitrogen atom of norephedrine derivatives **47** and **48**, was studied in the ZnEt_2 addition to benzaldehyde [48]. When aminoalcohol **47** was used, enantioselectivity of 76% ee of the alcohol with (*S*)-configuration was observed whereas for aminoalcohol **48**, generally lower enantioselectivity, 29% ee of the alcohol with (*R*)-configuration was observed. This result is probably because of an unfavorable steric interaction between CH_3 and Ph groups in the intermediate complex formed between the aminoalcohol and zinc.



p-fluorinated benzyl-ephedrine **49**, **50** and -norephedrine **51** have been used as chiral auxiliaries in the enantioselective ZnEt_2 addition to aromatic aldehydes to produce the corresponding (*R*)-alcohol in 87-95% yields and 77 to 84% ee's [49]. In the addition to aliphatic aldehydes, the ee's were lower (70%). The fluorinated chain allowed the recovery of **49-51** by a simple filtration through a fluorinated reverse phase silica gel and were reusable without further purification.



An optically active *N*-sulfamide-ephedrine **52** was examined [50] in the enantioselective ZnEt_2 addition to aryl, vinyl and alkylaldehydes. The suitability of the study was extended without using $\text{Ti}(\text{O}^i\text{Pr})_4$, to explore the reaction in hexane at room temperature to obtain the corresponding (*S*)-alcohol in high enantiomeric excess (up to >99%).



The pyrazole group in systems **53** and **54** derived from pseudoephedrine **2b** and ephedrine **1a**, incorporate fluxional chirality to the chiral auxiliary (table 13). They promoted the enantioselective ZnEt_2 addition to benzaldehyde giving the corresponding (*R*)-secondary alcohol: ligands **53** (88-99% yield) with medium to high enantioselectivities (50-93% ee), ligands **54** (60-94% yield) with low to medium enantioselectivities (15-50% ee) were reported [51]. The size of the fluxional group on the ligand correlated well with the level of enantioselectivity of the product. To understand the role of the pyrazole substituent as fluxional group, two pyridine derived ligands that mimic the basic structure were evaluated in ZnEt_2 additions. The same was observed for the pyridine-ephedrine ligand [52]. The reaction with the pseudoephedrine derived amino-alcohol **55b** showed to be more selective (68% ee) than that from ephedrine derived **56a** (41% ee) (Table 13).

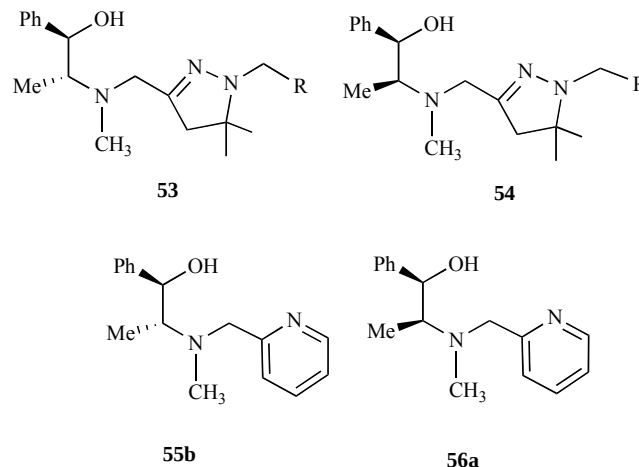
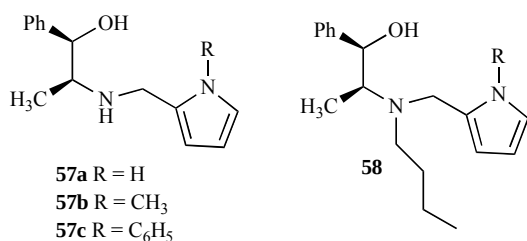


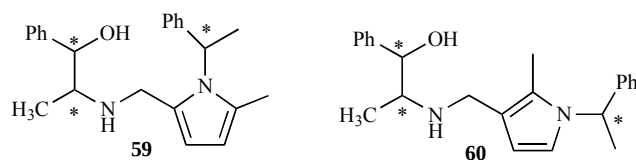
Table 13. Pyrazol **53**, **54** and pyridine **55b**, **56a** ligands derived from ephedrine as chiral auxiliaries in the ZnEt_2 addition to benzaldehyde.

Auxiliary	R	Yield (%)	Ee (%)	Config.
53	Methyl	96	50	<i>R</i>
53	Phenyl	97	77	<i>R</i>
53	Mesityl	88	81	<i>R</i>
53	Benzhydryl	98	76	<i>R</i>
53	2-Naphtyl	99	81	<i>R</i>
53	1-Naphtyl	99	89	<i>R</i>
53	2-Anthracenyl	99	93	<i>R</i>
54	Methyl	60	15	<i>R</i>
54	Phenyl	90	40	<i>R</i>
54	1-Naphtyl	94	50	<i>R</i>
55b	-----	98	68	<i>R</i>
56a	-----	97	41	<i>R</i>

(1*R*,2*S*)-*N*-substituted- and -*N*-disubstituted norephedrine ligands **57** and **58** were used as chiral auxiliaries in the enantioselective ZnEt_2 addition to benzaldehydes [53]. *N*-alkylated (1*R*, 2*S*)-norephedrine derivative **57a**, in toluene-hexane at 0° C, gave the highest enantioselectivity, for benzaldehyde, 78% ee and for *p*-bromobenzaldehyde, 84% ee of the corresponding (*R*)-alcohol. The effect of different parameters on the enantioselectivity of the product were investigated.

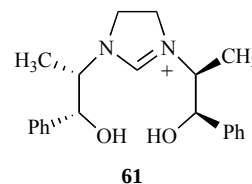


Norephedrine-based chiral auxiliaries with chiral *N*-substituted pyrrole group to have three stereogenic centers **59** and **60** [54], were used to promote the enantioselective ZnEt₂ addition to benzaldehydes in toluene at 0 °C. In this case, when 5% mol of the corresponding ligand was used, the reaction proceeded with good yields, **59** (up to 76%), **60** (up to 96%) and good enantioselectivities, **59** (up to 88%), **60** (up to 83%) of the (*R*)-alcohol derived from *p*-trifluoro-methylbenzaldehyde. The same auxiliaries produced the (*S*)-alcohol from other substituted benzaldehydes.



Enantiopure imidazolium carbene ligand **61** (X = BH₄, Cl, Br, I), derived from chiral norephedrine **3a** [55], has been investigated in the ZnEt₂ addition to aromatic aldehydes in toluene at room temperature with KO-*t*Bu in combination with different metallic salts. When Cu(OTf)₂ was used as metallic salt combined with 5% mol of ligand **61**, moderate yields (31-43%) and the best enantioselectivity was found for the tetraborate anion (84% ee) in the reaction with 1-naphthalaldehyde to get the corresponding (*R*)-

alcohol. When the metallic salt CrCl₃ was used with the tetraborate anion, the enantioselectivity was increased to 86% ee.



A family of (*1R,2S*)-*N*-alkylnorephedrine **6a**, **10a**, **62-84** (Table 14) and (*1S,2S*)-*N*-alkylnorpseudoephedrine **85-89** (Table 15) ligands were used in the enantioselective ZnEt₂ addition to benzaldehyde [56]. It was found that the norephedrine-based chiral auxiliaries **62-84** afforded an averaged enantiomeric excess of 60% of the alcohol with *R* configuration. The norpseudoephedrine-based chiral auxiliaries **85-89** favored the production of the corresponding (*S*)-alcohol. The absolute configuration of the alcohol was directed by the benzylic position of the *Ephedra* alkaloid, whereas the enantiomeric excess was influenced by the nitrogen substituent. However, the steric demand of substituents at the nitrogen position caused no enantioselectivity differences for two diastereomeric systems. From the aminoalcohols used, *N*-cyclooctylnorpseudoephedrine derivative **86** afforded the best enantiomeric excess (80%).

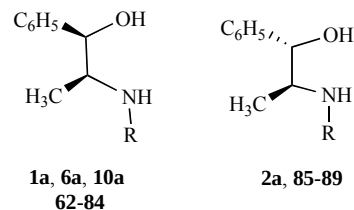


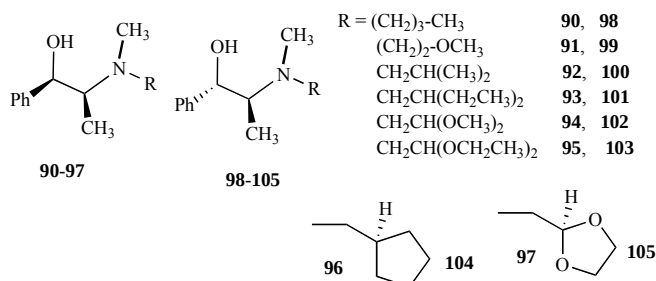
Table 14. *N*-Groups on (*1R,2S*)-norephedrine **3a**. Ee's are in favor of alcohols with *R*-configuration.

Auxiliary	N-R	Ee (%)	Auxiliary	N-R	Ee (%)
1a	CH ₃	57	72	<i>o</i> -CH ₂ biphenyl	48
6a	CH ₂ C ₆ H ₅	69	73	<i>p</i> -CH ₂ biphenyl	64
10a	CH(CH ₃) ₂	74	74	2,4,6-CH ₂ C ₆ H ₂ (CH ₃) ₃	60
62	(CH ₂) ₇ CH ₃	69	75	3,5-CH ₂ C ₆ H ₃ Br ₂	61
63	CH ₂ C(CH ₃) ₃	70	76	3,5-di- <i>tert</i> -butyl-2-MeObenzyl	80
64	1-CH ₂ naphthyl	50	77	cyclohexyl	69
65	2-CH ₂ naphthyl	70	78	2-Bn-3,5-di- <i>tert</i> -butylbenzyl	79
66	9-CH ₂ anthryl	35	79	CH(CH ₂ CH ₃) ₂	70
67	<i>o</i> -C ₆ H ₄ OH	8	80	2-indanyl	29
68	<i>o</i> -C ₆ H ₄ OMe	66	81	cyclooctyl	10
69	<i>m</i> -C ₆ H ₄ OMe	60	82	cyclopentyl	31
70	3,4-CH ₂ C ₆ H ₃ (OMe) ₂	65	83	C(C ₆ H ₅) ₃	20, <i>S</i>
71	<i>o</i> -C ₆ H ₄ C ₂ H ₅	65	84	2-1,7,7-trimethylnorbornano	47

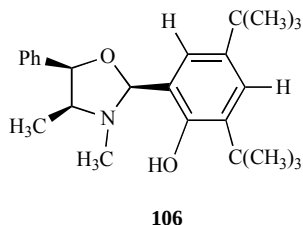
Table 15. *N*-Groups on (*1S,2S*)-norpseudoephedrine **4a**.

Auxiliary	N-R	Ee (%), conf.	Auxiliary	N-R	Ee (%), conf.
2a	CH ₃	15, <i>S</i>	87	CH ₂ anthryl	48, <i>R</i>
85	<i>o</i> -CH ₂ C ₆ H ₄ OMe	12, <i>S</i>	88	C(C ₆ H ₅) ₃	88, <i>S</i>
86	cyclooctyl	80, <i>S</i>	89	2-1,7,7-trimethyl-norbornano	47, <i>R</i>

The effect of an oxygen atom in the N-alkyl group in a series of N-alkylated (1*R*,2*S*)-ephedrine **90-97** (10% mol) and its corresponding analogous derived from (1*S*,2*S*)-pseudoephedrine **98-105** (10% mol) on the asymmetric ZnEt₂ addition to a variety of aromatic aldehydes in hexane-toluene, has been studied [57]. It was found that when the ephedrine-derived aminoalcohols were used, the (*R*)-configuration of the product was obtained, whereas the pseudoephedrine-derived ligands afforded the (*S*)-configuration of the product. Lower enantioselectivities were induced by the methoxyethyl side chain of **91** whereas the acetal bearing side chains of **94** and **97** yielded enantioselectivities comparable to their non-oxygenated side chain analogs. It was determined that the presence of oxygen could have a negative effect in terms of obtaining high levels of enantiomeric discrimination, but the effect is reduced with higher levels of substitution near the oxygen atom.



A series of oxazolidines derived from (1*R*,2*S*)-ephedrine **1a** or (1*S*,2*S*) pseudoephedrine **2a** with salicylaldehyde derivatives [58] were used as chiral auxiliaries in the ZnEt₂ additions to a variety of aromatic aldehydes. The (1*R*,2*S*)-ephedrine based oxazolidine derivative **106** (20% mol) produced the highest enantioselectivity (59% ee) in 83% yield. The results indicated that the absolute configuration of the addition product depends not only on the N-methyl, as an important stereocontrol element, but also on both the stereoelectronic effects of the substrates and oxazolidine ligand.



Ephedra-N-benzyl substituted aminoalcohols **6a**, **107** and **111-113** and the analogous N-pyridylmethyl-substituted aminoalcohols **108-110** and **114-118** derived from (1*R*,2*S*)-ephedrine **1a**, (1*S*,2*S*)-pseudoephedrine **2a**, (1*R*,2*S*)-norephedrine **3a**, and (1*S*,2*S*)-pseudonorephedrine **4a** respectively, were employed as chiral auxiliaries in the asymmetric ZnEt₂ addition to aldehydes in toluene, Table 16. [59] In the addition of ZnEt₂ to benzaldehyde, the comparative alcohol enantioselectivities, generated by ligands **107** and **111**, derived from secondary β-amino alcohols, **107** (69% ee favouring the (*R*)-enantiomer and **111** (11% ee favouring the (*S*)-enantiomer) showed the different capacity for asymmetric induction exhibited by the diastereomeric derivatives. The opposite configuration resulted in the products, may be resulted from the change in configuration at the benzylic position. In contrast, tertiary β-amino alcohols, **6a**, **110**, **112**, with the exception of **113** and **116**, showed higher enantiomeric ratios.

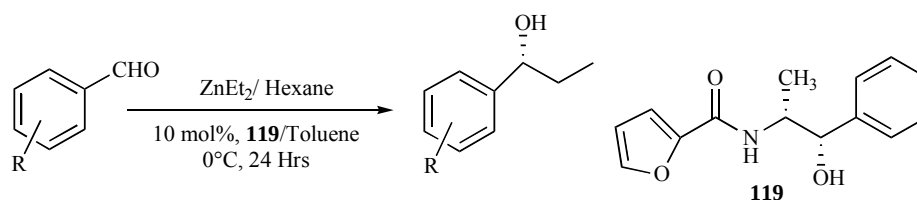
The more successful aminoalcohols (1*R*,2*S*)-N-benzylephedrine **6a**, (1*R*,2*S*)-N-pyridinylmethyl-ephedrine **110** and (1*R*,2*S*)-N-dibenzyl-norephedrine **113** were reacted with several aldehydes to demonstrate whether the reactivity is the same. Thus, the application of the N-benzylephedrine **6a** in the addition reaction afforded the (*R*)-alcohol in 82% ee. Surprisingly, the N-pyridylmethyl-ephedrine ligand **110** gives a comparable enantioselection when employed in the same reaction [75.5% ee, favoring the (*R*)-enantiomer].

Chiral amide **119** derived from (1*S*,2*R*)-norephedrine **3b** and furoic acid was found to promote the enantioselective ethylation of substituted benzaldehydes and heteroaromatic aldehydes to the secondary (*R*)-alcohols with 90% yield and 99.8% enantioselectivity at 0 °C without the addition of a promoter such as titanium tetraisopropoxide [60], Table 17. It was noted (entries 5, 10 and 12) that aldehydes containing an electron-donating oxygen atom have a significant negative influence on this reaction, especially in ee values (10 to 53%) which could be attributed to the additional coordination of the oxygen atom of the substituent with diethylzinc.

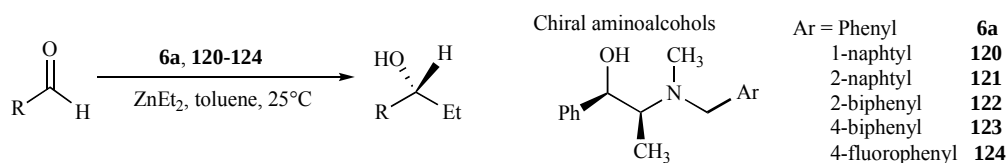
A collection of N-arylmethylephedrine derivatives **6a**, **120-124** [61] were tested as auxiliaries in the enantioselective ZnEt₂ addition to a variety of aldehydes, Table 18. The chiral ligands afforded product enantioselectivities comparable to those of N-benzylephedrine **6a**. The N-cyclohexylmethylephedrine was prepared and did not perform well in the ZnEt₂ addition to 2-

Table 16. N-benzyl and N-pyridinylmethyl ligand system derived from ephedrine.

 R⁴ X R⁵ % Ee, Conf. 6a CH₃ CH H 76, <i>R</i> 107 H CH H 69, <i>R</i> 108 H N H 33, <i>S</i> 109 H N CH₃ 14, <i>S</i> 110 CH₃ N H 69, <i>R</i>	 R⁴ X R⁵ % Ee, Conf. 111 H CH H 11, <i>S</i> 112 CH₃ CH H 60, <i>S</i> 113 benzyl CH H 24, <i>R</i> 114 H N H 4, <i>S</i> 115 H N CH₃ 32, <i>S</i> 116 CH₃ N H 10, <i>S</i>	 R⁴ R⁵ % Ee, Conf. 117 Ph H 19, <i>S</i> 118 H Ph 25, <i>R</i>
--	--	---

**Table 17.** Diethylzinc addition to different aldehydes using **119**^a.

Entry	Aldehyde	Yield, % ^b	Ee, % ^c
1	Benzaldehyde	90	99.8
2	<i>o</i> -Nitro benzaldehyde	70	68
3	<i>m</i> -Nitro benzaldehyde	82	95
4	<i>p</i> -Nitro benzaldehyde	92	85
5	<i>p</i> -Methoxy benzaldehyde	90	30
6	<i>p</i> -Methyl benzaldehyde	90	76
7	<i>p</i> -Cloro benzaldehyde	95	99.8
8	Salicylaldehyde	23	99.8
9	5-Cloro salicylaldehyde	46	99
10	4-Hydroxy-3-methoxy benzaldehyde	20	20
11	Pyrrole-2-carboxaldehyde	99	53
12	Furfural	96	10

^a The reaction was carried out in toluene, for 24 h at 0 °C. ^b Isolated yield by column chromatography. ^c Determined by HPLC analysis using chiralcel OD-H column.**Table 18.** Asymmetric 1, 2 addition of ZnEt₂ to aldehydes using chiral aminoalcohols **6a**, **120-124**. Ee's are in favor of alcohols with (*R*)-configuration^a.

Entry	Auxiliary	R-CHO	Yield (%) ^b	Ee (%) ^c
1	6a	2-C ₁₀ H ₇	88	82
2	6a	<i>t</i> -CH=CHC ₆ H ₅	82	70
3	120	2-C ₁₀ H ₇	50	81
4	120	<i>p</i> -C ₆ H ₄ OCH ₃	66	71
5	120	<i>t</i> -CH=CHC ₆ H ₅	81	64
6	121	2-C ₁₀ H ₇	78	84
7	121	1-C ₁₀ H ₇	69	87
8	121	<i>t</i> -CH=CHC ₆ H ₅	59	71
9	122	2-C ₁₀ H ₇	50	80
10	123	2-C ₁₀ H ₇	74	84
11	123	<i>t</i> -CH=CH-Ph	78	68
12	123	<i>p</i> -C ₆ H ₄ OCH ₃	72	76
13	124	2-C ₁₀ H ₇	75	84
14	124	1-C ₁₀ H ₇	37	86
15	124	<i>t</i> -CH=CHC ₆ H ₅	61	62

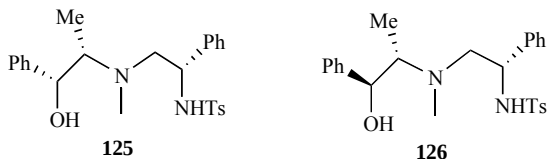
^a Absolute configurations (*R*) were determined from the literature. ^b Yield was reported after purification via flash chromatography. ^c Enantiomeric excess was determined using chiral cell HPLC with OD column in 98:2 hexanes/IPA.

naphthaldehyde, thus suggesting that the aromatic system is key in maintaining the asymmetric induction.

Sulfamide-ephedrine aminoalcohols **125**, **126** (10% mol) derived from ephedrine **1a** and pseudo-ephedrine **2a** [62] with three

stereogenic centers were applied to promote asymmetric addition of ZnEt₂ to substituted benzaldehydes in hexane at 10 °C, providing (*S*)-products in 30-99% yields and good enantio-selectivities, 81-83% ee. These sulfamide-aminoalcohols together with Ti(OiPr)₄

afforded the (*R*)-products in significant enantiomeric excesses (13–47%) and high yields (69–99%). In addition, sulfamide-ephedrine **125** and **126** were superior to the original chiral respective sources, which afforded no enantioselectivity in the $\text{Ti}(\text{O}i\text{Pr})_4$ -mediated reaction.

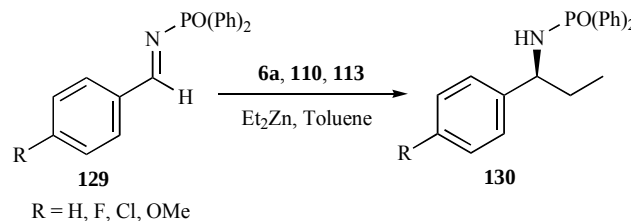


1.3. DIALKYLZINC ADDITIONS TO IMINE COMPOUNDS

The dialkylzinc addition to aromatic *N*-diphenylphosphinoylimines **127** in the presence of a catalytic or stoichiometric amount of β -aminoalcohol **46b**, as chiral auxiliary, provided chiral *N*-diphenylphosphoramides **128** with good yields 56–89 % and excellent enantioselectivities (85–91% ee) [63]. Optically active (*S*)-amines were easily obtained without any racemisation by the acidic hydrolysis of the phosphoramides **128**.

The ephedrine-based chiral auxiliaries benzylephedrine **6a**, pyridylmethylephedrine **110**, and dibenzylnorephedrine **113** were employed in the enantioselective ZnEt_2 addition to *N*-(*P*,*P*-diphenyl-phosphinoyl)imines **129**, to understand the contribution of the *N*-pyridinemethyl group [59]. The *N*-benzylephedrine **6a** and *N*-

dibenzylnorephedrine **113** performed well in these experiments and afforded products **130** in very good enantiomeric excess [(**6a**, 88–94% ee, (*R*); **113**, 49–93% ee, (*R*)]. In contrast, the ZnEt_2 addition to the phosphinoylimines mediated by the *N*-pyridylmethyl aminoalcohol **110**, yielded the corresponding chiral amine in poor enantiomeric excess [(9–24% ee, (*S*)). It has been proposed that the aminoalcohols **6a** and **110** promote the enantioselective addition via different transition states. Similar results were obtained with a collection of *N*-arylmethylephedrine derivatives **6a**, **120–124** in the enantioselective dialkylzinc to diphenylphosphinoylimines **129** (*R* = *m*-OMe) [61].



1.4. Enantioselective and Chemoselective Dialkyl Zincs Addition to Aldehydes Containing Functional Groups

1.4.1. Addition to Formyl Esters

If ZnEt_2 was added to ethyl 4-formylbutanoate (Table 19) using aminoalcohol **7b**, the respective (*S*)-5-hydroxyheptanoic acid ethyl

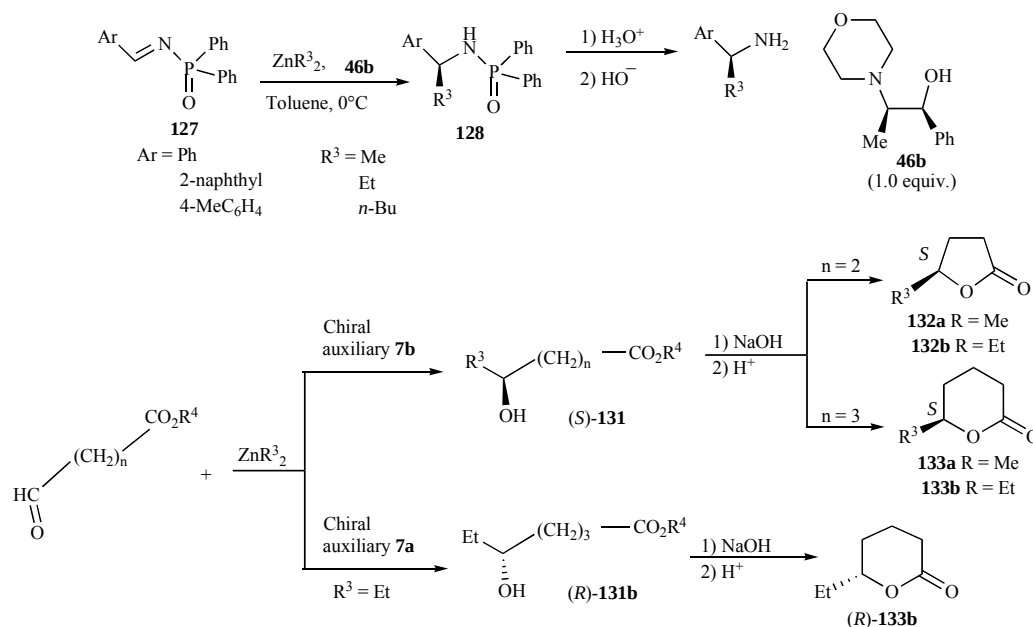


Table 19. Enantioselective synthesis of hydroxy esters and lactones^a.

Entry	Formyl Ester		R ³	Hydroxyl ester	Lactone	Yield (%)	Ee (%)	Config.
	n	R ⁴		Yield (%)				
1	2	Me	Et	88	132b	96	85	<i>S</i>
2	2	Et	Me	78	132a	95	90	<i>S</i>
3	2	Et	Et	90	132b	97	92	<i>S</i>
4	3	Me	Et	83	133b	97	88	<i>S</i>
5	3	Et	Me	88	133a	97	91	<i>S</i>
6	3	Et	Et	87	133b	98	95	<i>S</i>
7 ^b	3	Et	Et	85	133b	97	95	<i>R</i>

^aUnless otherwise noted, **7b** was used. ^b**7a** was used.

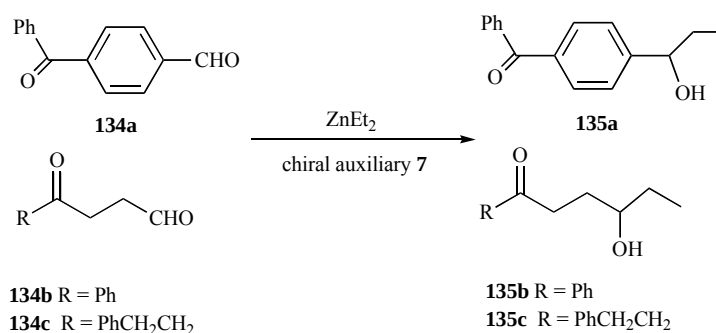
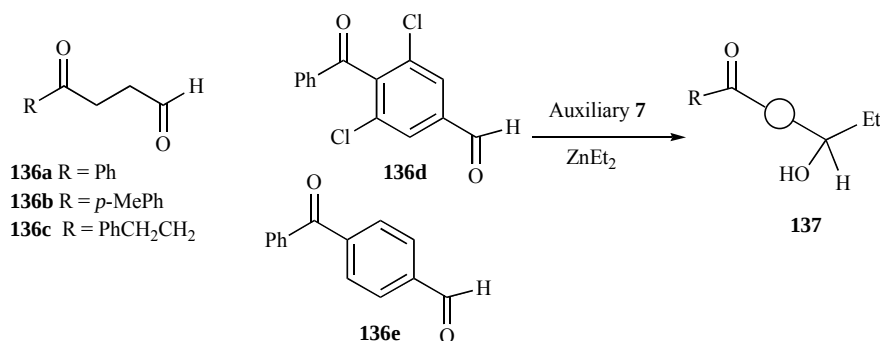


Table 20. Asymmetric synthesis of hydroxy ketones by using 7a or 7b as chiral auxiliaries.

Entry	Ketoaldehyde	Auxiliary	Hidroxyketones 135		
	134		Yield (%)	Ee (%)	Config. ^a
1	a	7b	84	91	S
2	b	7b	52	87	-
3	b	7a	48	85	-
4	c	7b	47	81	-

^aTentatively assigned.

ester (**131a**, *n* = 3) was obtained in 87% yield (entry 6) [64]. This result demonstrates that the aldehyde reacts enantio- and chemoselectively with ZnEt₂ in the presence of an ester group. The (*S*)-(-)-5-Heptanolide **133a**, a key intermediate in a steroid synthesis [65], with 98% yield and 95% ee (entry 6) was obtained after the hydrolysis of the ester group and its spontaneous cyclization. When **7a** was used as a chiral auxiliary, (*R*)-**133b** in 97 % yield with 95 % ee was obtained (entry 7). Similarly, (*S*)-(-)-4-hexanolide **132** was isolated in 92% ee (entry 3) [66].

1.4.2. Addition to Keto Aldehydes

When 4-benzoylbenzaldehyde (**134a**) was treated with ZnEt₂ using **7b** as chiral auxiliary (table 20), 4-(1-hydroxypropyl)benzo-phenone (**135a**) in 84% yield with 91% ee was enantioselectively and chemoselectively obtained (entry 1) [22]. Both enantiomers of γ -hydroxy ketone **135b** were isolated with similar ee's using **7b** and **7a** (entries 2 and 3). The method is a unique approach to optically active γ -hydroxy ketones **135b,c**.

Aminoalcohols **7a** and **7b** were used to promote enantioselective ZnEt₂ additions to keto aldehydes **136a-e** to give the chiral hydroxyketones **137a-c** [67]. When **7a** was used in the ZnEt₂ addition to **136a**, (-)-**137a** with 85% ee was obtained in 48% yield. However, when **7b** was employed, (+)-**137a** with 87% ee was ob-

tained in 52% yield. When 12% mol of either **7a** or **7b** is used, diethylzinc is added to **136e** with 91% ee and 82% yield at 0 °C in toluene. The other substrates gave 81-87% ee without reaction of the keto group.

1.4.3. Addition to Furyl Aldehydes

Both enantiomers of aminoalcohol **7b** were used to promote the enantioselective dialkylzinc additions to aldehydes derived from furane, thiophene, and pyrrole, which afforded the respective chiral alcohols with 70-86% ee [34b].

Chiral 2- and 3-furyl alcohols **139a-c** were synthesized in high ee's by the enantioselective dialkylzinc addition to 2-furyl aldehydes **138a,b** or 3-furyl aldehyde **138c** using **7** as chiral auxiliary (Table 21) [68]. Independently, the enantioselective synthesis of 5-substituted-2-furyl alcohols using **7** was reported [69].

1.4.4. Addition to Pyridyl Aldehydes

The diisopropylzinc addition to pyridine-3-carbaldehyde **140** in the presence of aminoalcohol **7b** afforded (-)-2-methyl-1-(3-pyridyl)-propanol (**141**, R³ = *i*-Pr) in 86% ee [70]. Other pyridylalkyl alcohols were obtained in medium to good ee's. In contrast to the use of benzaldehyde, isopropylation of nicotinaldehyde afforded the respective alcohol in higher enantioselectivity than other dialkylzincs with straight carbon chains.

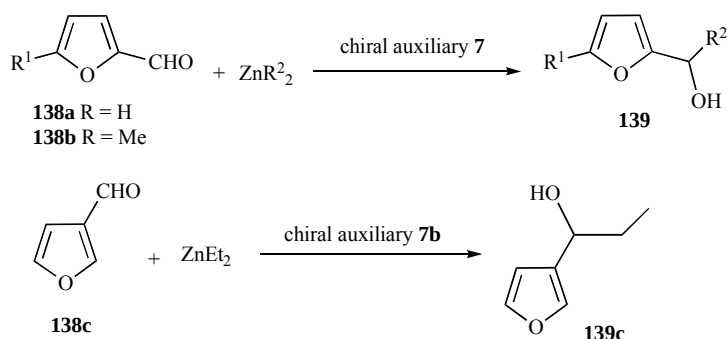
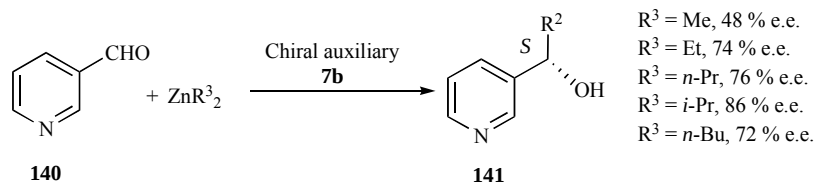


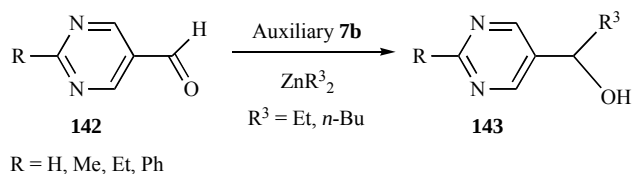
Table 21. Synthesis of furyl alcohols by the enantioselective alkylation of furyl aldehydes.

Entry	Aldehyde 138	R ³	Auxiliary	Yield (%)	Ee (%)	Config.
1	a	Et	7b	83	89	S
2	a	Et	7a	74	88	R
3	a	Et	7b	82	93	S
4	a	<i>n</i> -Bu	7b	58	90	-
5	b	Et	7b	79	87	S
6	b	Et	7b	88	91	S
7	c	Et	7b	52	94	-

Reactions are run in hexane at 0 °C.



The enantioselective ZnEt_2 addition to pyrimidine-5-carboxaldehydes **142**, in the presence of chiral aminoalcohol **7b**, generates the corresponding (S)-5-pyrimidyl alkanols **143** with up to 94% ee [71].



1.4.5. Addition to Aromatic Dialdehydes

When **7b** was used in the ZnEt_2 addition to terephthalaldehyde **144c**, the corresponding diol [1,4-bis(1-hydroxypropyl) benzene] **145c** was obtained in 74% yield (ratio of optically active **145c** to meso-**145c** = 80 : 20) (ratio *dl*/meso = 80:20) when the reaction was performed by refluxing a mixture of THF-hexane solution for 30

min. When the same reaction was carried out at room temperature, the diol with 100% ee was obtained (54%, *dl*/meso = 84:16) [72]. Configuration of diol **145c** was tentatively assigned as (*S,S*) based on the results for the formation of (S)-1-phenylpropanol from benzaldehyde and ZnEt_2 in the presence of **7b**. The opposite enantiomer of **145c** was obtained in 100% e.e. (62% yield) when **7a** was used.

On stirring THF-hexane or benzene-hexane for 30 minutes at 0 °C, chiral aminoalcohol **7b** promotes the enantioselective dialkylzinc additions to Phthalaldehydes **144a-c** to give the optically active diols **145a-c** [73].

The reaction of phthalaldehyde **144a** using diethylzinc in the presence of **7b** gave lactol **146a** as a major product by the *in situ* cyclization of monoalkylated alcohol.

In the case of iso- and terephthalaldehydes **144b,c**, the corresponding diols **145b,c** were obtained with very high enantioselectivity, as shown in Table 22. In most cases, formation of the meso products was observed. The chiral diols produced from these reac-

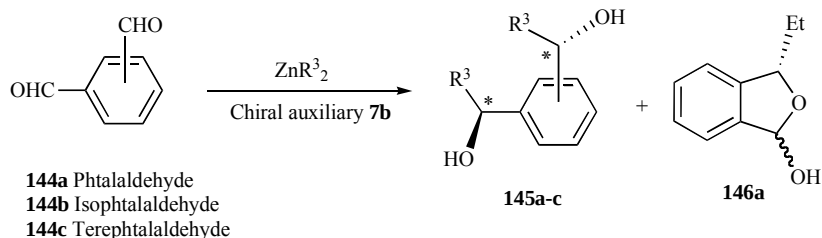
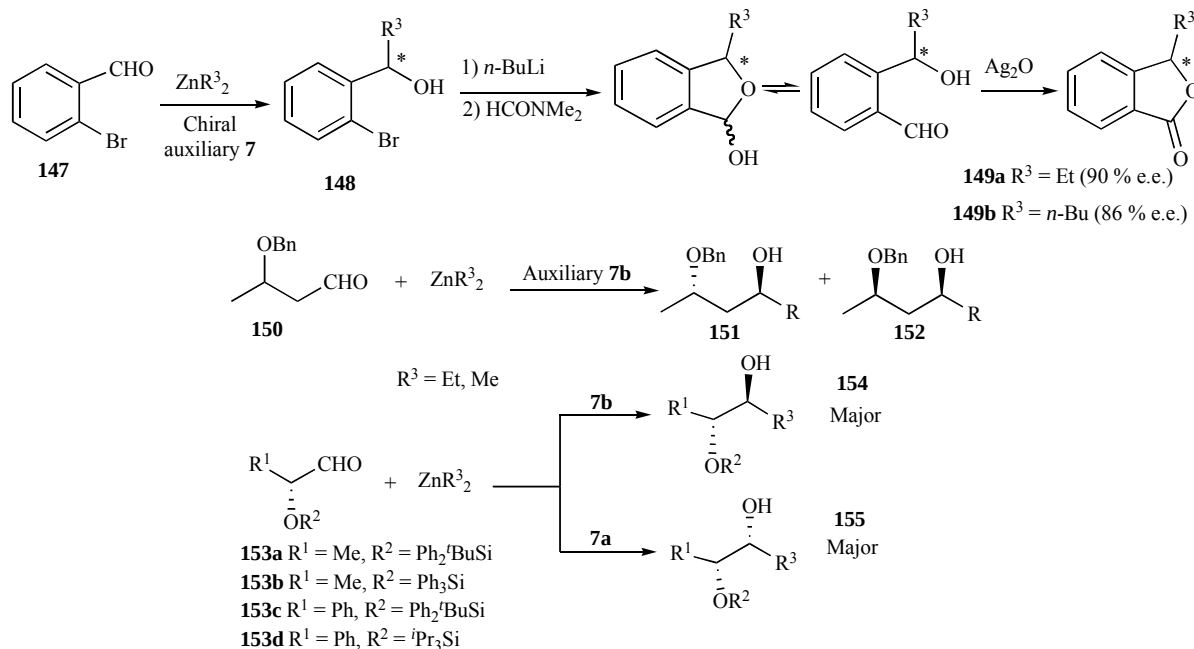


Table 22. Dialkylzinc additions to iso- and terephthalaldehydes **144b,c** promoted by the chiral auxiliary **7b**.

Entry	ZnR ³ ₂	Aldehyde 144	Solvent	T °C	Equivalents 7b	Yield (%)	Ee (%)	Ratio <i>dl</i> / <i>meso</i>
1	ZnEt ₂	b	THF-hexane	r.t.	0.2	74	>99	80/20
2	Zn ^{<i>i</i>} Pr ₂	b	Toluene-hexane	0	0.5	66	>99	91/9
3	Zn ^{<i>i</i>} Pr ₂	c	Toluene-hexane	0	0.2	70	>99	89/11
4	Zn ^{<i>i</i>} Pr ₂	c	Toluene-hexane	0	0.2	60	>99	>99/1

Table 23. Dialkylzinc additions to **153a-d** promoted by **7b** and **7a** as chiral auxiliaries^a.

Entry	ZnR ₂	Aldehyde 153	Auxiliary (%mol)	T (°C)	Time (h)	Yield (%) 154+155	Ratio 154/155
1	ZnEt ₂	a	7b (5)	0	16	46	95/5
2	ZnEt ₂	a	7a (5)	0	16	56	4/96
3	ZnMe ₂	a	7b (5)	0	26	24	85/15
4	ZnMe ₂	a	7a (5)	0	26	32	7/93
5	ZnEt ₂	b	7b (20)	r.t.	20	46	98/2
6	ZnEt ₂	b	7a (20)	r.t.	20	50	7/93
7	ZnEt ₂	c	7b (20)	r.t.	22	46	97/3
8	ZnEt ₂	c	7a (20)	r.t.	23	46	8/92
9	ZnEt ₂	d	7b (5)	r.t.	22	57	96/4
10	ZnEt ₂	d	7a (20)	r.t.	21	68	7/93

^aThe reactions were carried out in toluene solution

tions were observed to carry out an autocatalytic diethylzinc addition to the dialdehydes **144b,c** in the presence of Ti(O-*i*-Pr)₄ with up to 91% yield but only ≤44% ee.

1.4.6. Addition to Other Functionalized Aldehydes

Aminoalcohols **7a,b** as chiral auxiliaries were used in the enantioselective dialkylzinc addition to 2-bromobenzaldehyde **147** for the synthesis of optically active 3-alkylphthalides **149** (86-90% ee) [74]. The optically active alcohols **148** were converted into opti-

cally active lactone **149** without racemization. When **7a** was used in the enantioselective Zn(*n*-Bu)₂ addition to aldehyde **147**, (*R*)-2-*n*-butylphthalide **149b**, with 86% ee was obtained. The enantiomer (*S*)-2-*n*-butylphthalide was obtained from aminoalcohol **7b**. The enantioselective ZnEt₂ addition gave (*S*)-3-ethylphthalide **149a** with 90% ee.

Aminoalcohol **7b** was used in dialkylzinc additions to functionalized aldehydes such as the β-alkoxy aldehyde **150** [75] and the α-alkoxy aldehydes **153a-d** (Table 23) [76]. In the reaction of alde-

hyde **150**, both *erithro* **151** and *threo* **152** products preferred an (S)-configuration for the hydroxyl carbon. The ee's of these products were in the range of 61-85% [75].

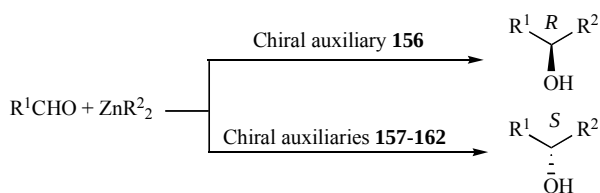
The reaction of **153a-d** led to **154** with (S)-configuration, as the major product, when the chiral auxiliary **7b** was used. Nevertheless, when chiral aminoalcohol **7a** was used, **155** with (R)-configuration in the new chiral center, was preferred as the major product. The ratio of the diastereomeric products was from 85/15 to 98/2 [76]. Hence, in the reactions of **150** and **153**, the configuration of the chiral center of both products was controlled by the chiral auxiliary rather than by the chiral center already present in the substrates.

1.5. Ephedrine Immobilized on Supports

The use of polymer-based chiral auxiliaries is very attractive for enantioselective catalysis because of the easy recovery and reuse of the often expensive chiral auxiliary. Polymer-based chiral auxiliaries also make possible to carry out the enantioselective reactions in a flow system for continuous production. Additionally, the isolation site may increment the catalyst activity and lifetime because it stops the aggregation of the catalytically active species. Three class of heterogeneous chiral auxiliaries were found in the asymmetric dialkylzinc additions: (1) polymers anchored with chiral auxiliaries, (2) chiral auxiliaries supported on silica gel or alumina, and (3) dendritic chiral auxiliaries.

1.5.1. Polymers Anchored with Chiral Auxiliaries

Polymer-supported *ephedra* chiral auxiliaries have been reported in the enantioselective ZnR_2 addition to aldehydes. Fréchet, Itsuno and Soai were the first to use polymer-anchored chiral auxiliaries **156**, **157-162** for the asymmetric dialkylzinc addition to aldehydes [20, 77]. It was found that, in general, the enantioselectivity of the polymer-supported chiral auxiliaries was lower than the respective chiral aminoalcohol.



Chiral alcohols were produced in high ee's by the enantioselective dialkylzinc addition to benzaldehydes in presence of the polymer-supported-ephedrine **156**, as a chiral auxiliary (Table 24) [77]. The chiral polymer was recovered and reused with a little change in enantioselectivity (entries 5-7). The ethylation of benzaldehyde in the presence of the chiral polymer **156** give (R)-1-phenylpropanol in 83% yield with 89% ee (entry 1). When polymer-bound-pseudoephedrine **160a**, a diastereoisomer of **156**, was used as auxiliary for the addition reaction, (S)-1-phenylpropanol in 84% yield with 44% ee was obtained (entry 2).

Although polymer auxiliary **156** was good in the ethylation reaction of aromatic aldehydes, the enantioselectivity was low in the same reactions with aliphatic aldehydes. However, the use of polymer-supported (1*S*,2*R*)-*N*-ethylnorephedrine **157** increased the enantioselectivity in the ethylation reaction of aliphatic aldehydes [78]. In summary, polymer auxiliary **156** was the most enantioselective (90% ee) in the alkylation of aromatic aldehydes and polymer auxiliary **157** (63% ee) in aliphatic aldehydes.

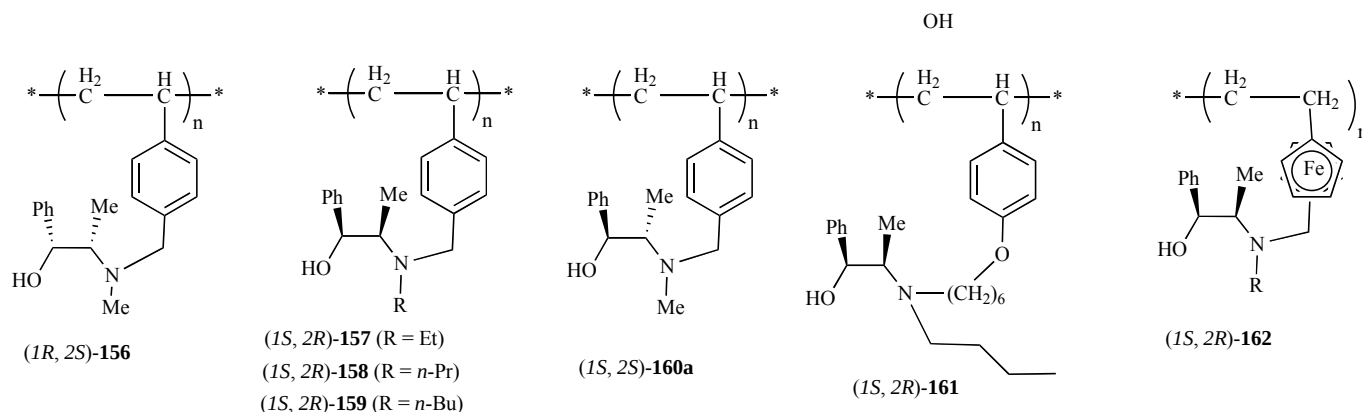
On the other hand, polymer-supported **161** containing a hexamethylene spacer, was probed to be an effective chiral auxiliary in the enantioselective alkylation of both aromatic and aliphatic aldehydes (Table 25) [79]. The polymer-supported auxiliary **161** was the most enantioselective in the ZnEt_2 addition to aliphatic aldehydes (69-70% ee's) (entries 2 and 3), probably due to the presence

Table 24. Enantioselective addition of ZnR_2 to aldehydes using **156a** and **160a** as polymer-supported auxiliaries^a.

Entry	Auxiliary	R ¹	R ³	Yield (%)	Ee (%)	Config.
1	156	Ph	Et	83	89	<i>R</i>
2	160a	Ph	Et	84	44	<i>S</i>
3	156	Ph	<i>n</i> -Pr	57	86	<i>R</i>
4	156	Ph	<i>n</i> -Bu	21	68	<i>R</i>
5	156	4-Cl-C ₆ H ₄	Et	78	83	<i>R</i>
6	156^b	4-Cl-C ₆ H ₄	Et	77	77	<i>R</i>
7	156^c	4-Cl-C ₆ H ₄	Et	69	76	<i>R</i>

^aChloromethylated polystyrene cross-linked with 1% divinylbenzene was used as a support. The reactions were run in hexane at room temperature using 10% mol (calculated as the content of the amino alcohol part) of auxiliary.

^bFirst recycle. ^cSecond recycle.

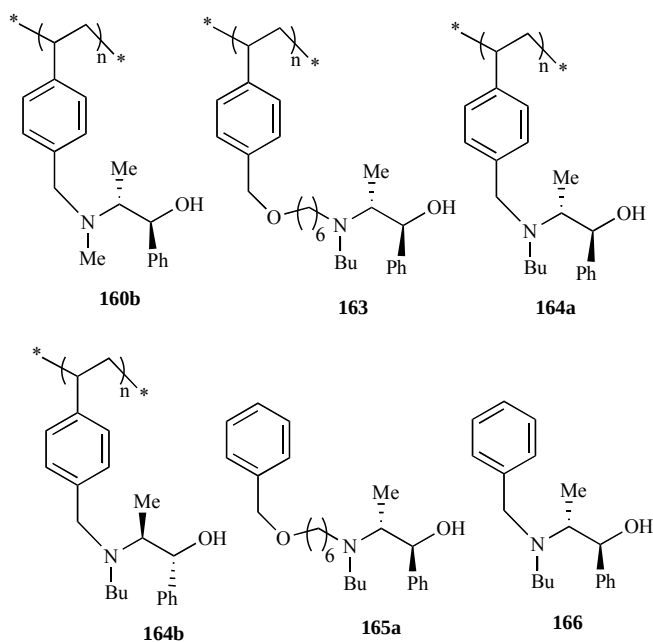


of the hexamethylene spacer. This polymer **161** was easily recovered without any loss of the enantioselectivity (entry 3).

Table 25. Enantioselective addition of diethylzinc to aldehydes using polymer-supported auxiliary **161**.

Entry	R ¹	(S)-alcohol	
		Yield (%)	Ee (%)
1	Ph	91	82
2	<i>n</i> -octyl	75	69
3 ^a	<i>n</i> -octyl	80	70
4	PhCH ₂ CH ₂	77	75

^aRecycled polymer auxiliary **161** was used.



Polymer-supported chiral auxiliary possessing *N*-ferrocenyl-ephedrine **162** also works in the enantioselective ZnEt₂ addition to benzaldehyde [80]. When 5% mol of **162** and 33% mol of (*1S,2R*)-ephedrine unit were used, (*S*)-1-phenylpropanol with 72% ee in 85% yield was obtained.

The use of the polystyrene-supported *N*-disubstituted ephedrine and norephedrine ligands **163b**, **163**, **164a**, showed 89%, 82%,

and 17% ee, respectively, for the ZnEt₂ addition to benzaldehyde [81]. In general, it was found that, the enantioselectivity of the polymer-supported chiral auxiliary was lower compared to that of their corresponding monomers **6a** (92% ee), **165a** (99% ee), **166** (89% ee). Polymer-supported chiral auxiliary **163** gave the highest enantioselectivity among the amino alcohols-based polymers for the reaction of aliphatic aldehydes; it promoted the enantioselective diethylzinc addition to nonyl aldehyde with 71% ee, its monomer ligand **157** gave 73% ee. If hexane is used as solvent in these reactions, the chiral polymer can be recycled by simple filtration, the recycled polymers showed the same enantioselectivity as the original polymers.

Polymer-supported chiral auxiliary **164b** was studied in a flow system [82a]. However, after a period of reaction time, a decrease in enantioselectivity due to the gradual chemical degradation of the auxiliary sites was observed. Under typical reaction conditions, polystyrene-supported ephedrine ligand **164b** gave higher ee of the product compared with the use of the monomeric (*1R,2S*)-*N*-benzylephedrine **6a**. The high ee could be due to the fact that the flow system can produce a high molar ratio of the auxiliary compared with ZnEt₂ to substrate. On the other hand, the flow system could continuously remove the initial alkoxide formed that might act as auxiliary to form a product with low ee. Highly enantioselectivities (up to 97% ee) by using polymer auxiliaries were obtained in the ZnEt₂ addition to benzaldehyde [82b].

1.5.2. Chiral Auxiliaries Supported on Silica Gel or Alumina

Chiral auxiliary ligands were also anchored to inorganic supports to be used in the dialkylzinc additions to aldehydes. The use of alumina or silica gel supported-ephedrine **167a**, **168a**, **169a**, **169b** and (*1S,2R*)-*N*-alkylnorephedrine **170b**, in this reaction gave low enantioselectivities (< 60% ee) [83].

The heterogeneous chiral auxiliary **168a** was obtained by the binding of a chiral (*1R,2S*)-ephedrine on mesoporous micelle-templated silicas (MTS) [84a]. It gave low ee (35-37%) in the ZnEt₂ addition to benzaldehyde. The heterogeneous chiral auxiliary containing high surface densities of the chiral aminoalcohol ligand and organics was shown to have enantioselectivity up to 76% ee [84b].

Chiral secondary (*R*)-alcohols were obtained in moderate ee's using (*1R,2S*)-**167a** (Table 26) and (*1R,2S*)-**168a** (Table 27) as heterogeneous chiral auxiliaries [83]. (*1R,2S*)-ephedrine immobilized on silica gel coated with polystyrene **169a** was found to be the first example used to afford optically active secondary alcohol in moderate ee's (Table 28) [84].

Chiral (*1R,2S*)-(-)-norephedrine amide, immobilized onto a nanoporous material **171**, has been studied for the enantioselective

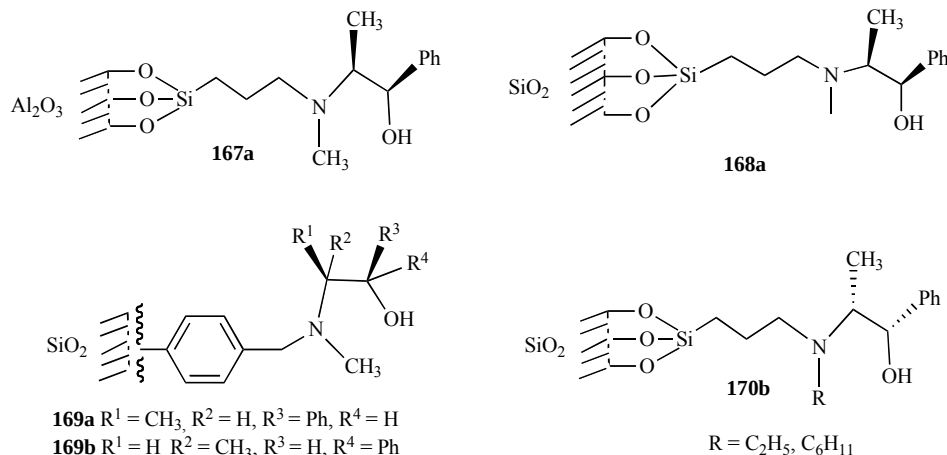


Table 26. Enantioselective addition of R^3_2Zn to aromatic aldehydes using a heterogeneous chiral auxiliary immobilized on alumina 167a^a.

Entry ^b	R ¹	R ³	Solvent	Time (days)	Yield (%)	Ee (%)	Config.
1	Ph	Et	hexane	2	81	43	<i>R</i>
2	Ph	Et	hexane	5	79	59	<i>R</i>
3	Ph	<i>n</i> -Bu	hexane	5	13	40	<i>R</i>
4	4-CH ₃ -C ₆ H ₄	Et	hexane	5	58	55	<i>R</i>
5	Ph	Et	hexane:benzene	3	66	50	<i>R</i>

^aReactions were run at 0 °C. ^bAmounts of aldehyde, auxiliary, and dialkylzinc are 0.98 mmol, 0.130 g, and 2.2 mmol, respectively.

Table 27. Enantioselective addition of ZnR^3_2 to aromatic aldehydes using a heterogeneous chiral auxiliary immobilized on silica gel 168a^a.

Entry	R ¹	R ³	Solvent	Yield (%)	Ee (%)	Config.
1	Ph	Et	hexane	47	37	<i>R</i>
2	Ph	<i>n</i> -Pr	hexane	12	38	<i>R</i>
3	2-naphthyl	Et	hexane	56	33	<i>R</i>
4	4-CH ₃ -C ₆ H ₄	Et	hexane	33	40	<i>R</i>
5	Ph	Et	hexane:benzene	69	39	<i>R</i>

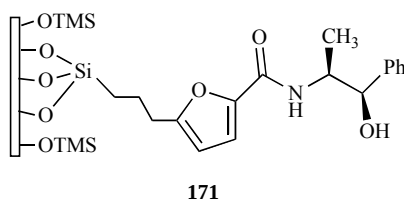
^aReactions were run at 0 °C for 4-5 days.

Table 28. Enantioselective addition of ZnR^3_2 to aldehydes using ephedrine 1a immobilized on silica gel coated with polystyrene as chiral auxiliary 169a.

Entry	R ¹	R ³	Solvent	T (°C)	Time (days)	Yield (%)	Ee (%)	Config.
1	Ph	Et	hexane	0	5	56	36	<i>R</i>
2	Ph	Et	hexane:benzene	0	5	64	56	<i>R</i>
3	2-naphthyl	<i>n</i> -Bu	hexane:benzene	rt	3	66	49	<i>R</i>
4 ^a	4-CH ₃ -C ₆ H ₄	Et	hexane:benzene	rt	3	78	49	<i>R</i>
5	Ph	Et	benzene	rt	5	63	48	<i>R</i>
6	<i>n</i> -octyl	Et	hexane:benzene	rt	5	40	48	<i>R</i>

^aRecycled auxiliary is used.

ZnEt₂ addition to benzaldehydes [85]. The immobilized ligand was more enantioselective than the homogeneous catalyst (40% ee), affording the respective chiral alcohols in 92% yield with a 95% ee. Immobilized chiral auxiliary **171** was also investigated for benzaldehydes with electron donating and withdrawing groups, and was found to give excellent yields and moderated to high enantioselectivities (47-96% ee). It was found that, electronic and steric effects of the substrates affects the activity and the enantioselectivity of the auxiliary. In addition, the ancillary was stable enough and could be recyclable.



1.5.3. Dendritic Chiral Auxiliaries

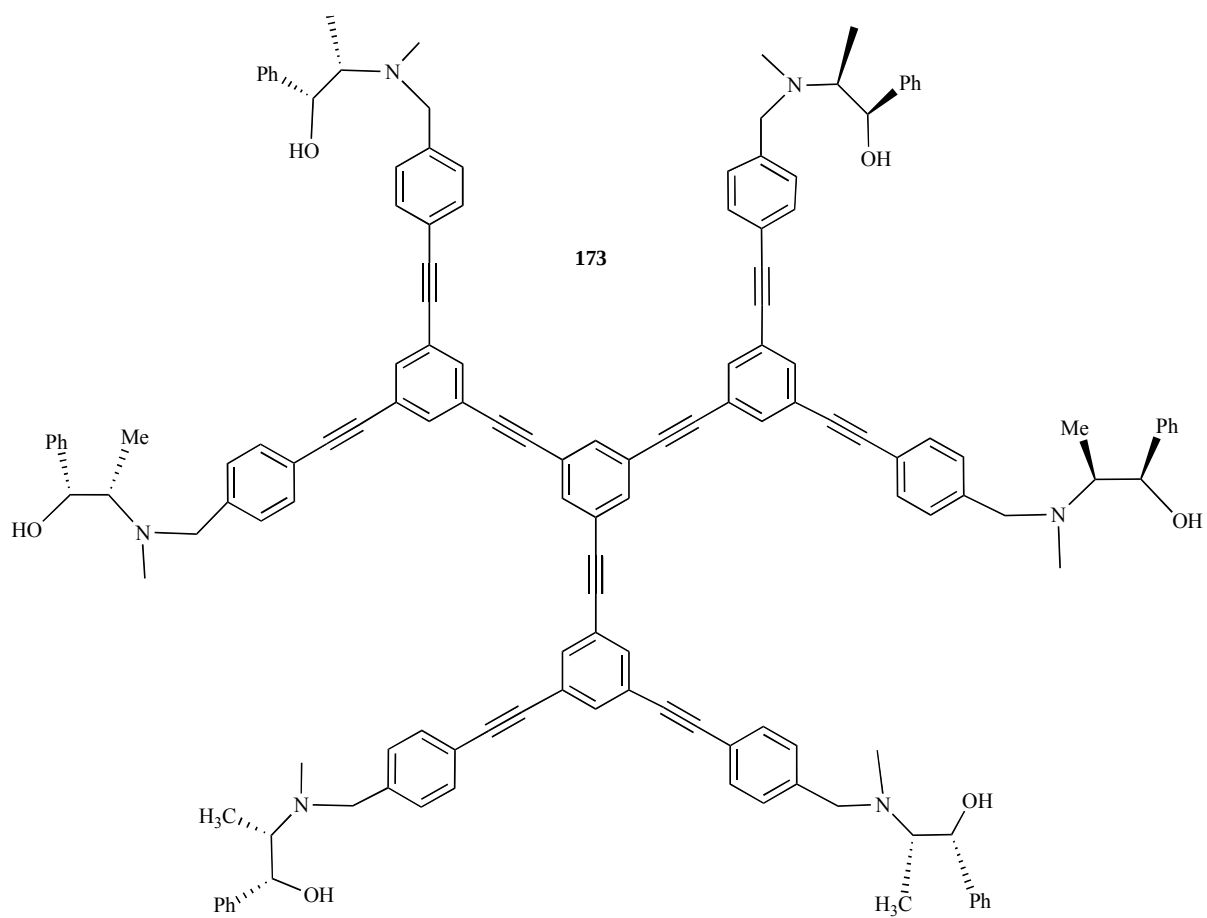
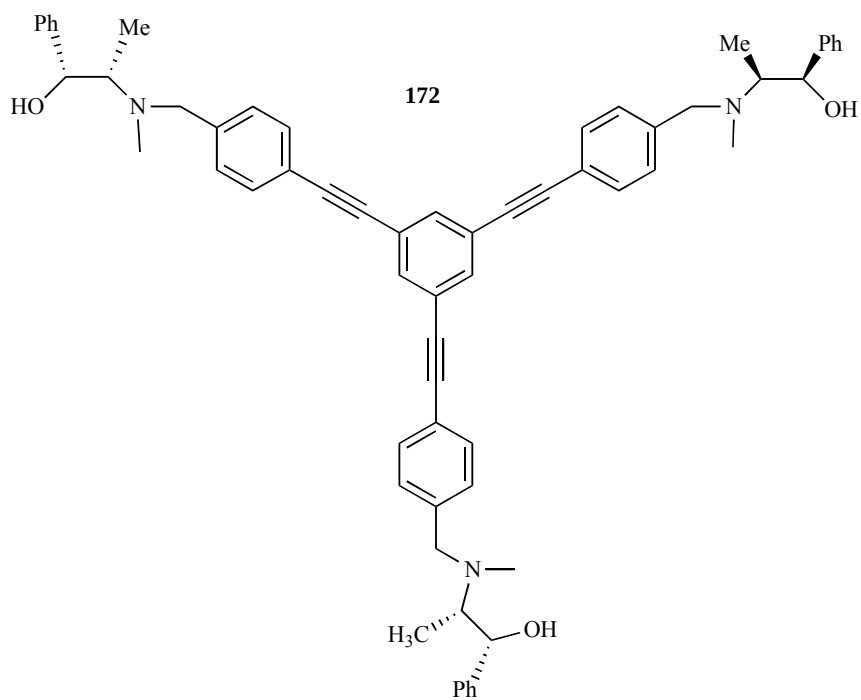
(1*R*,2*S*)-Ephedrine-*N*-dendrimers **172** and **173** having rigid branches and ephedrine groups on the periphery were used as chiral auxiliaries in dialkylzinc addition to aromatic aldehydes [86]. Both dendrimers showed similar enantioselectivities in the ZnPr₂ additions, **172** (77-86% ee) and **173** (80-86%).

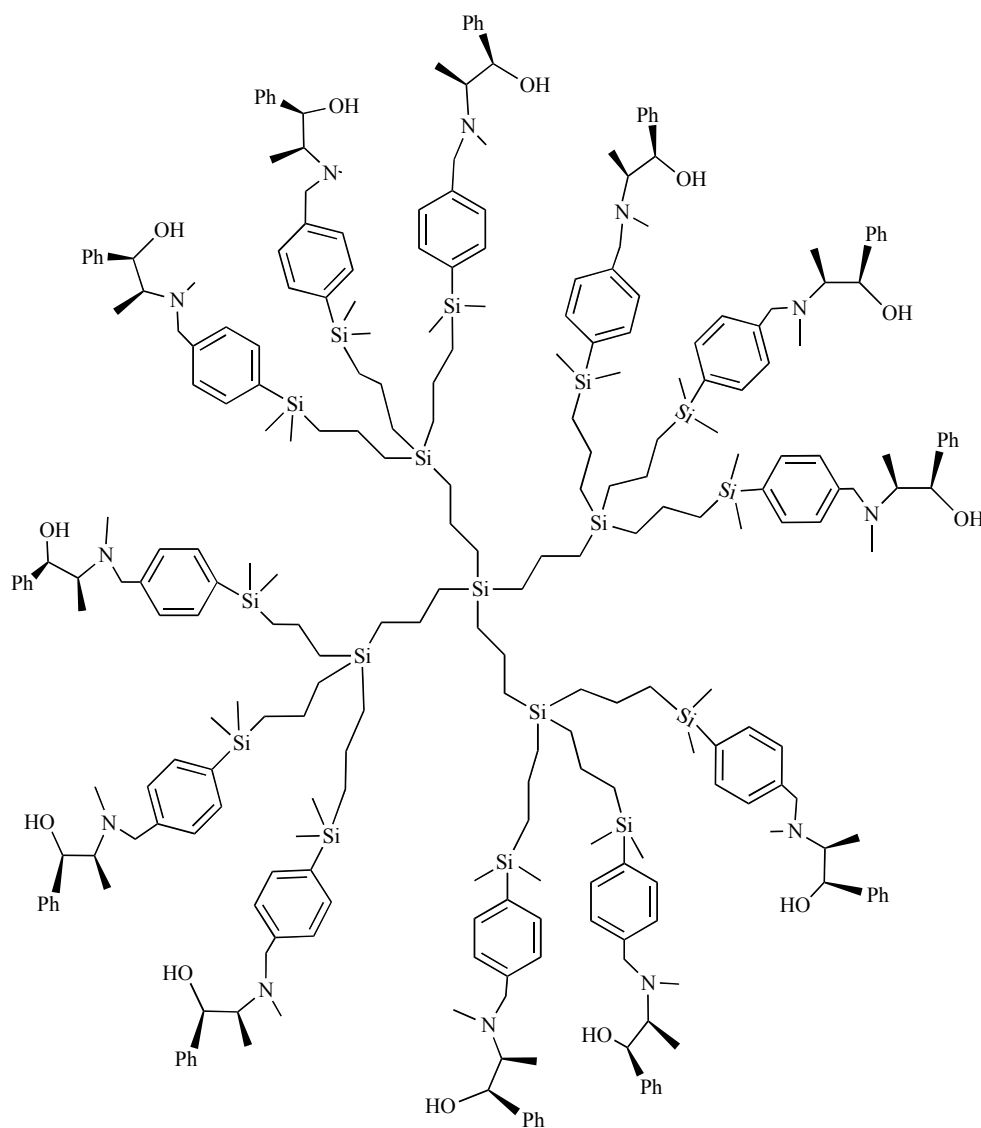
(1*R*,2*S*)-Ephedrine-*N*-dendrimer **174**, bearing two, four or twelve chiral ephedrine groups on the hyperbranched flexible carbosilane chain-ends, was good as chiral auxiliary for the enantioselective ZnEt₂ and Zn*i*-Pr₂ addition to sixteen aromatic aldehydes in toluene at 0 °C to give chiral (*R*)-*sec*-alcohols with 74-93% ee [87].

1.6. Enantioselective Addition of Various Organozinc Reagents to Aldehydes

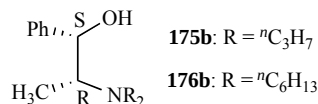
1.6.1. Alkynylzinc Reagents

The first report of an enantioselective dialkynylzinc addition to an aldehyde was carried out by using chiral *N*-dialkyl-*N*-ephedrine **7b**, **8b**, **16b**, **17b**, **45b**, **175b** and **176b** as chiral auxilia-



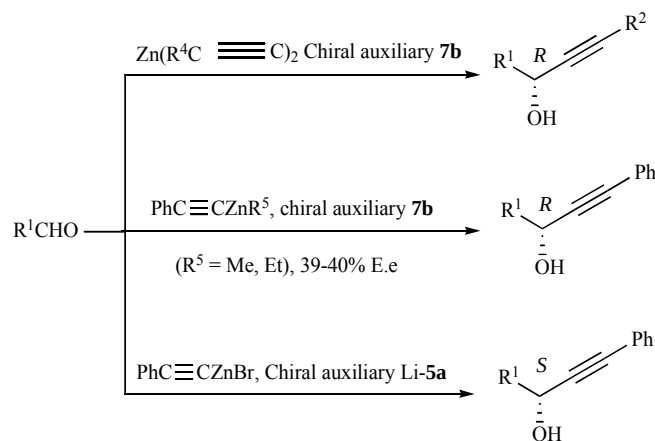
Dendritic chiral auxiliary **174**

ries [88]. The addition reactions showed good yields but lower ee's (7-34%).



The best results were obtained in the presence of amino alcohol **7b** as auxiliary (5% mol), the corresponding chiral alkynyl alcohols in moderate ee's were obtained (Table 29) [88]. The addition of bis(2-phenylethynyl)zinc to benzaldehyde in the presence of 5% mol of **7b** gave (*R*)-1,3-diphenyl-1-propynyl-3-ol in 99% yield and 34% ee (entry 1). The enantioselectivity is increased when 20% mol of **7b** was used (43% ee) (entry 2). Alkylalkynylzinc reagents react with aldehydes in the same form as the ZnR_2 reagents did [21]. The enantioselectivity is increased if ethyl- or methyl-(2-phenylethynyl)-zinc was used instead of the dialkynylzinc in the presence of 5% mol of **7b**.

The use of stoichiometric quantities of the optically pure methylephedrine **5a** and its lithium salt to carry out alkynylzinc additions to aldehydes has been studied [89]. In this work, the authors



reported that by using one mol of **5a**, the addition of 2 mol of phenylacetyleneethylzinc to (3-phenoxy-4-fluorophenyl)-benzaldehyde gave the respective propargyl alcohol with only 42% ee. However, when the lithium salt Li-**5a** with phenylzinc bromide was used, the

Table 29. Enantioselective addition of alkynylzincs to aldehydes using **7b as auxiliary^a.**

Entry	R ¹	R ⁴	Time (h)	Yield (%)	Ee (%)	Config.
1	Ph	Ph	14	99	34	<i>R</i>
2	Ph	Ph	14	99	43 ^b	<i>R</i>
3	<i>trans</i> -PhCH=CH	Ph	14	97	10	-
4	Ph	<i>n</i> -hexyl	44	81	22	-
5	Ph	<i>n</i> -Bu	52	93	20	-
6	Ph	Me ₃ Si	168	36	21	<i>R</i>
7	<i>n</i> -octyl	Me ₃ Si	48	80	24	<i>R</i>

^aThe reactions were carried out at room temperature in hexane/THF (2:3) in the presence of **7b**. Unless otherwise stated, molar ratio R¹CHO/dialkylzinc/**7b** = 1.0:2.0:0.05.^bMolar ratio: R¹CHO/dialkylzinc/**7b** = 1.0:2.0:0.2.**Table 30. Enantioselective addition of 2-phenylethynylzinc bromide to aldehydes using lithium salt of **5a** as a chiral auxiliary.**

Entry	Aldehyde	T (°C)	Time (h)	Yield (%)	Ee (%)	Config.
1	3-PhO-4-F-C ₆ H ₃ CHO	0-5	20	80	88	<i>S</i>
2	PhCHO	-30	19	70	80	<i>S</i>
3	<i>t</i> -BuCHO	-30	24	50	67	-
4	<i>n</i> -BuCHO	-30	20	90	19	-

Table 31. Alkynylzinc additions to aldehydes using **5b-Zn(OTf)₂-Et₃N.**

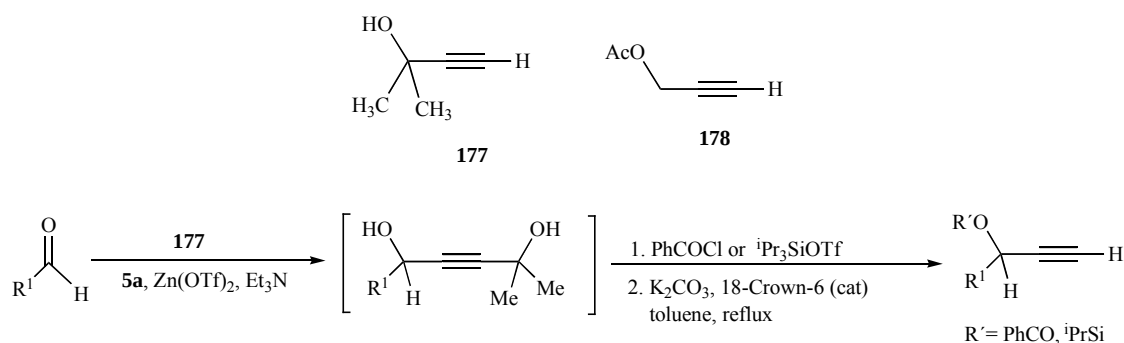
Entry	Alkyne	Aldehyde	Time (h)	Yield (%)	Ee (%)
1	PhC≡CH	C ₆ H ₁₁ CHO	1	99	96
2	Ph(CH ₂) ₂ C≡CH	C ₆ H ₁₁ CHO	4	98	99
3	Ph(CH ₂) ₂ C≡CH	Me ₂ CHCHO	2	90	99
4	PhC≡CH	Me ₂ CHCHO	2	95	90
5	Ph(CH ₂) ₂ C≡CH	<i>trans</i> -PhCH=CHCHO	20	39	80
6	Ph(CH ₂) ₂ C≡CH	Me ₃ CCHO	2	84	99
7	PhC≡CH	Me ₃ CCHO	2	99	94
8	Ph(CH ₂) ₂ C≡CH	PhCHO	20	52	96
9	PhC≡CH	PhCHO	20	53	94
10	Me ₃ SiC≡CH	C ₆ H ₁₁ CHO	2	93	98
11	Ph(CH ₂) ₂ C≡CH	Me ₃ CCH ₂ CHO	2	72	99
12	PhC≡CH	Me ₃ CCH ₂ CHO	2	90	97
13	Me ₃ SiCH ₂ C≡CH	C ₆ H ₁₁ CHO	4	84	98
14	TBDMISOCH ₂ C≡CH	C ₆ H ₁₁ CHO	5	83	98
15	(EOT) ₂ CHC≡CH	C ₆ H ₁₁ CHO	8	90	98
16	CH ₂ =C(Me)C≡CH	C ₆ H ₁₁ CHO	3	94	98
17	(CH ₃) ₂ C(OH)C≡CH	Me ₂ CHCHO	4	97	98

ee of the respective (*S*)-alcohol was increased to 88% (80% yield), Table **30**.

Enantioselective alkynylzinc additions to aldehydes were carried out using stoichiometric quantities of chiral aminoalcohol **5b** to control the stereoselective addition of terminal alkynes [90]. When **5b** was used in the presence of Zn(OTf)₂ and Et₃N in alkyne additions to several aliphatic aldehydes and benzaldehyde, the corre-

sponding chiral propargyl alcohols were obtained with up to 99% ee. The results are shown in Table **31**. In the case of the addition reaction of an α,β-unsaturated aldehyde, the enantioselectivity was significantly lower (entry 5).

The enantioselective addition reactions of 2-methyl-3-butyne-2-ol **177** to various aldehydes showed very high enantioselectivities, in the presence of 1.1-1.2 equiv. of the chiral *N*-Methyl ephedrine **5a**,

Table 32. Reactions of 177 with aldehydes using 5a-Zn(OTf)₃-Et₃N.

Entry	Aldehyde	Yield (%)	Ee (%)
1	<i>i</i> C ₃ H ₇ CHO	97	98
2	C ₆ H ₁₁ CHO	89	99
3	Me ₃ CCHO	82	98
4	ⁿ C ₅ H ₁₁ CHO	81	51
5	ⁿ C ₅ H ₁₁ CHO (1.2 eq. 5b)	81	98
6	ⁿ C ₅ H ₇ CHO (1.2 eq. 5b)	77	99
7	PhCHO	96	98
8	<i>trans</i> -PHCH=CHCHO	47	75
9	<i>trans</i> -PHCH=CHCHO (3 eq. 5b)	99	88
10	<i>i</i> Pr ₃ SiO(CH ₂) ₂ CHO	82	97

Table 33. Reactions of 178 with aldehydes using 5b-Zn(OTf)₂-Et₃N.

Entry	Aldehyde	Yield (%)	Ee (%)
1	<i>i</i> C ₃ H ₇ CHO	95	96
2	C ₆ H ₁₁ CHO	88	97
3	Me ₃ CCHO	68	97
4	(S)-CH ₃ CH(OTBS)CHO	70	97
5	PhCHO (3.2 eq. 5b)	57	97
6	TBSOCH ₂ CHO (2.2 eq. 5b)	54	88

Zn(OTf)₂ and Et₃N at room temperature, (Table 32) [91]. Terminal (S)-alkynyl alcohols were obtained after the protection of the propargylic alcohol products with benzoyl or triisopropylsilyl groups, followed by refluxing in toluene in the presence of 18-crown-6 and K₂CO₃.

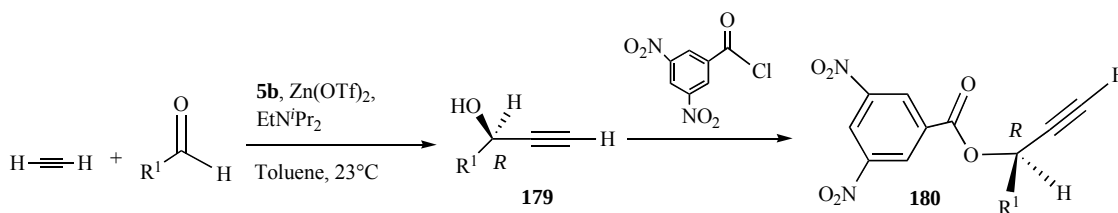
The addition reaction of **178** to several aldehydes using 1.1-1.2 equiv. of *N*-Methyl ephedrine **5b** in the presence of Zn(OTf)₂ and Et₃N, showed very good enantioselectivities of the corresponding (R)-propargyl alcohol, Table 33.

The addition of acetylene to aliphatic and aromatic aldehydes when **5b** is used in the presence of Zn(OTf)₂ proceeded enantioselectively [92]. The isolated yields were modest, because some of the substrates studied afforded low-molecular weight volatile alcohol adducts **179**. However this problem was solved when this unpurified products were treated with 3,5-dinitrobenzoyl chloride (DMPAP, Et₃N) to afford the corresponding propargylic ester **180**.

The results are shown in Table 34, in spite of the high ee's, the yields of several reactions were low.

On the other hand, enantioselective alkynylzinc additions in the presence of **5b** were carried out at 60 °C instead of at room temperature [93]. In this condition, catalytic quantities of the zinc reagent and the chiral aminoalcohol were used. High enantioselectivities and yields of the corresponding (R)-propargylic alcohol were obtained from the reactions of several alkynes with aliphatic aldehydes (Table 35). These reactions proceeded in solventless conditions and showed tolerancy to air and moisture [93b].

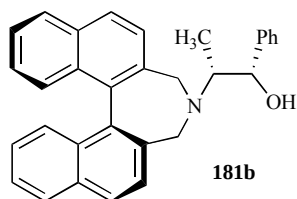
The structures of both 1,1'-binaphthyls and norephedrine were combined in the amino-alcohol **181b** [94]. This compound was used to catalyze the alkynylzinc addition to 2-chlorobenzaldehyde in high yield (95%) but low enantioselectivity was obtained (24% ee).

Table 34. Reactions of acetylene with aldehydes using **5b**- $\text{Zn}(\text{OTf})_2$ - Et_3N .

Entry	Aldehyde	Yield (%)	Ee (%)
1	${}^n\text{C}_5\text{H}_{11}\text{CHO}$	30 (179)	97
2	${}^i\text{C}_3\text{H}_7\text{CHO}$	76 (180)	98
3	$\text{C}_6\text{H}_{11}\text{CHO}$	70 (179)	98
4	Me_3CCHO	92 (180)	98
5	PhCHO	35 (179)	97
6	<i>trans</i> - $\text{PHCH}=\text{CHCHO}$	34 (179)	92
7	<i>trans</i> - $\text{PHCH}=\text{CMeCHO}$	28 (179)	91

Table 35. Alkynylzinc additions to aldehydes catalyzed by **5b**- $\text{Zn}(\text{OTf})_2$ - Et_3N .

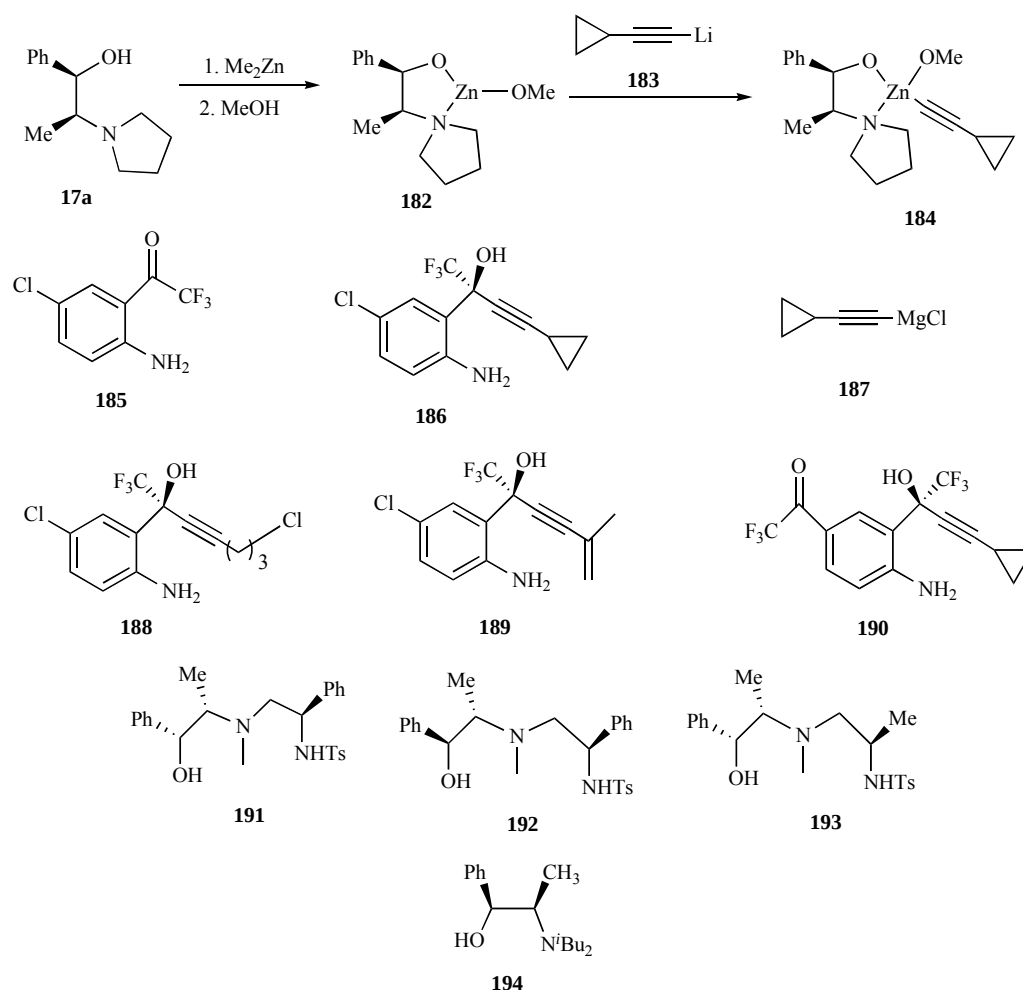
Entry	Alkyne	Aldehyde	Time (h)	Yield (%)	Ee (%)
1	$\text{Bn}_2\text{NCH}_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	2	91	97
2	$\text{Ph}(\text{CH}_2)_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	4	89	94
3	$\text{PhC}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	5	94	86
4	$\text{TMSOCMe}_2\text{C}\equiv\text{CH}$	Me_3CHCHO	5	77	98
5	$\text{Ph}(\text{CH}_2)_2\text{C}\equiv\text{CH}$	$\text{C}_7\text{H}_{15}\text{CHO}$	20	45	92
6	$\text{Ph}(\text{CH}_2)_2\text{C}\equiv\text{CH}$	Me_3CCHO	24	77	93
7	$\text{TBSOCH}_2\text{C}\equiv\text{CH}$	Me_3CCHO	5	81	93
8	$(\text{EtO})_2\text{CHC}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	9	88	94
9	$n\text{-C}_4\text{H}_9\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	6	81	93
10	$\text{TMSOCMe}_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	4	80	99
11	$\text{Et}_3\text{SiCH}_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	7	85	96
12	$\text{Bn}_2\text{NCH}_2\text{C}\equiv\text{CH}$	$\text{Me}_3\text{CCH}_2\text{CHO}$	5	80	95
13	$\text{TBSOCH}_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	5	88	90
14	$\text{Bn}_2\text{NCH}_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	5	81	94
15	$\text{Ph}(\text{CH}_2)_2\text{C}\equiv\text{CH}$	$\text{C}_6\text{H}_{11}\text{CHO}$	6	80	93
16	$\text{Bn}_2\text{NCH}_2\text{C}\equiv\text{CH}$	$\text{C}_7\text{H}_{15}\text{CHO}$	24	55	91



The asymmetric alkynylzinc addition to activated ketones, in stoichiometric amounts, was reported [95]. The *N*-disubstituted-norephedrine **17a** was reacted with one equiv. of ZnMe_2 and one equiv. of methyl alcohol, to generate the zinc oxazolidine **182**. The addition of alkynyllithium **183** formed an alkynylzinc heterocycle

184, which reacted with ketone **185** to give the tertiary-propargylic alcohol **186** with 83% yield and 83% ee. When the Grignard reagent **187** was used instead of **183**, the ee was increased to 87%. When methanol was replaced for neopentyl alcohol, under the optimized conditions, the addition of 1.5 equivalents of the zinc reagent **184**, obtained from **187**, to **185** in THF/toluene at room temperature of 18°C , gave **186** in up to 98% ee. Propargylic alcohols such as **188-190** were also obtained with high yields and high enantioselectivities.

Ephedrine and pseudoephedrine-derived sulfamide-amino-alcohols **52**, **125**, **126**, **191-193** were found to be effective auxiliaries for the enantioselective phenylacetylene addition without using $\text{Ti}(\text{OiPr})_4$ and $\text{Zn}(\text{OTf})_2$ [96]. Propargylic alcohols with high yields

Table 36. Diisopropenylzinc additions to aldehydes promoted by chiral aminoalcohol 194^a.

Entry	Aldehyde	Yield (%)	Ee (%)	Config.
1 ^b	Benzaldehyde	73	88	S
2	<i>p</i> -Chlorobenzaldehyde	84	89	S
3	<i>p</i> -Methoxybenzaldehyde	68	92	S
4	1-Naphthalaldehyde	85	79	S
5	Cinnamaldehyde	75	73	S
6	3-Phenylpropionaldehyde	81	73	S

^aThe reactions were carried out in toluene at 0 °C in the presence of 5% mol of the chiral aminoalcohol and 3 equiv. of diisopropenylzinc unless indicated otherwise. The reaction was quenched in 2 h. ^bThe reaction was performed at -10 °C.

(70-99%) and up to 84% ee for *p*-nitrobenzaldehyde were obtained; in comparison with *N*-methylephedrine, under the same conditions, the ee's were higher. It was proposed, that the coordinative sulfonamide group plays an important role in the stereochemistry and rate of the alkynylation reaction. Among these aminoalcohols, **192** was found to be the more effective for the alkynylzinc addition to aromatic aldehydes, avoiding the use of other metallic reagents, giving 81- 92% ee and up to 99% yields [97].

1.6.2. Divinylzinc and Alkenylzinc Bromides

The alkenylzinc bromides addition to both benzaldehyde and 2,2-dimethyl-propanal, was reported by using the lithium salt of **5b** as chiral auxiliary in stoichiometric quantities [98]. The chiral allyl alcohols were obtained in 86 and 98% ee's, respectively.

The *N*-dialkylnorephedrine **17b**, **175b** and **194** (5-20% mol) were used in the enantioselective diisopropenylzinc addition to benzaldehydes to afford the corresponding optically active allyl alcohol with 80-90% ee [99, 100]. Aminoalcohol **194** was found to be the best to promote the enantioselective addition reaction to other aromatic aldehydes in 73-92% ee (Table 36). Benzaldehyde in the presence of **194** and bis-(1,2-dimethyl-1-propenyl)zinc afforded the corresponding chiral allylic alcohol in 69% ee.

1.6.3. Enantioselective Phenylation

The organo phenylmetal reagent (MPh), generated from the mixture of MgPhBr and zinc chloride, was added to aliphatic, aromatic, and α,β -unsaturated aldehydes in high ee's (up to 82% ee) in the presence of chiral aminoalcohol **7b** (Table 37) [101]. The chiral

complex formed between MPh and chiral aminoalcohol **7b** (0°C, 20 s), as the kinetic complex, is responsible for the enantioselectivity. Aldehydes react with MPh from the *si* face in the presence of **7a**. This enantioselectivity is contrary to that observed in the reaction of aldehydes with ZnR_2 reagent, which attacks from *re* face, in the presence of **7b**.

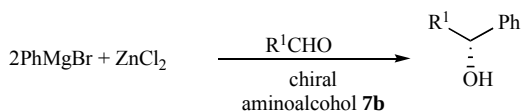


Table 37. Enantioselective phenylation of aldehydes using a kinetically formed chiral complex between Grignard-Zn halide reagent and chiral aminoalcohol **7b**^a.

Entry	R ¹	Yield (%)	Ee (%)	Config.
1	2-Naphthyl	83	79	NR
2	4-MeO-C ₆ H ₄	41	68	S
3	<i>trans</i> -PhCHCH	90	82	R
4	<i>i</i> -Pr	44	78	R
5	<i>n</i> -Hexyl	43	79	R

^aReactions are run in a mixed solvent of THF and hexane at room temperature for 30 min.

The enantioselective arylation reaction of various aldehydes using arylboronic acids as a source for transfer aryl groups to give several chiral diaryl methanols has been studied [102]. In the presence of *N*-alkylephedrine **1a**, **5a**, **8a**, **92**, **195a-199a** and *N*-dialkylnorephedrine **7a** and **200a**, excellent yields (up to 98%) and high enantioselectivities (up to 87% ee) for *p*-tolualdehyde, were reported (Table 38). Phenylboronic acids are the sources of the reactive arylzinc species, with yields of 89-98% and selectivities of 96% ee. The same reaction using aminoalcohol **7a** in the addition of ZnEt_2 to various substituted benzaldehydes gave major enantioselectivities (up to 96% ee).

1.6.4. Enantioselective Furylation

Stoichiometric or catalytic quantities of the lithium salt of (*1S,2R*)-*N*,*N*-di(4-phenylbutyl)-norephedrine **201**, allow the addition of difurylzinc (prepared *in situ* from furyllithium and ZnCl_2) to aromatic and α,β -unsaturated aldehydes to give chiral furyl alcohols **202** in good ee's (up to 73% ee) (Table 39, entry 2) [103]. The presence of LiCl increased the enantioselectivity, which could be due to the coordination of lithium cation with the oxygen atom of the furane ring.

In the presence of aminoalcohol **7b**, difurylzinc adds to benzaldehyde from the *re* face to produce the respective (*S*)-furyl alcohol. In this case, the stereochemical course is the reverse of that for dialkylzinc where the aldehyde is attacked from the *si* face to give the (*S*)-alcohol [31].

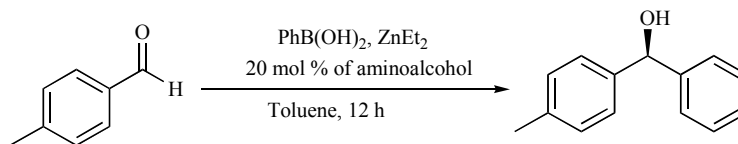


Table 38. *N*-alkylephedrine catalytic enantioselective arylation of aldehydes.

	1a	5a	7a	8a	92a	195a	196a	197a	198a	199a	200a
R ¹	CH ₃	CH ₃	ⁿ Bu	CH ₃	CH ₃	CH ₃	CH ₃	CH ₃	CH ₃	CH ₃	-(CH ₂) ₅
R ²	H	CH ₃	ⁿ Bu	C ₂ H ₅	C ₄ H ₉	C ₃ H ₇	C ₅ H ₁₁	C ₈ H ₁₇	C ₁₂ H ₂₅	CH ₂ Ph	
% ee	36	55	87	63	65	56	46	56	43	83	80

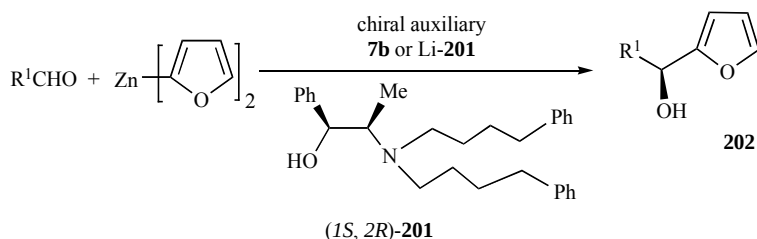


Table 39. Enantioselective furylation of aldehydes using the lithium salt of (*1S,2R*)-*N*,*N*-di(4 phenyl-butyl) norephedrine **201** as chiral auxiliary in the presence of lithium chloride^a.

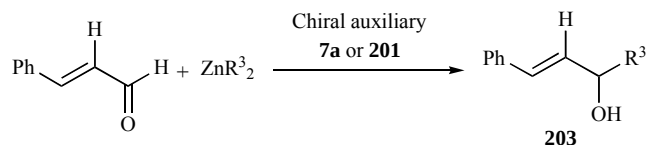
Entry	R ¹	Yield (%)	Ee (%)	Config.
1 ^b	Ph	62	72	S
2 ^b	4-MeO-C ₆ H ₄	58	73	-
3 ^b	2-Naphthyl	68	59	-
4 ^b	<i>trans</i> -PhCHCH	79	45	-
5 ^c	Ph	53	30	S

^aReactions are run in a mixed solvent of THF and hexane at 0 °C; ^bMolar ratio: R¹CHO/difurylzinc/chiral auxiliary/LiCl = 1:2.5:1:4.9-5.0; ^cMolar ratio: R¹CHO/difurylzinc/ chiral auxiliary/LiCl = 1:2.5:0.3:5.7.

1.7. Addition to α , β -unsaturated Aldehydes

Several chiral allyl alcohols **203** were obtained in high ee's from the enantioselective 1,2-addition of ZnR^3_2 reagents to cinnamaldehyde, by using chiral auxiliaries **7a** [101] and **201** [103], Table 40. This method complements the enantioselective alkenylation reaction of aldehydes.

Table 40. Asymmetric synthesis of allyl alcohols **203** by the enantioselective arylation of cinnamaldehyde.



Entry	R ³	Auxiliary	Yield (%)	Ee (%)	Config.
1	Ph	7a	90	82	R
2	2-furyl	201	79	45	NR

1.8. Ephedra Ligands with N-heteroatom Group Derivatives

The thiophosphoramidate norephedrine aminoalcohols **204b-206b** in mixture with $\text{Ti}(\text{O}^i\text{Pr})_4$ were used in the enantioselective

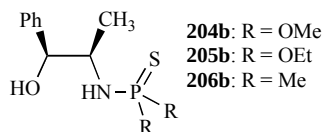


Table 41. Enantioselective diethylzinc additions to aldehydes promoted by **204b-206b**^a.

Entry	ZnR ₂	Aldehyde	Auxiliary	T (°C)	Time (h)	Yield (%)	Ee (%)	Config.
1	Et_2Zn	PhCHO	204b	-50	2.3	93	95	S
2	Me_2Zn	PhCHO	204b	-30	2.9	80	97	S
3	Et_2Zn	PhCHO	204b	-30 to 0	3.8	52	95	S
4	Et_2Zn	<i>o</i> -MeO-PhCHO	204b	-50 to 0	5.0	82	95	S
5	Et_2Zn	cinnamaldehyde	204b	-50	2.2	75	90	S
6	Et_2Zn	PhCHO	205b	-35 to -20	1.3	97	88	S
7	Et_2Zn	PhCHO	206b	-35 to -20	3.0	98	74	S

^aThe reactions were carried out in toluene in the presence of 0.8 equiv. $\text{Ti}(\text{O}^i\text{Pr})_4$ and 15% mol of the chiral auxiliary was used.

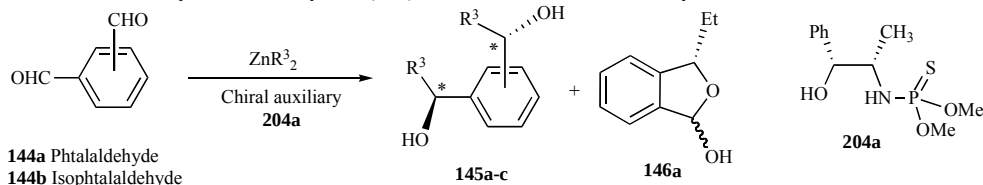


Table 42. Enantioselective diisopropylzinc additions to aldehydes in the presence of **204a** and $\text{Ti}(\text{O}^i\text{Pr})_4$ ^a.

Entry	Aldehyde	Time (h)	Yield (%)	Ee (%)	Config.
1	PhCHO	2	93	96	R
2	<i>p</i> -MeOPhCHO	3	99	97	NR
3	<i>p</i> -ClPhCHO	2	85	91	NR
4	1-Naphthaldehyde	2	89	96	NR
5	Cinnamaldehyde	1.5	94	64	NR

^aThe reactions were carried out in toluene at -30 °C in the presence of 1.2 equiv. of $\text{Ti}(\text{O}^i\text{Pr})_4$ and 1.5% mol of aminoalcohol **204a**.

ZnEt_2 addition to aldehydes [104]. The use of **204b** (15% mol) in mixture with 0.8 equiv of $\text{Ti}(\text{O}^i\text{Pr})_4$ in the reaction with benzaldehydes and α,β -unsaturated aldehydes gave chiral alcohols with 90-97% ee and good yields (Table 41). The reaction with benzaldehyde gave (*S*)-1-phenylethanol with 95% ee and 52% yield (entry 3). Compounds **205b** and **206b** gave lower ee's (88% and 74%, respectively) for the same reaction with benzaldehyde (entries 6 and 7).

Chiral thiophosphoramidate aminoalcohol **204b**, combined with $\text{Ti}(\text{O}^i\text{Pr})_4$, was used in the ZnEt_2 reagent addition to phthalaldehyde **144a** and isophthalaldehyde **144b** [73]. Diethylzinc adds to isophthalaldehyde in the presence of **204b** (0.3 equiv.) in mixture with $\text{Ti}(\text{O}^i\text{Pr})_4$ (1.6 equiv.) at 0 °C in toluene/hexane (1:1), to give the respective chiral diol in 94% yield with >99% ee. The 92:8 *dl*/meso proportion, under the same conditions, was obtained. Phthalaldehyde **144a** reacted with ZnEt_2 in the presence of **204b** to give the chiral diol **145a** as the principal product with 92% ee and 64% yield in a mixture with lactol **146a** in 21% yield. The *dl*/meso ratio was 9:91 with the meso diol as principal product.

In the presence of $\text{Ti}(\text{O}^i\text{Pr})_4$, the thiophosphoramidate aminoalcohol **204a**, the enantiomer of **204b**, was used in the addition of Zn^iPr_2 addition to aldehydes [105]. By using **204a** in mixture with 1.2 equiv of $\text{Ti}(\text{O}^i\text{Pr})_4$, the reaction gave up to 97% ee's for the additions to aromatic aldehydes, Table 42. The enantioselectivity for the addition to cinnamaldehyde was low (64% ee, entry 5).

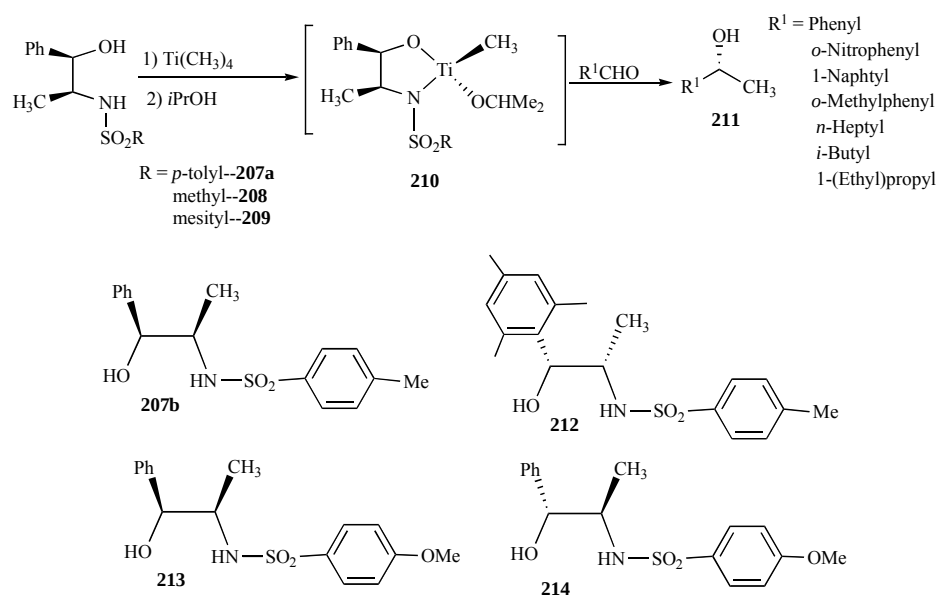


Table 43. Enantioselective diethylzinc additions to aldehydes promoted by chiral aminoalcohol **213** in the presence of $\text{Ti}(\text{O}^i\text{Pr})_4$ and CaH_2^a .

Entry	Aldehyde	Yield (%)	Ee (%)	Config.
1	PhCHO	99	97	<i>S</i>
2	Cinnamaldehyde	95	90	<i>S</i>
3	<i>p</i> -MeOPhCHO	98	94	<i>S</i>
4	$\text{C}_6\text{H}_{11}\text{CHO}$	77	90	<i>S</i>

^aThe reactions were carried out in the presence of 20% mol of aminoalcohol **213**, 1.2 equiv. $\text{Ti}(\text{O}^i\text{Pr})_4$ and 0.5 equiv. of CaH_2 at 0 °C over 24 h using 1M hexane solution of diethylzinc.

N-sulfonylated norephedrine derivatives **207a**, **208** and **209** were tested to be excellent auxiliaries for $\text{Ti}(\text{Me})_4$ additions to aldehydes [106]. The chiral intermediate **210** was generated *in situ* from the corresponding sulfonamides when treated with $\text{Ti}(\text{CH}_3)_4$ followed by the addition of *i*PrOH. The oligomeric form of **210**, was enantioselectively reacted with aromatic and aliphatic aldehydes to form the corresponding (*R*)-carbinols **211**. When the R group at the sulfur is mesityl, ligand **209**, pronounced stereoselectivity was observed (88-90% ee) to form compound **211** from benzaldehydes but if R is methyl, ligand **208**, ee is the lowest (62%). In contrast, unacceptable ee values were generally obtained in case of aliphatic aldehydes (30-60% ee). The effect of the type of alcohols was tested, resulting that in all cases enantioselectivity turned out to be lower than 60% ee.

Chiral norephedrinesulfonamides **207b**, **212-214** [107] were tested as chiral auxiliaries. They could not directly catalyze the ZnEt_2 addition to benzaldehyde, but when they were treated with excess of $\text{Ti}(\text{O}^i\text{Pr})_4$ in toluene at 0 °C, 6 to 84% ee's were obtained. In the same conditions, chiral aminoalcohols **207b** and **213** catalysed the reaction with 81% and 83% ee, respectively. In contrast, chiral aminoalcohol **214** showed no enantioselectivity under the same reaction conditions. When the solvent was changed to methylene chloride, the enantioselectivity was greatly increased when **207b** and **213** were used. In the presence of $\text{Ti}(\text{O}^i\text{Pr})_4$, aminoalcohol **213** carried out the enantioselective ZnEt_2 additions to a few aldehydes with 90-97% ee, Table 43.

Using *N*-tolysulfonylnorephedrine **207a** as auxiliary ligand, the enantioselectivity of the (*S*)-BINOL- $\text{Ti}(\text{O}^i\text{Pr})_4$ catalyst was improved [108]. **207a** (10% mol) combined with (*S*)-BINOL (10% mol), $\text{Ti}(\text{O}^i\text{Pr})_4$ (15% mol) and ZnMe_2 reagent (2 equiv.), in the reactions of phenylacetylene (one equiv.) and arylaldehydes (one

equiv.) in THF at 0° C, the reactions gave good yields and enantioselectivities of the (*R*)-alcohols, the results are shown in Table 44.

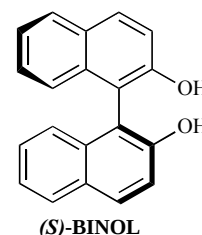


Table 44. Phenylacetylene additions to aldehydes catalyzed by (*S*)-BINOL- $\text{Ti}(\text{O}^i\text{Pr})_4$ - ZnMe_2 and the sulfonamide **207a**.

Entry	Aldehyde	Yield (%)	Ee (%)
1	PhCHO	83	96
2	<i>o</i> -NO ₂ PhCHO	83	88
3	<i>m</i> -NO ₂ PhCHO	82	99
4	<i>p</i> -NO ₂ PhCHO	82	99
5	<i>p</i> -BrPhCHO	85	99
6	<i>m</i> -ClPhCHO	84	97
7	<i>p</i> -ClPhCHO	86	95
8	2-naphthylaldehyde	81	95
9	<i>p</i> -MeOPhCHO	78	95
10	<i>p</i> -MePhCHO	79	92

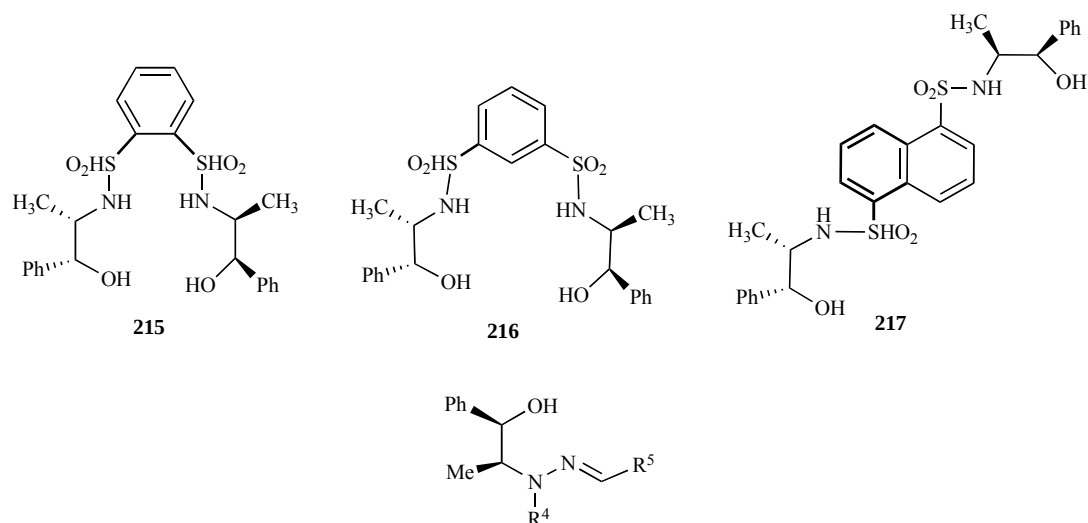
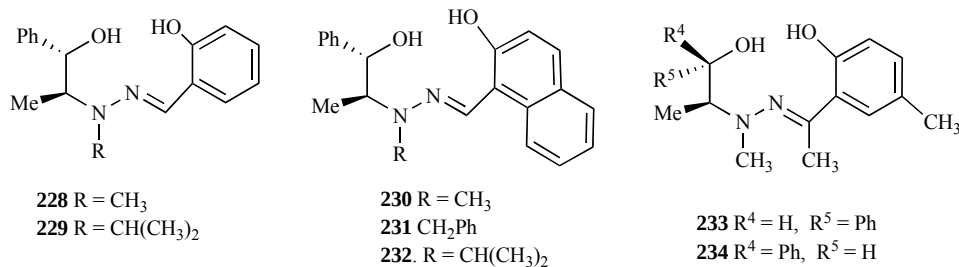


Table 45. Tridentate β -hydroxysalicylhydrazone chiral auxiliaries 218-227, derived from ephedrine 1a and norephedrine 3a in the Et_2Zn addition to benzaldehyde.

Entry	Ligand	R ⁴	R ⁵	Ee	Config.
1	218	CH ₃	Phenyl	-	-
2	219	CH ₃	2-Hydroxy-1-naphthyl	88	S
3	220	<i>i</i> -Pr	2-Hydroxy-1-naphthyl	65	S
4	221	CH ₃	2-Hydroxyphenyl	65	S
5	222	CH ₃	2-hydroxy-3-methylphenyl	11	R
6	123	CH ₃	3- <i>t</i> -Buthyl-2-hydroxyphenyl	-	-
7	124	CH ₃	3,5-Di-(<i>t</i> -Buthyl)-2-hydroxyphenyl	-	-
8	125	CH ₃	2-Hydroxy-5-nitrophenyl	78	S
9	126	CH ₃	2-Hydroxy-5-methoxyphenyl	22	S
10	227	<i>i</i> -Pr	2-Hydroxyphenyl	65	S



In the presence of $\text{Ti}(\text{O}^i\text{Pr})_4$, benzene-1,2- and 1,3-disulfonamides **215**, **216** and naphthalene-1,5-disulfonamides **217** derived from norephedrine **3b**, were used in the enantioselective dialkylzincs additions to aldehydes [109]. There are differences in enantioselectivity depending on the position of the amino alcohol moieties in the chiral auxiliary, being the benzene 1,3-disulfonyl derivative, the best system (95% yield, 78% ee).

A series of tridentate β -hydroxysalicylhydrazone chiral auxiliaries **218-227**, derived from (1*R*,2*S*)-ephedrine **1a** and (1*R*,2*S*)-norephedrine **3a**, were used in the asymmetric ZnEt_2 addition to benzaldehyde in toluene at room temperature [110]. The isolated chemical yields of the addition products ranged from 50% to 84%. The reactions exhibited very good enantioselectivities with 11 to 78% ee, Table 45. Ligand **219** proved to be the most effective catalyst in the family of β -hydroxysalicylhydrazones and was tested

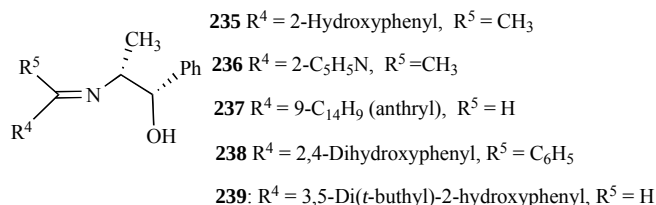
with a series of aldehydes in toluene at 25°C. The enantiomeric excess ranging from 88 to 92% ee of the corresponding (S)-alcohol.

β -Hydroxysalicylhydrazones **228**, **230** and **233** derived from (1*S*,2*S*)-pseudoephedrine, **2a** and **234** derived from (1*R*,2*S*)-ephedrine **1a** and **229**, **231** and **232** derived from (1*S*,2*S*)-pseudonorephedrine **4a** have been used in the enantioselective ZnEt_2 addition to 2-naphthaldehyde in toluene as solvent to give the (S)-alcohol in 4 to 56% ee [111].

1. 9. Imine Ephedra Derivatives as Chiral Auxiliaries

The activity of chiral iminealcohols **235-239** derived from (1*S*,2*R*)-norephedrine **3b** in the enantioselective addition of ZnEt_2 to aldehydes was determined [112]. Imine ligands revealed medium enantioselectivity, with 30% ee. The highest ee (97%) was obtained

for imine **237** with 9-anthryl substituent, when *p*-methoxybenzaldehyde was used as substrate. The yields of the reaction obtained after 24-72 h of reaction, reached 76-96%. The absolute configuration of the addition product depends on the geometry of the transition states and π - π stacking interactions between the iminealcohols and the benzaldehyde substrate. Intermediates for the iminealcohol-ZnEt₂-based auxiliaries and probable mechanisms were proposed.



1.10. Thiolephedrine Derivatives as Chiral Auxiliaries

The hydrochloride of the chiral thiolephedrine **240**HCl and ephedrinedisulfides **241-243** were tested as chiral auxiliaries for the ZnEt₂ addition to benzaldehyde [113, 114]. They exhibited good enantioselectivities (80-90% ee, Table 46).

The *N*-disubstituted norephedrinethiols **244-252**, containing acyclic and cyclic amino groups, were used for the ZnEt₂ addition to aldehydes (Table 47) [115]. In toluene at 0°C, the ZnEt₂ additions to some aldehydes in the presence of 5% mol of **244**, proceeded with 78-99% ee (table 46, entries 1, 8, 18, and 25) [115a,b]. Chiral auxiliaries **244-250**, in the ZnEt₂ additions to cyclohexanecarboxaldehyde, gave very good enantioselectivities (96-100% ee). However, in the same reactions with cinnamaldehyde the enantioselectivities were decreased (66-81% ee). In general, the chiral

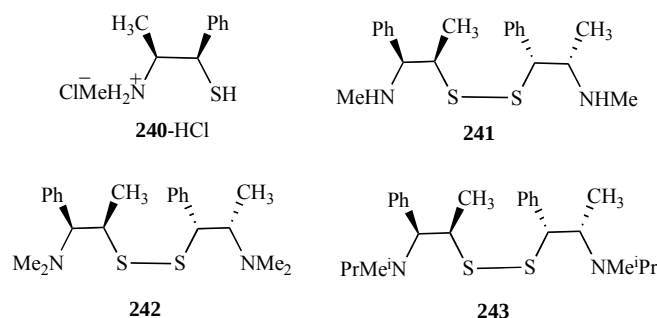


Table 46. Enantioselective diethylzinc additions to benzaldehyde promoted by the amino thiol **240**HCl and the aminodisulfides **241-243**^a.

Entry	Auxiliary ^b	Time (h)	Yield (%)	Ee (%)	Config.
1	240 -HCl	40	95	80	<i>R</i>
2	241	40	75	86	<i>R</i>
3	242	20	96	90	<i>R</i>
4	243	20	97	89	<i>R</i>

^aThe reactions were carried out in toluene at room temperature.

^b5% mol of the auxiliary was used.

aminothiols exhibited good enantioselectivity for several aldehydes except for cinnamaldehyde.

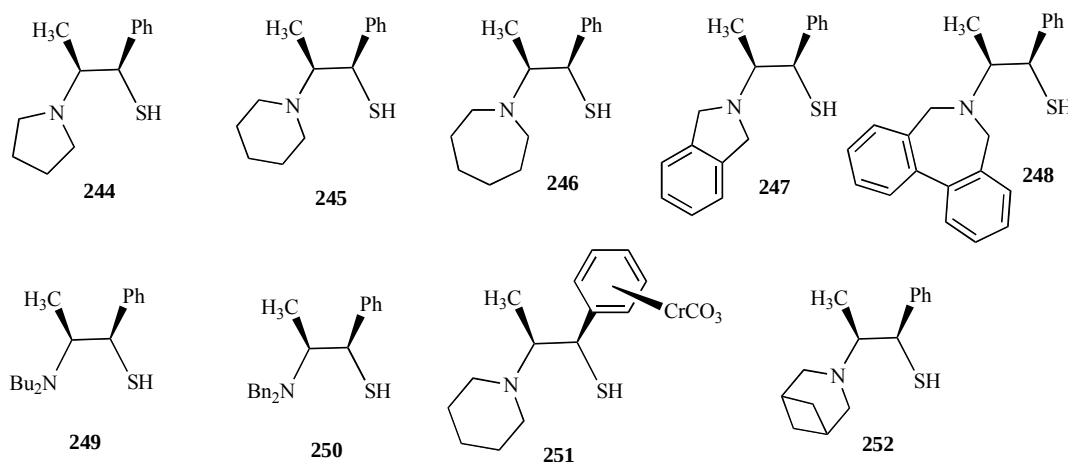


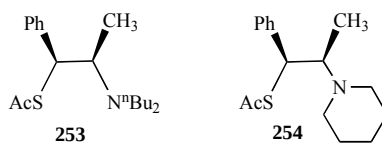
Table 47. Enantioselective diethylzinc additions to aldehydes promoted by chiral aminothiols **244-252**^a.

Entry	Aldehyde	Auxiliary	T (°C)	Yield (%)	Ee (%)	Config.
1	PhCHO	244	0	94	99	<i>R</i>
2	PhCHO	245	0	94	100	<i>R</i>
3	PhCHO	246	0	94	86	<i>R</i>
4	PhCHO	247	0	94	97	<i>R</i>
5	PhCHO	248	0	94	73	<i>R</i>
6	PhCHO	249	0	94	99	<i>R</i>
7	PhCHO	250	0	94	72	<i>R</i>
8	PhCH=CHCHO	244	0	94	78	<i>R</i>

Table 47. contd...

Entry	Aldehyde	Auxiliary	T (°C)	Yield (%)	Ee (%)	Config.
9	PhCH=CHCHO	245	-50	92	68	R
10	PhCH=CHCHO	245	0	98	77	R
11	PhCH=CHCHO	246	0	98	74	R
12	PhCH=CHCHO	247	0	98	72	R
13	PhCH=CHCHO	248	0	98	66	R
14	PhCH=CHCHO	249	0	98	81	R
15	PhCH=CHCHO	250	0	98	72	R
16	PhCH=CHCHO	251	0	98	69	R
17	PhCH=CHCHO	252	0	98	75	R
18	C ₆ H ₁₁ CHO	244	0	98	99	R
19	C ₆ H ₁₁ CHO	245	0	97	100	R
20	C ₆ H ₁₁ CHO	246	0	97	97	R
21	C ₆ H ₁₁ CHO	247	0	97	96	R
22	C ₆ H ₁₁ CHO	248	0	97	97	R
23	C ₆ H ₁₁ CHO	249	0	97	100	R
24	C ₆ H ₁₁ CHO	250	0	97	97	R
25	2-naphthaldehyde	244	0	97	99	R
26	2-naphthaldehyde	245	0	98	99	R
27	2-naphthaldehyde	246	0	98	96	R
28	2-naphthaldehyde	247	0	98	97	R
29	2-naphthaldehyde	248	0	98	96	R
30	2-naphthaldehyde	249	0	98	98	R
31	2-naphthaldehyde	250	0	98	95	R

^aThe reactions were carried out in toluene over 12 h using 1M hexane solution of diethylzinc and 5% mol of the auxiliary.

Table 48. Diethylzinc additions to aldehydes catalyzed by 253 and 254^a.

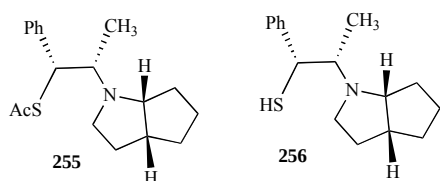
Entry	Aldehyde	Auxiliary	Time (h)	Yield (%) ^b	Ee (%)	Config.
1	PhCHO	253	6	100	99	S
2	PhCHO	254	6	100	99	S
3	<i>p</i> -ClPhCHO	253	6	100	99	S
4	<i>p</i> -ClPhCHO	254	6	100	99	S
5	<i>p</i> -MeOPhCHO	253	6	99	99	S
6	<i>p</i> -MeOPhCHO	254	6	99	99	S
7	<i>o</i> -MeOPhCHO	253	6	100	99	S
8	<i>o</i> -MeOPhCHO	254	6	99	99	S
9	2-naphthaldehyde	253	6	99	99	S
10	2-naphthaldehyde	254	6	99	99	S
11	C ₆ H ₁₁ CHO	254	10	99	99	S

^aThe reactions were carried out in hexane at 0-20 °C and 5% mol of the auxiliary.

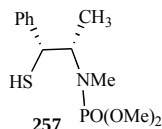
^bDetermined by GC.

The *N*-disubstituted norephedrinethioacetates **253** and **254**, generated very high enantioselectivities for the ZnEt₂ addition to cyclohexanecarboxaldehyde and aromatic aldehydes (98-99% ee, Table 48) [116]. In spite of the alkyl substituent on the nitrogen, the enantioselectivities of these auxiliaries were similar.

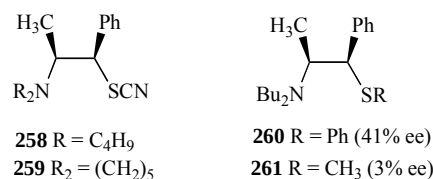
N-bicyclic norephedrinethioacetate **255** and thiol **256** showed high enantioselectivities for the ZnEt₂ addition to benzaldehyde. The reaction in the presence of 5% mol of **255** or **256** in toluene at room temperature gave (*R*)-1-phenylpropanol with 90% ee and 100% yield [117].



A stoichiometric quantity of the *N*-phosphorilated-ephedrinethiol **257** was used in the ZnEt_2 addition to benzaldehyde with over 98% ee and over 98% yield with or without excess of $\text{Ti}(\text{O}^i\text{Pr})_4$ [118]. However, degradation of **257** was observed. In contrast, the use of ephedrinethiol **240** significantly increased the enantioselectivity from 36% to 89% ee, without addition of $\text{Ti}(\text{O}^i\text{Pr})_4$.



Optically active amino thiocyanates **258** and **259** derived from (-)-norephedrine **3a** were effective aprotic auxiliaries for enantioselective ZnEt_2 addition to aromatic aldehydes to afford optically active secondary alcohols with ee up to 96% [119]. On the other hand, the use of amino sulfides **260** and **261** in the ZnEt_2 reagent addition to benzaldehyde, the reaction proceeded smoothly and poor ee's were observed.



2. ORGANOMETALLIC 1, 4 CONJUGATED ADDITIONS TO A,B-CONJUGATED KETONES

2.1. Dialkylzinc Reagents and Auxiliary-Nickel Complex Catalysts

Soai *et al.* reported that 1,4-conjugated addition of dialkylzincs to prochiral enones was possible using $\text{Ni}(\text{acac})_2$ reagent, in the presence of a chiral aminoalcohol as auxiliary [23, 120]. The auxiliary-nickel catalyst was obtained by stirring one equiv of $\text{Ni}(\text{acac})_2$ or NiBr_2 with the chiral aminoalcohol **7b** in toluene at 80 °C (Table 49, entries 1-6). The substrate was added, the reaction mixture was cooled to -30 °C, and then the dialkylzinc was added. The ee's increased with the ratio of aminoalcohol to substrate. The enantioselectivity of the reactions was significantly increased when a combination of $\text{Ni}(\text{II})$ -2,2'-dipyridyl in MeCN/toluene to form the catalyst was used [121]. ZnR_3^2 additions to aryl substituted enones produce β -substituted ketones in up to 90% ee (entries 15 and 16). In addition to the chiral aminoalcohol and $\text{Ni}(\text{II})$, MeCN was essential to obtain the product with high ee.

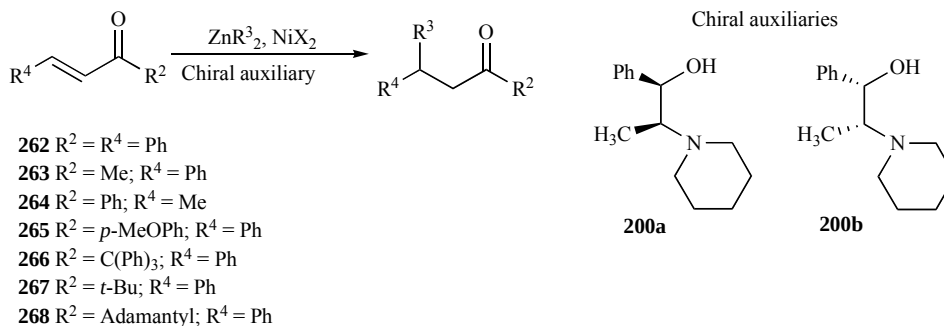


Table 49. Enantioselective conjugate addition of dialkylzinc to enones 262-268.

Entry	NiX ₂	Auxiliary	Substrate	R ³	Ni:L*:S	Yield (%)	Ee (%)	Config.
1	Ni(acac) ₂	7b	262	Me	1:1.2:1.7	72	40	<i>R</i>
2	Ni(acac) ₂	7b	262	Et	1:1.2:2.0	75	45	<i>R</i>
3	Ni(acac) ₂	7b	262	Et	1:1.2:17	94	20	<i>R</i>
4	Ni(acac) ₂	7a	262	Et	1:1.2:17	89	22	<i>R</i>
5	Ni(acac) ₂	7b	263	Et	1:1.2:1.7	63	12	<i>R</i>
6	Ni(acac) ₂	7b	264	Et	1:1.2:2.0	78	44	<i>R</i>
7	NiBr ₂	7b	262	Et	1:1.2:4.0	32	48	<i>R</i>
8	NiBr ₂	7b	262	Et	1:1.2:33	NR	36	<i>R</i>
9	NiBr ₂	7b	262	Et	1:1.2:2.5	NR	32	<i>R</i>
10	NiCl ₂	7b	262	Et	1:1.2:4.0	NR	30	<i>R</i>
11	NiI ₂	7b	262	Et	1:1.2:4.0	NR	3	<i>R</i>
12	NiBr ₂	7b	262	<i>n</i> -Bu	1:2.4:14	16	43	-
13	NiBr ₂	7b	263	Et	1:2.4:14	32	18	<i>S</i>
14	NiBr ₂	7b	264	Et	1:2.4:17	40	14	<i>S</i>

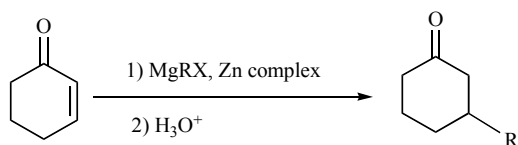
Table 49. contd....

Entry	NiX ₂	Auxiliary	Substrate	R ³	Ni:L*:S	Yield (%)	Ee (%)	Config.
15	Ni(acac) ₂	7b /2,2'-bipyridyl	262	Et	1:2.4:17	47	90	<i>R</i>
16	Ni(acac) ₂	17b /2,2'-bipyridyl	262	Et	1:2.4:17	63	82	<i>R</i>
17	Ni(acac) ₂	7b /piperazine	262	Et	1:2.4:17	44	87	<i>R</i>
18	Ni(acac) ₂	7b /1,10-phenantroline	262	Et	1:2.4:17	72	81	<i>R</i>
19	Ni(acac) ₂	7b /2,2'-bipyridyl	262	<i>n</i> -Bu	1:2.4:17	47	74	<i>R</i>
20	Ni(acac) ₂	7b /2,2'-bipyridyl	265	Et	1:2.4:17	58	80	<i>R</i>
21	Ni(acac) ₂	7b /2,2'-bipyridyl-ketone	262	Et	1:2.4:17	87	71	<i>R</i>
22	Ni(acac) ₂	7b /2,2'-biquinoline	262	Et	1:2.4:17	92	0	<i>R</i>
23	Ni(acac) ₂	7b /1,2-DPPE	262	Et	1:2.4:17	89	71	<i>R</i>
24	Ni(acac) ₂	7b /morpholine	262	Et	1:2.4:17	84	71	<i>R</i>
25	Ni(acac) ₂	7b /quinuclidine	262	Et	1:2.4:17	85	72	<i>R</i>
26	Ni(acac) ₂	7b /pyridine	262	Et	1:2.4:17	84	71	<i>R</i>
27	Ni(acac) ₂	200a	266	Et	0:1.0:4.0	81	80	(+)
28	Ni(acac) ₂	200a	266	Et	0:1.0:1.0	96	94	(+)
29	Ni(acac) ₂	200b	266	Et	0:1.0:1.0	95	93	(-)
30	Ni(acac) ₂	200a	267	Et	0:1.0:4.0	82	72	<i>R</i>
31	Ni(acac) ₂	200a	267	Et	0:1.0:1.0	84	81	<i>R</i>
32	Ni(acac) ₂	200a	268	Et	0:1.0:4.0	88	60	-
33	Ni(acac) ₂	200a	268	Et	0:1.0:1.0	84	82	-
34	Ni(acac) ₂	200a	264	Et		34	82	<i>R</i>

The use of stoichiometric quantities of chiral 1-phenyl-2-(1-piperidinyl) propan-1-ols **200a** and **200b** to catalyze the ZnEt₂ addition to enones **262-268** was carried out. An increase in the enantioselectivities of the corresponding ketones was observed. In general longer reaction times than those employing nickel catalysts were reported [120b].

2.2. Grignard Reagents with Auxiliary-Zinc Complex Catalysts

Ephedrine chiral auxiliaries **1a**, **11a**, **269** and **270** (1 % mol) were used to form auxiliary chiral zinc complexes to promote the addition of Grignard reagents to 2-cyclohexenone with the purpose of make this reaction enantioselective (Table 50) [122]. Counterions, rate of addition, temperature and solvent influence were studied with ligand **11a**, giving 98% yield but 26 % ee for BrMg(*i*-Pr) in stirring THF at -90 °C. In the same conditions, the reactions of **1a**, **269** and **270** proceed with high chemical yields and selectivities in favor of 1,4-addition, but with very low ee's (14 to 17%). *In situ* preparation of the catalyst, from the lithium salt of the chiral diamino alcohol in THF, gave the best enantioselectivities. 92 % yield, 33% ee.



The suggested mechanism for this reaction goes *via* the complexation of 2-cyclohexenone and Grignard reagent to the chiral tetracoordinated alkylzinc complex **271** through carbonyl and alkoxide oxygen atoms, respectively. The alkyl group of Grignard moiety is in the proper position in complex **272** to be transferred to

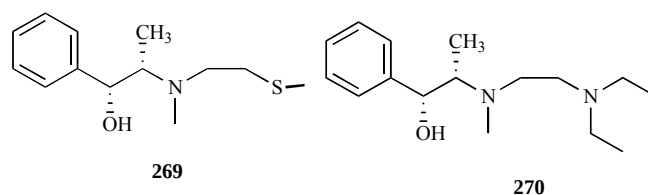


Table 50. Enantioselective conjugate addition of Grignard reagents to 2-cyclohexenone using chiral zinc complexes.

Entry	Auxiliary	R ³	X	Yield (%)	Ee (%)
1	11a	<i>i</i> -Pr	Br	90	26
2	11a	<i>i</i> -Pr	Cl	97	26
3	269	<i>i</i> -Pr	Cl	87	14
4	270	<i>i</i> -Pr	Cl	83	17
5	1a	<i>i</i> -Pr	Cl	93	17

the β -position of the enone, this mechanism does not explain the stereochemistry of the reaction.

2.3. Chiral Lithium and Magnesium Alkylcuprate Catalysts

The induction of asymmetry of prochiral 2-cyclohexenone with chiral organocuprates, prepared with equimolar quantities of each of the alcoholate of **5a** (L*Li) with LiR³ and CuI, Table 51, was investigated for the first time by Crabbè *et al.* [17]. In the 1,4 addition of *n*-butylcuprate to cyclohexenone, the optical activity of the product was found to depend upon the reaction temperature.

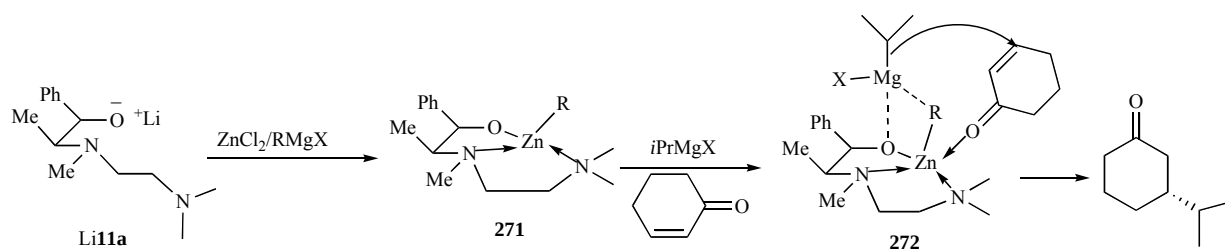
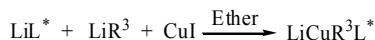


Fig. (16). Mechanism of Zn-catalyzed conjugate addition of Grignard reagents.

Table 51. Enantioselective 1,4-additions of organocuprates to aldehydes and 2-cyclohexenone using chiral organocuprates and 5a-Li^a.

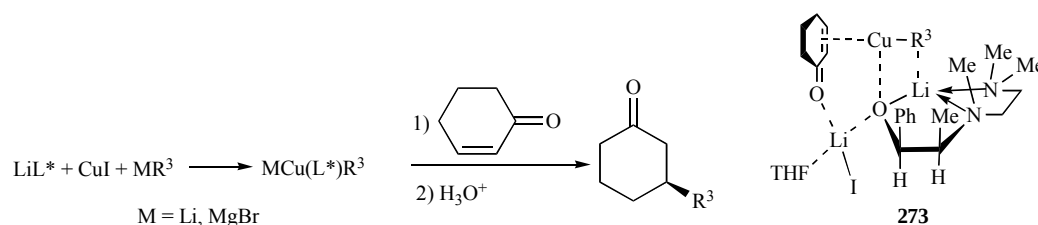
Entry	LiR ³	T (°C)	[α] _D	Ee (%)	Config.
1	Me	0	-0.29°	2.0	NR
2	<i>n</i> -Bu	0	-0.56°	NR	NR
3	<i>n</i> -Bu	-20	-0.45°	NR	NR
4	<i>n</i> -Bu	-30	-0.38°	NR	NR
5	<i>t</i> -Bu	0	-0.12°	NR	NR
6	Ph	0	-0.54°	NR	NR

^aThe reactions were carried out by using ether-HMPT.

The lithium alkylcuprate catalyst, LiCuR³L*, where L* is the alcoholate of the chiral diaminoalcohol (*1R,2S*)-**11a** derived from ephedrine **1a** was reported by Corey, Naef, and Hannon [123]. The alkyl cuprate catalyst was prepared *in situ* from the reaction of the lithium alkoxide of the ligand (LiL*) with CuI in THF/DMSO to give the copper alkoxide, and one equiv of the alkyl lithium (LiR³) to form the reagent, then the enone is added at -78 °C. In order to explain the enantioselectivity for the (*R*)-enantiomer, the formation of complex **273** has been proposed in the case of the use of LiR³. Other chiral auxiliaries such as optically active ephedrine **1a**, pseudoephedrine **2b** and norephedrine **3a** have been used by other groups with modest results, Table 52 [124].

II.4. Asymmetric Allylic Alkylation Catalyzed by Palladium

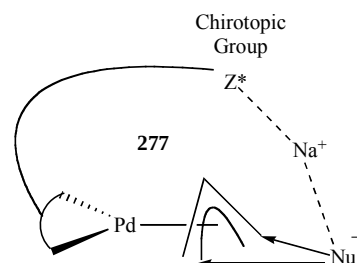
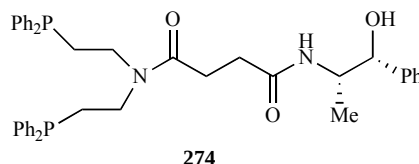
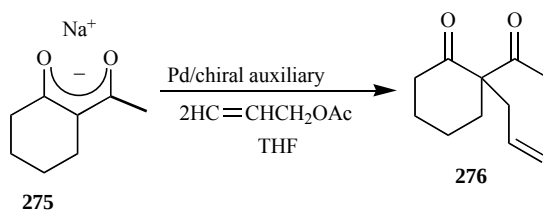
The chiral diphosphine auxiliary with an *N*-substituted norephedrine **274** as chirotopic functional group at an appropriate dis-

Table 52. Enantioselective conjugate addition to enones with cuprate chiral complexes, MCuR³L*.

Entry	Substrate	L*	MR ³	Solvent	Yield (%)	Ee (%)	Config.
1	2-cyclopentenone	11a	LiEt	THF	68	77	<i>R</i>
2	2-cyclopentenone	11a	Lin-Bu	THF	60	72	<i>R</i>
3	2-cyclopentenone	11a	Li <i>t</i> -BuOCH ₂	THF	52	81	<i>R</i>
4	2-cyclohexenone	11a	LiEt	THF	90	92	<i>R</i>
5	2-cyclohexenone	11a	Lin-Bu	THF	90	89	<i>R</i>
6	2-cyclohexenone	11a	Li <i>t</i> -BuOCH ₂	THF	73	85	<i>R</i>
7	2-cyclohexenone	1a	LiPh	THF	94	50	<i>R</i>
8	2-cyclohexenone	1a	LiPh	THF	91	0	<i>R</i>
9	2-cyclohexenone	1a	LiPh	THF	48	40	<i>R</i>
10	2-cyclohexenone	1a	LiPh	THF	72	20	<i>R</i>
11	2-cyclohexenone	1a	LiPh	THF	34	50	<i>R</i>
12	2-cyclohexenone	1a	LiPh	THF	46	10	<i>R</i>
13	2-cyclohexenone	1a	LiPh	THF	39	50	<i>R</i>
14	2-cyclohexenone	1a	LiPh	THF	38	0	<i>NR</i>
15	2-cyclohexenone	1a	LiPh	ether	65	20	<i>R</i>
16	2-cyclohexenone	1a	BrMgPh	ether	41	10	<i>S</i>
17	2-cyclohexenone	1a	LiPh	THF	39	50	<i>R</i>

Table 52. cond...

Entry	Substrate	L*	MR ³	Solvent	Yield (%)	Ee (%)	Config.
18	2-cyclohexenone	1a	BrMgPh	THF	7	50	S
19	2-cyclohexenone	1a	LiPh	DME	30	30	R
20	2-cyclohexenone	1a	LiPh	(i-Pr) ₂ O	45	20	R
21	2-cyclohexenone	1a	LiPh	Me ₂ O	39	0	NR
22	2-cyclohexenone	1a	LiPh	dioxane	12	0	NR
23	2-cyclohexenone	1a	LiPh	Me ₂ S	98	0	NR
24	2-cyclohexenone	2b	LiPh	ether	73	0	NR
25	2-cyclohexenone	3a	LiPh		44	0	NR



tance from an achiral diphosphine moiety, was demonstrated to be fairly effective for the asymmetric allylic alkylation [125]. Phosphine auxiliary **274** was examined for the stereoselectivity in the palladium-catalyzed reaction of sodium enolate of 2-acetylcyclohexanone **275** with allyl acetate. The reaction at -30 °C gave optically active allylation product **276** in only 31% ee. The linker chain length in the coordinative interaction between the chiral functional groups and the sodium enolate has been proposed for the low stereoselection, as is represented by complex **277**.

CONCLUSION

The *N*-disubstituted *ephedra* aminoalcohols react with organometallic compounds to form chiral organometallic complexes, which are responsible for the asymmetric alkylation of carbonyl compounds. Among several organometallic compounds (cuprates, aluminates and magnesium) used in the alkylation reaction of carbonylic compounds, dialkylzincs demonstrated to be the best.

The organozinc complex, as responsible catalyst, is formed when one equivalent of the *N*-disubstituted *ephedra* ligand reacts with two equivalents of the corresponding dialkylzinc. The labile hydrogen atom of the chiral amino alcohol, which produces the alkylzinc alkoxide, is essential for this reaction.

Relevant literature reports, strongly support that *N,N*-dialkylnorephedrine ligands are excellent chiral auxiliaries for the enantioselective diorganozinc addition to aldehydes and ketones. On the other hand, mono-*N*-alkylated norephedrine ligands exhibit modest enantioselectivities.

In the case of *N,N*-dialkylnorephedrine ligands, the enantioselectivity of the addition product is increased as the chain length of the *N*-*n*-alkyl substituent increases and it is optimal at a chain length of four carbons. Thus, *N,N*-dibuthylnorephedrine **7** gave the best results. In contrast, no enantioselectivity was observed for *N*-alkyl substituents branched at the β -position.

To ensure enantioselectivity, the *N*-alkyl substituent must have one β -carbon atom, but if the *N*-alkyl substituents are larger than *n*-buthyl, they block the approach of the attacking species to one of the enantiotopic faces of the aldehyde. However, bulkier *N*-alkyl substituents than *n*-buthyl, prevent the formation of the catalyst complex-aldehyde.

Most of the addition reactions of ZnRX (R = alkynyl, alkenyl, X = halogen) and ZnR₂ (R = phenyl and furyl) reagents require the use of a stoichiometric quantity of a chiral auxiliary for best enantioselectivities.

Diastereoselective dialkylzinc addition to racemic aldehydes in the presence of an aminoalcohol as chiral auxiliary produces the stereoselective chiral alcohols. In general, enantioselective 1,2-addition reactions of dialkylzincs to functionalized aldehydes and ketones are highly enantio- and chemoselective.

The configuration of the chiral auxiliary is more determinant than the pro-chirality of the aldehydes or ketones on the stereoselectivity of catalyzed organoadditions.

Auxiliary-metal complex catalysis is an excellent methodology used in 1,4-conjugated additions to α,β -conjugated ketones. The relationship between the enantioselectivities of chiral *N*-substituted *ephedra* auxiliaries and the ee's of the products is no linear. In some cases, the ee's of the products are higher than those of the auxiliaries.

CONFLICT OF INTEREST

The authors confirm that this article content has no conflict of interest.

ACKNOWLEDGEMENTS

Authors thank Secretaría de Investigación y Posgrado and Comisión de Operación y Fomento de Actividades Académicas del IPN, México, D. F. for financial support.

REFERENCES

- [1] Noyori, R.; Kitamura, M. In: *Modern Synthetic Methods*, Schelfold, R., Ed.; Springer, Berlin, **1989**; Vol. 5, pp. 115-198.
- [2] Frankland, E. Liebig's. *Ann. Chem. Pharm.* **1849**, 71, 171-213.
- [3] Rathke, M. W. *Org. React.* (NY). The Reformatsky Reaction. **1975**, 22, 423-460.
- [4] Simmons, H.E.; Cairns, T.L.; Vladuchick, S.A. Cyclopropanes from Unsaturated Compounds, Methylene Iodide, and Zinc-Copper Couple. *Org. React.* **1973**, 20, 1-132.
- [5] Grignard, V.C.R. *Hebdom. Sceances Acad. Sci.* **1900**, 130, 1322.
- [6] (a) Wittig, G.; Pockels, U.; Droge, H. Über die Austauschbarkeit von aromatisch gebundenem Wasserstoff gegen Lithium mittels Phenyllithiums. *Chem. Ber.* **1938**, 71, 1903; (b) Gilman, H.; Jacoby, A.L. Dibenzothiophene: orientation and derivatives. *J. Org. Chem.* **1938**, 3, 108-119; (c) Gilman, H.; Langham, W.; Jacoby, A.L. Metalation as a Side Reaction in the Preparation of Organolithium Compounds. *J. Am. Chem. Soc.* **1939**, 61, 106-109.
- [7] Lindsell, W.E. In Wilkinson, G.; Stone, F.G.A.; Abel, E.W. (Eds): *Comprehensive Organometallic Chemistry*, Pergamon, New York **1982**, Vol. 1, pp. 155-252.
- [8] Langer, A.W. Polyamine-Chelated Alkali Metal Compounds. *Adv. Chem. Ser.* **1974**, 130.
- [9] Seebach, D. Structure and reactivity of lithium enolates. From pinacolone to selective C-alkylations of peptides. Difficulties and opportunities afforded by complex structures. *Angew. Chem. Int. Ed. Engl.* **1988**, 27, 1624-1654.
- [10] Noyori, L.; Kitamura, M. Enantioselective addition of organometallic reagents to carbonyl compounds: chirality transfer, multiplication, and amplification. *Angew. Chem. Int. Ed. Engl.* **1991**, 30, 49-69.
- [11] Soai, K.; Niwa, S. Enantioselective Addition of Organozinc Reagents to Aldehydes. *Chem. Rev.* **1992**, 92, 833-856.
- [12] Soai, K.; Shibata, T. In: *Comprehensive Asymmetric Catalysis II*; "Alkylation of carbonyl groups". Jacobsen, E.N.; Pfaltz, A.; Yamamoto, H.; Eds. Springer: Berlin, **1999**, pp. 911-922.
- [13] Pu, I.; Yu, H.-B. Catalytic Asymmetric Organozinc Additions to Carbonyl Compounds. *Chem. Rev.* **2001**, 101, 757-824.
- [14] Cruz, A.; and Juárez-J. M. Heterocyclic compounds derived from ephedrine. *Curr. Org. Chem.* **2004**, 8, 671-692.
- [15] Cruz, A.; Bautista-Ramírez, M.E. Ephedrine and their Acyclic Derivatives. *Curr. Org. Synth.* **2011**, 8, 901-928.
- [16] Battioni, J.-P.; Chodkiewicz, W. *Bull. Soc. Chim. Fr.* **1972**, 74, 2068-2069.
- [17] Zweig, J.S.; Luche, J.L.; Barreiro, E.; Crabbé, P. Asymmetric induction in additions of organocuprates. *Tetrahedron Lett.* **1975**, 16, 2355-2358.
- [18] Boireau, G.; Abenham, D.; Bourdais, J.; Henry-Basch, E. Asymmetric alkylation of carbonyl compounds with chiral lithium alkoxy-tributyl aluminate. *Tetrahedron Lett.* **1976**, 17, 4781-4782.
- [19] Muchow, G.; Vannorenbergh, Y.; Buono, G. Use of alkaloids and aminoalcohols in catalytic asymmetric induction: Temperature effect on the addition of diethylzinc to benzaldehyde. *Tetrahedron Lett.* **1987**, 28, 6163-6166.
- [20] Itsuno, S.; Fréchet, J.M.J. Enantioselective addition of diethylzinc to aldehydes catalyzed by polymer-supported chiral amino alcohols. Evidence for a two zinc species mechanism. *J. Org. Chem.* **1987**, 52, 4140-4142.
- [21] Soai, K.; Yokoyama, S.; Ebihara, K.; Hayasaka, T. A new chiral catalyst for the highly enantioselective addition of dialkylzinc reagents. *J. Chem. Soc., Chem. Commun.* **1987**, 1690-1691. DOI: 10.1039/C39870001690
- [22] Soai, K.; Watanabe, M.; Koyano, M. Catalytic asymmetric synthesis of hydroxyketones by chemo- and enantioselective alkylation of ketoaldehydes. *J. Chem. Soc. Chem. Commun.* **1989**, 534-536. DOI: 10.1039/C3989000534
- [23] Soai, K.; Yokoyama, S.; Hayasaka, T.; Ebihara, K. Catalytic asymmetric induction in enantioselective conjugate addition of dialkylzinc to enones. *J. Org. Chem.* **1988**, 53, 4148-4149.
- [24] Noyori, R.; Suga, S.; Kawai, K.; Okada, S.; Kitamura, M.; Oguni, N.; Hayashi, M.; Kaneko, T.; Matsuda, Y. Enantioselective addition of diorganozinc to aldehydes catalyzed by β -amino alcohols. *J. Organomet. Chem.* **1990**, 382, 19-37.
- [25] Chaloner, P.A.; Perera, S.A.R. Enantioselective addition of diethylzinc to benzaldehyde in the presence of ephedrine derivatives. *Tetrahedron Lett.* **1987**, 28, 3013-3014.
- [26] Chaloner, P.A.; Langadianou, E. Catalytic asymmetric alkylation of aldehydes by diethylzinc in the presence of *N*-iso-propylephedrine; dependence of enantioselectivity on $\text{Et}_2\text{Zn}:\text{RCHO}$. *Tetrahedron Lett.* **1990**, 31, 5185-5188.
- [27] Corey, E.J.; Hannon, F.J. Chiral catalysts for the enantioselective addition of organometallic reagents to aldehydes. *Tetrahedron Lett.* **1987**, 28, 5233-5236.
- [28] Corey, E.J.; Naef, R.; Hannon, F.J. Enantioselective conjugate addition of rationally designed chiral cuprate reagents to 2-cycloalkenones. *J. Am. Chem. Soc.* **1986**, 108, 7114-7116.
- [29] Corey, E.J.; Hannon, F.J. Zinc complexes of chiral phenols as catalysts for enantioselective addition of organozinc reagents to aldehydes. *Tetrahedron Lett.* **1987**, 28, 5237-5240.
- [30] Soai, K.; Nishi, M.; Ito, Y. Enantioselective addition of diethylzinc to aldehydes catalyzed by chiral diamino-diols derived from ephedrine. *Chem. Lett.* **1987**, 16, 2405-2406.
- [31] Soai, K.; Yokoyama, S.; Hayasaka, T. Chiral *N,N*-dialkylnorephedrine as catalysts of the highly enantioselective addition of dialkylzinc to aliphatic and aromatic aldehydes. The asymmetric synthesis of secondary aliphatic and aromatic alcohols of high optical purity. *J. Org. Chem.* **1991**, 56, 4264-4268.
- [32] Soai, K.; Hirose, Y.; Niwa, S.J. Asymmetric synthesis of optically active fluorine-containing alcohols using chiral catalysts. *Fluorine Chem.* **1992**, 59, 5-8.
- [33] Soai, K.; Hirose, Y.; Sakata, S. Catalytic Asymmetric Synthesis of Optically Active Secondary Alcohols Deuterated on the Secondary Carbon by the Enantioselective Alkylation of Aldehydes. *Bull. Chem. Soc. Jpn.* **1992**, 65, 1734-1735.
- [34] (a) Soai, K.; Konishi, T.; Shibata, T. Highly Enantioselective Addition of Dialkylzinc to Aromatic Aldehydes Using 1-Phenyl-2-(1-pyrrolidinyl)-1-propanol as a Chiral Catalyst. *Heterocycles* **1999**, 51, 1421-1426; (b) Sato, I.; Saito, T.; Omiya, D.; Takizawa, Y.; Soai, K. 1-Phenyl-2-(1-pyrrolidinyl)-1-propanol as a chiral catalyst for the highly enantioselective addition of dialkylzinc to five-membered heterocyclic aldehydes. *Heterocycles* **1999**, 51, 2753-2758.
- [35] Price, R.H. In: *Comprehensive Coordination Chemistry*; Wilkinson, G., Ed.; Pergamon Press: Oxford, **1987**; Chapter 56.1.
- [36] Mukaiyama, T.; Soai, K.; Sato, T.; Shimizu, H.; Suzuki, K. Enantioface-differentiating (asymmetric) addition of allyllithium and dialkylmagnesium to aldehydes by using (2*S*, 2'*S*)-2-hydroxymethyl-1-[(1-alkylpyrrolidin-2-yl)methyl]pyrrolidines as chiral ligands. *J. Am. Chem. Soc.* **1979**, 101, 1455-1460.
- [37] Hursthouse, M.B.; Motevalli, M.; O'Brien, P.; Waleh, J.R.; Jones, A.C. X-Ray crystal structure of a triazine adduct of dimethylzinc; an important precursor for the deposition of II/VI materials. *J. Mater. Chem.* **1991**, 1, 139-140.
- [38] Soai, K.; Watanabe, M. Chiral quaternary ammonium salts as solid state catalysts for the enantioselective addition of diethylzinc to aldehydes. *J. Chem. Soc. Chem., Commun.* **1990**, 43-44. DOI: 10.1039/C39900000043
- [39] Soai, K.; Niwa, S.; Hatanaka, T. Asymmetric synthesis of alcohols with two quiral centres from a racemic aldehyde by the selective addition of dialkylzinc reagents using (1*S*, 2*R*)-(-)-*N*, *N*-dibutylnorephedrine and (5)-(+)-diphenyl-(-1-methylpyrrolidin-2-yl) methanol as chiral catalysts. *J. Chem. Soc., Chem. Commun.* **1990**, 709-711. DOI: 10.1039/C3990000709
- [40] Niwa, S.; Hatanaka, T.; Soai, K. Asymmetric synthesis of optically active alcohols with two chiral centres from a racemic aldehyde by the selective addition of dialkylzinc reagents using chiral catalysts. *J. Chem. Soc., Perkin Trans. 1* **1991**, 2025-2027. DOI: 10.1039/P1991000205
- [41] Andrés, J.M.; Martínez, M.A.; Pedroza, R.; Pérez-Encabo, A. Enantioselective ethylation of aldehydes catalyzed by chiral *C*₂-symmetrical β -hydroxy-*m*-xylylene diamines. *Tetrahedron: Asymmetry* **1994**, 5, 67-72.
- [42] Williams, D.; Fromhold, M.G. Enantioselective Alkylation of Aldehydes with Diethylzinc Catalyzed by *C*₂-Symmetric Ligands. *Synlett* **1997**, (Supp. 1), 523-524.
- [43] Bastin, S.; Agbossou-Niedercorn, F.; Brocard, J. and Pélineski, L. Enantioselective alkylation of benzaldehyde with diethylzinc catalyzed by 1,1'- and 1,2-disubstituted ferrocenyl amino alcohols. *Tetrahedron: Asymmetry* **2001**, 12, 2399-2408.
- [44] Blay, G.; Fernández, I.; Marco-Aleixandre, A.; Pedro, J.R. Chiral bis-(amino alcohol)-oxalamides as ligands for asymmetric catalysis. Ti (IV) catalyzed enantioselective addition of diethylzinc to aldehydes. *Tetrahedron: Asymmetry* **2005**, 16, 1207-1213.
- [45] (a) Heaton, S.B.; Jones, G.B. Chiral arene chromium tricarbonyl complexes as enantioselective catalysts: highly selective 1, 2 alkyl additions to aldehydes. *Tetrahedron Lett.* **1992**, 33, 1693-1696; (b) Jones, G.B.; Heaton, S.B. Catalytic asymmetric induction Part 2. Chiral tricarbonyl (η^6 arene) chromium (0) complexes as enantioselective catalysts. *Tetrahedron: Asymmetry* **1993**, 4, 261-272; (c) Jones, G.B.; Huber, R.S.; Chapman, B.J. Catalytic enantioselective synthesis of macrolides via asymmetric alkylation. *Tetrahedron: Asymmetry* **1997**, 8, 1797-1809.
- [46] Jones, J.B.; Guzel, M.; Chapman, B.J. Refined enantioselective methylation catalysts: improved routes to bifunctional C5 synthons. *Tetrahedron: Asymmetry* **1998**, 9, 901-905.
- [47] (a) Soai, K.; Hayase, T.; Takai, K.; Sugiyama, T. Asymmetric synthesis of chiral secondary alcohols with isopropyl substituents by the catalytic enantioselective addition of diisopropylzinc to aldehydes using *N*, *N*-dialkyl norephedrine as chiral catalysts. *J. Org. Chem.* **1994**, 59, 7908-7909; (b) Sato, I.; Saito, T.; Soai, K. Solvent-free catalytic enantioselective addition of diethylzinc to aldehydes. *J. Chem. Soc. Chem. Commun.* **2000**, 2471-2472.
- [48] Paleo, M.R.; Cabeza, I.; Sardina, F.J. Enantioselective Addition of Diethylzinc to Aldehydes Catalyzed by *N*-(9-Phenylfluoren-9-yl) β -Amino Alcohols. *J. Org. Chem.* **2000**, 65, 2108-2113.
- [49] Nakamura, Y.; Takeuchi, S.; Okamura, K. and Ohgo, Y. Enantioselective addition of diethylzinc to aldehydes catalyzed by fluorine β -aminoalcohols. *Tetrahedron* **2001**, 57, 5565-5571.
- [50] Mao, J.; Wan, B.; Wang, R.; Wu, F. and Lu, S. Concise Synthesis of Novel Practical Sulfamide-Amine Alcohols for the Enantioselective Addition of Diethylzinc to Aldehydes. *J. Org. Chem.* **2004**, 69, 9123-9127.
- [51] Sibi, M.P. and Stanley, L.M. Enantioselective diethylzinc additions to aldehydes catalyzed by chiral relay ligands. *Tetrahedron: Asymmetry* **2004**, 15, 3353-3356.

- [52] (a) Wu, Y.; Yun, H.; Wu, Y.; Ding, K.; Zhou, Y. Synthesis of *N*-pyridylmethyl amino alcohols and application in catalytic asymmetric addition of diethylzinc to aromatic aldehydes. *Tetrahedron: Asymmetry* **2000**, *11*, 3543-3552; (b) Yun, H.; Wu, Y.; Wu, Y.; Ding, K.; Zhou, Y. Chiral ligands for asymmetric synthesis: enantioselective addition of diethylzinc to aromatic aldehydes catalyzed by chiral *N*-pyridylmethyl amino alcohols. *Tetrahedron Lett.* **2000**, *41*, 10263-10266.
- [53] Tasgin, D.I.; Unalerglu, C. Enantioselective addition of diethylzinc to aldehydes catalyzed by β -aminoalcohols derived from (*1R*, *2S*)-norephedrine. *Appl. Organomet. Chem.* **2010**, *24*, 33-37.
- [54] Unalerglu, C.; Aydin, A.E.; Demir, A.S. Synthesis of novel norephedrine-based chiral ligands with multiple stereogenic centers and their application in enantioselective addition of diethylzinc to aldehyde and chalcone. *Tetrahedron: Asymmetry* **2006**, *17*, 742-749.
- [55] Gilani, M. and Wilhelm, R. New enantiopure imidazolium carbene ligands incorporating two hydroxy groups for Lewis acid-catalyzed diethyl zinc addition to aldehydes. *Tetrahedron: Asymmetry* **2008**, *19*, 2346-2352.
- [56] Parrott II, R.W. Hitchcock, S.R. β -Amino alcohols derived from (*1R*, *2S*)-norephedrine and (*1S*, *2S*)-pseudonorephedrine as catalysts in the asymmetric addition of diethylzinc to aldehydes. *Tetrahedron: Asymmetry* **2008**, *19*, 19-26.
- [57] Dean, M. A. and Hitchcock, S.R. A structural examination of the impact of oxygenated side chains in *Ephedra* compounds in the catalytic asymmetric addition of diethylzinc to aldehydes. *Tetrahedron: Asymmetry* **2009**, *20*, 2351-2356.
- [58] (a) Parrott, II, R.W. Hitchcock, S.R. Intramolecular chiral relay at stereogenic nitrogen: oxazolidine catalysts derived from *Ephedra* alkaloids. *Tetrahedron: Asymmetry* **2007**, *18*, 377-382; (b) Blocka, E.; Jaworska, M.; Kozakiewicz, A.; Welniak, M.; Wojtczak, A. Structural and electronic effects of oxazolidine ligands derived from (*1R*, *2S*)-ephedrine in the asymmetric addition of diethylzinc to aldehydes. *Tetrahedron: Asymmetry* **2010**, *21*, 571-577.
- [59] Banerjee, S.; Groeper, J.A.; Standard, Jean M. and Hitchcock, S. R. *N*-Pyridylmethylephedrine derivatives in the catalytic asymmetric addition of diethylzinc to aldehydes and diphenyl-phosphinoylmines. *Tetrahedron: Asymmetry* **2009**, *20*, 2154-2161.
- [60] Ananthi, N.; Balakrishnan, U.; Vinu, A.; Ariga, K.; Velmathi, S. Chiral amide from (*1S*, *2R*)-(+)-norephedrine alkaloid in the enantioselective addition of diethylzinc to aryl and hetero-aryl aldehydes. *Tetrahedron: Asymmetry* **2009**, *20*, 1731-1735.
- [61] Banerjee, S.; Camodeca, A.J.; Griffin, G.G.; Hamaker, Ch. G.; Hitchcock, Sh. R. Aromatic motifs in the design of *Ephedra* ligands for application in the asymmetric addition of diethylzinc to aldehydes and diphenylphosphinoyl-imines. *Tetrahedron: Asymmetry* **2010**, *21*, 549-557.
- [62] Mao, J.; Wan, B.; Zhang, Z.; Wang, R.; Wu, F.; Lu, S. Reversal of enantioselectivity by adding $\text{Ti}(\text{O}i\text{Pr})_4$: novel sulfamide-amine alcohol ligands for the catalytic asymmetric addition of diethylzinc to aldehydes. *J. Mol. Cat. A: Chemical*, **2005**, *225*, 33-37.
- [63] Soai, K.; Hatanaka, T.; Miyazawa, T. Highly enantioselective alkylation of carbon-nitrogen double bonds. Catalytic and stoichiometric asymmetric synthesis of optically active amines by the enantioselective addition of dialkylzinc reagents to *N*-diphenylphosphinoylmines. *J. Chem. Soc., Chem. Commun.* **1992**, 1097-1098. DOI: 10.1039/C39920001097
- [64] Soai, K.; Yokoyama, S.; Hayasaka, T.; Ebihara, K. Catalytic asymmetric synthesis of alkyl substituted lactones by enantioselective and chemoselective alkylation of formylesters with dialkylzinc using *N,N*-dibutylnorephedrine. *Chem. Lett.* **1988**, *17*, 843-846.
- [65] (a) Saucy, G.; Borer, R. Steroid Total Synthesis, Part II; (–)-17 β -Hydroxydes-A-androst-9-en-5-one. *Helv. Chim. Acta* **1971**, *54*, 2121-2132. b) Rosenberger, M.; Borer, R.; Saucy, G. Steroid total synthesis. 11-(+)-Estr-4-ene-3, 17-dione from a chiral lactone. *J. Org. Chem.* **1978**, *43*, 1550-1555.
- [66] Smith, L.R.; Williams, H.J.; Silverstein, R. M. Facile synthesis of optically active 2-ethyl-1, 6-dioxaspiro[4, 4]nonane, component of the aggregation pheromone of the beetle *pityogenes chalcographus* L. *Tetrahedron Lett.* **1978**, *19*, 3231-3232.
- [67] Watanabe, M.; Soai, K. Catalytic asymmetric synthesis of γ -hydroxyketones and aromatic hydroxyl ketones by the chemo- and enantioselective alkylation of keto aldehydes with dialkylzinc. *J. Chem. Soc. Perkin Trans. 1* **1994**, 3125-3128. DOI: 10.1039/P19940003125
- [68] Soai, K.; Kawase, Y.; Niwa, S. Catalytic Asymmetric Synthesis of 2- and 3-Furylcarbinols. *Heterocycles* **1989**, *29*, 2219-2223.
- [69] Van Overen, A.; Menge, W.; Feringa, B.L. Enantioselective synthesis of furyl alcohols by catalytic asymmetric addition of diethylzinc to furanaldehydes. *Tetrahedron Lett.* **1989**, *30*, 6427-6430.
- [70] Soai, K.; Hori, H.; Niwa, S. Enantioselective addition of dialkylzinc to pyridinecarbaldehyde in the presence of chiral aminoalcohols: asymmetric synthesis of pyridylalkyl alcohols. *Heterocycles* **1989**, *29*, 2065-2067.
- [71] Shibata, T.; Hayase, T.; Aiba, Y.; Tabira, H.; Soai, K. Asymmetric Synthesis of Various Chiral 5-Pyrimidylalkanols by the Enantioselective Alkylation of Pyrimidine-5-carbaldehydes with Dialkylzinc in the Presence of Chiral Amino Alcohols. *Heterocycles* **1997**, *46*, 235-240.
- [72] Soai, K.; Hori, H.; Kawahara, M. Nonlinear relationship between the enantioselectivities for asymmetric reactions of monofunctional and bifunctional substrates. Synthesis of practically optically pure diols by the catalytic enantioselective diethylation of terephthalaldehyde. *J. Chem. Soc., Chem. Commun.* **1992**, 106-108. DOI: 10.1039/C39920000106
- [73] Soai, K.; Inoue, Y.; Takahashi, T.; Shibata, T. Asymmetric synthesis of chiral diols by the catalytic enantioselective dialkylation of tere-, iso-, and phthalaldehydes and by a catalytic enantioselective autoinductive reaction. *Tetrahedron* **1996**, *52*, 13355-13362.
- [74] Soai, K.; Hori, H.; Kawahara, M. Catalytic enantioselective synthesis of optically active phthalides. *Tetrahedron: Asymmetry* **1991**, *2*, 253-254.
- [75] Soai, K.; Hatanaka, T.; Yamashita, T. Stereoselective synthesis of optically active syn- and anti-1,3-diols by the catalytic alkylation of a β -alkoxy aldehyde. *J. Chem. Soc. Chem. Commun.* **1992**, 927-929. DOI: 10.1039/C39920000927
- [76] Soai, K.; Shimada, C.; Takeuchi, M.; Itabashi, M. Chiral catalyst controlled addition of dialkylzinc to chiral α -siloxyaldehydes; asymmetric synthesis of either diastereoisomer of mono-protected vicinal diols. *J. Chem. Soc. Chem. Commun.* **1994**, 567-568. DOI: 10.1039/C39940000567
- [77] Soai, K.; Niwa, S.; Watanabe, M. Polymer-bound ephedrine as an efficient chiral catalyst for the enantioselective addition of dialkylzinc to aldehydes. *J. Org. Chem.* **1988**, *53*, 927-928.
- [78] Soai, K.; Niwa, S.; Watanabe, M. Carbonium ion rearrangements controlled by the presence of a silyl group. *J. Chem. Soc. Perkin Trans. 1*, **1989**, 109-114. DOI: 10.1039/P19890000109
- [79] Soai, K.; Watanabe, M. New polymer bound chiral catalyst with methylene spacer for the enantioselective addition of diethylzinc to aldehydes. *Tetrahedron: Asymmetry* **1991**, *2*, 97-100.
- [80] Watanabe, M.; Araki, S.; Butsugan, Y.; Uemura, M. *Chem. Express* **1990**, *5*, 761.
- [81] Watanabe, M.; Soai, K. Enantioselective addition of diethylzinc to aldehydes using chiral polymer catalysts possessing a methylene spacer. *J. Chem. Soc. Perkin Trans. 1* **1994**, 837-842. DOI: 10.1039/P19940000837
- [82] (a) Hodge, P.; Sung, D.W.L.; Stratford, P.W. Asymmetric synthesis of 1-phenylpropanol using polymer-supported chiral catalysts in simple bench-top flow system. *J. Chem. Soc. Perkin Trans. 1* **1999**, 2335-2342. DOI: 10.1039/A902967B; (b) Sung, D.W.L.; Hodge, P.; Stratford, P.W. Reactions of aldehydes with diethylzinc catalysed by polymer-supported ephedrine and camphor derivatives: comparisons of enantiomeric excesses achieved with various supports: optimization of support parameters to enable high. *J. Chem. Soc. Perkin Trans. 1* **1999**, 1463-1472. DOI: 10.1039/A901455A
- [83] Soai, K.; Watanabe, M.; Yamamoto, A. Enantioselective addition of dialkylzinc to aldehydes using heterogeneous chiral catalysts immobilized on alumina and silica gel. *J. Org. Chem.* **1990**, *55*, 4832-4835.
- [84] (a) Laspéras, M.; Bellocq, N.; Brunel, D.; Moreau, P. Chiral mesoporous templated silicas as heterogeneous inorganic-organic catalysts in the enantioselective alkylation of benzaldehyde. *Tetrahedron: Asymmetry* **1998**, *9*, 3053-3064; (b) Abramson, S.; Laspéras, M.; Galameau, A.; Desplandier-Giscard, D.; Brunel, D. Best design of heterogenized β -aminoalcohols for improvement of enantioselective addition of diethylzinc to benzaldehyde. *J. Chem. Soc. Chem. Commun.* **2000**, 1773-1774.
- [85] Balakrishnan, U.; Ananthi, N.; Velmathi, S.; Benzigar, M.R.; Talapaneni, S.N.; Aldeyab, S.S.; Ariga, K.; Vinu, A. Immobilization of chiral amide derived from (*1R*, *2S*)-(-)-norephedrine over 3D nanoporous silica for the enantioselective addition of diethylzinc to aldehydes. *Microporous Mesoporous Mater.* **2012**, *155*, 40-46.
- [86] Sato, I.; Shibata, T.; Ohtake, K.; Kodaka, R.; Hirokawa, Y.; Shirai, N.; Soai, K. Synthesis of chiral dendrimers with a hydrocarbon backbone and application to the catalytic enantioselective addition of dialkylzinc to aldehydes. *Tetrahedron Lett.* **2000**, *41*, 3123-3126.
- [87] Sato, I.; Kodaka, R.; Hosoi, K. and Soai, K. Highly enantioselective addition of dialkylzinc to aldehydes using dendritic chiral catalysts with flexible carbosilane backbones. *Tetrahedron: Asymmetry* **2002**, *13*, 805-808.
- [88] Niwa, S.; Soai, K. Catalytic asymmetric synthesis of optically active alkynyl alcohols by enantioselective alkylation of aldehydes and by enantioselective alkylation of alkynyl aldehydes. *J. Chem. Soc., Perkin Trans. 1* **1990**, 937-943. DOI: 10.1039/P19900000937
- [89] Tombo, G.M.R.; Didier, E.; Loubinoux, B. Amino Alcohol Mediated Enantioselective Addition of 2-Phenylethynylzinc Bromide to Aldehydes. *Synlett* **1990**, 9, 547-548.
- [90] (a) Frantz, D.E.; Fässler, R.; Carreira, E.M. Facile Enantioselective synthesis of propargylic alcohols by direct addition of terminal alkynes to aldehydes. *J. Am. Chem. Soc.* **2000**, *122*, 1806-1807; (b) Frantz, D.E.; Fässler, R.; Tomooka, C.S.; Carreira, E. M. The discovery of novel reactivity in the development of C-C bond-forming reactions: *In Situ* Generation of Zinc Acetylides with $\text{Zn}^{\text{II}}/\text{R}_3\text{N}$. *Acc. Chem. Res.* **2000**, *33*, 373-381.
- [91] Boyall, D.; López, F.; Sasaki, H.; Frantz, D.; Carreira, E.M. Enantioselective addition of 2-methyl-3-butyne-2-ol to aldehydes: Preparation of 3-Hydroxy-1-butyne. *Org. Lett.* **2000**, *2*, 4233-4236.
- [92] Sasaki, H.; Boyall, D.; Carreira, E.M. Facile, Asymmetric addition of acetylene to aldehydes: *in situ* generation of reactive zinc acetylide. *Helv. Chim. Acta* **2001**, *84*, 964-971.
- [93] (a) Anand, N.K.; Carreira, E.M. A simple, mild, catalytic, enantioselective addition of terminal acetylenes to aldehydes. *J. Am. Chem. Soc.* **2001**, *123*, 9687-9688; (b) Boyall, D.; Frantz, D.E.; Carreira, E.M. Efficient enantioselective

- lective additions of terminal alkynes and aldehydes under operationally convenient conditions. *Org. Lett.* **2002**, 15, 2605-2606.
- [94] Lu, G.; Li, X. -S.; Zhou, Z.-Y.; Chan, W.L.; Chan, A.S.C. Enantioselective alkynylation of aromatic aldehydes catalyzed by new chiral amino alcohol-based ligands. *Tetrahedron: Asymmetry* **2001**, 12, 2147-2152.
- [95] Tan, L.S.; Chen, C.Y.; Tillyer, R.D.; Grabowsky, E.J.J.; Reider, P.J. A novel, highly enantioselective ketone alkynylation reaction mediated by chiral zinc aminoalkoxides. *Angew. Chem. Int. Ed.* **1999**, 38, 711-713.
- [96] Mao, J.; Wan, B.; Wu, F.; Lu, S. Readily available sulfamide-amine alcohols for enantioselective phenylacetylene addition to aldehydes in the absence of $\text{Ti}(\text{O}^i\text{Pr})_4$. *Chirality* **2005**, 17, 245-249.
- [97] Li, H.; Huang, Y.; Jin, W.; Xue, F.; Wan, B. Tf-based sulfamide-amine alcohol-catalyzed enantioselective alkynylation of aldehydes. *Tetrahedron Lett.* **2008**, 49, 1686-1689.
- [98] Oppolzer, W.; Radinov, R.N. Enantioselective addition of (Z)- and (E)-alkenylzinc bromides to aldehydes: asymmetric synthesis of sec-allyl alcohols. *Tetrahedron Lett.* **1991**, 32, 5777-5780.
- [99] Soai, K.; Takahashi, K. Asymmetric alkenylation of chiral and prochiral aldehydes catalysed by chiral or achiral amino alcohols: catalytic diastereoselective synthesis of protected *erythro*-sphingosine and enantioselective synthesis of chiral diallyl alcohols. *J. Chem. Soc., Perkin Trans. 1* **1994**, 1257-1258. DOI: 10.1039/P19940001257.
- [100] Shibata, T.; Nakatsui, K.; Soai, K. Highly enantioselective catalytic isopropenylation of aldehydes using diisopropenylzinc. *Inorg. Chim. Acta* **1999**, 296, 33-36.
- [101] Soai, K.; Kawase, Y.; Oshio, A. Enantioselective phenylation of prochiral aldehydes using a kinetically formed chiral complex between Grignard-zinc halide reagent and *N*, *N*-dibutylnorephedrine. *J. Chem. Soc., Perkin Trans. 1* **1991**, 1613-1614. DOI: 10.1039/P19910001613.
- [102] Paixao, M.W.; De Godoi, M.; Rhoden, C.R.B.; Westermann, B.; Wessjohann, L.A.; Diogo S. Lüdtk, D.S.; Antonio L. Braga, A.L. The application of chiral, non-racemic *N*-alkylnorephedrine and *N*, *N*-dialkylnorephedrine as ligands for the enantioselective aryl transfer reaction to aldehydes. *J. Mol. Cat. A: Chem.* **2007**, 261, 120-124.
- [103] Soai, K.; Kawase, Y. Enantioselective furylation of p-chiral aldehydes by difurylzinc in the presence of a chiral amino alcohol: asymmetric synthesis of 2-furylmethanols. *J. Chem. Soc., Perkin Trans. 1* **1990**, 3214-3215. DOI: 10.1039/P19900003214.
- [104] Soai, K.; Hirose, Y.; Ohno, Y. Chiral dialkyl thiophosphoramidates as highly enantioselective catalysts for the alkylation of aldehydes. *Tetrahedron: Asymmetry* **1993**, 4, 1473-1474.
- [105] Shibata, T.; Tabira, H.; Soai, K. A catalytic and enantioselective cyclopropylation of aldehydes using dicyclopropylzinc. *J. Chem. Soc., Perkin Trans. 1* **1998**, 177-178. DOI: 10.1039/A707315A.
- [106] Reetz, M.T.; Kükenhöner, T.; Weinig, P. Enantioselective addition of chirally modified methyltitanium reagents to aromatic aldehydes *Tetrahedron Lett.* **1986**, 27, 5711-5714.
- [107] Ito, K.; Kimura, Y.; Okamura, H.; Katsuki, T. Enantioselective Addition of Diethylzinc to Aldehydes in the Presence of Chiral Titanium Reagent Modified with *N*-Sulfonylated Amino Alcohol. *Synlett* **1992**, 7, 573-574.
- [108] Li, X. -S.; Lu, G.; Kwok, W.H.; Chan, A.S.C. Highly enantioselective alkynylzinc addition to aromatic aldehydes catalyzed by self-assembled titanium catalysts. *J. Am. Chem. Soc.* **2002**, 124, 12636-12637.
- [109] Yus, M.; Ramón J.D. and Prieto, O. Synthesis of C_2 -symmetrical bis(1,2-hydroxy sulfonamide) ligands and application in the enantioselective addition of dialkylzinc to aldehydes. *Tetrahedron: Asymmetry* **2002**, 13, 1573-1579.
- [110] Parrott II, R.W.; Dore, D.D.; Chandrashekar, S.P.; Bentley, J.T.; Morgan, B.S.; Hitchcock, S.R. Enantioselective diethylzinc addition to aldehydes via chiral, non-racemic β -hydroxysalicylhydrazone catalysts. *Tetrahedron: Asymmetry* **2008**, 19, 607-611.
- [111] Banerjee, S.; Ferrence, G.M. and Hitchcock, S.R. Catalytic asymmetric addition of diethylzinc to aldehydes via chiral, non-racemic β -hydroxy and β -methoxy salicylhydrazone catalysts. *Tetrahedron: Asymmetry* **2010**, 21, 837-845.
- [112] Jaworska, M.; Łączkowski, K.Z.; Welniak, M.; Welke, M.; Wojtczak, A. Imines derived from (1*S*, 2*R*)-norephedrine as catalysts in the enantioselective addition of diethylzinc to aldehydes. *Applied Catalysis A: General* **2009**, 357, 150-158.
- [113] (a) Poelert, M.A.; Hof, R.P.; Peper, N.C.M.W.; Kellogg, R.M. Enantiomerically pure β -amino sulfides and β -amino thiols from ephedrine. *Heterocycles* **1994**, 37, 461-475; (b) Hof, R.P.; Poelert, M.A.; Peper, N.C.M.W.; Kellogg, R.M. Sulfur derivatives of ephedra alkaloids; new and highly efficient chiral catalysts. *Tetrahedron: Asymmetry* **1994**, 5, 31-34.
- [114] Fitzpatrick, K.; Hulst, R.; Kellogg, R.M. Thiol and disulfide derivatives of ephedra alkaloids 2: A mechanistic study of their effect on the addition of diethyl zinc to benzaldehyde. *Tetrahedron: Asymmetry* **1995**, 6, 1861-1864.
- [115] (a) Kang, J.; Kim, J.W.; Lee, J.W.; Kim, D.S.; Kim, J.I. Chiral β -Amino Thiol Catalysts for the Enantioselective Addition of Diethylzinc to Aldehydes. *Bull. Korean Chem. Soc.* **1996**, 17, 1135-1142; (b) Kang, J.; Kim, D.S.; Kim, J.I. Enantioselective Addition of Diethylzinc to Aldehydes Catalyzed by A Drug-Unrelated Chiral Amino Thiol and the Corresponding Disulfide. *Synlett* **1994**, 842-844; (c) Kang, J.; Lee, J.W.; Kim, J.I. Enantioselective addition of diethylzinc to α -branched aldehydes. *J. Chem. Soc. Chem. Commun.* **1994**, 2009-2010. DOI: 10.1039/C39940002009.
- [116] Jin, M.-J.; Ahn, S.-J.; Lee, K.-S. New chiral catalysts for the highly enantioselective addition of diethylzinc to aldehydes. *Tetrahedron Lett.* **1996**, 37, 8767-8770.
- [117] Kossensjans, M.; Soeberdt, M.; Wallbaum, S.; Harms, K.; Martens, J.; Aurich, H.G. Utilization of industrial waste materials. Part 14. synthesis of β -aminoalcohols and thiols with a 2-azabicyclo[3.3.0]octane backbone and their application in enantioselective catalysis. *J. Chem. Soc., Perkin Trans. 1* **1999**, 2353-2365. DOI: 10.1039/A902362C.
- [118] Hulst, R.; Heres, H.; Fitzpatrick, K.; Peper, N.C.M.W.; Kellogg, R.M. Catalytic enantioselective alkylation of benzaldehyde with diethylzinc using chiral nonracemic (thio)-phosphoramidates. *Tetrahedron: Asymmetry* **1996**, 7, 2755-2760.
- [119] Jin, M.-J.; Kim, Y.-M.; Lee K.-S. Enantioselective addition of diethylzinc to aldehydes in the presence of chiral aprotic ligands. *Tetrahedron Lett.* **2005**, 46, 2695-2696.
- [120] (a) Soai, K.; Hayasaka, T.; Ugajin, S.; Yokoyama, S. Catalytic enantioselective conjugate addition of dialkylzinc to α , β -unsaturated ketones using nickel bromide-*N,N*-dibutylnorephedrine. *Chem. Lett.* **1988**, 17, 1571-1572; (b) Soai, K.; Okudo, M.; Okamoto, M. Enantioselective conjugate addition of diethylzinc to enones using chiral β -aminoalcohol as chiral catalyst or ligand. *Tetrahedron Lett.* **1991**, 32, 95-96.
- [121] Soai, K.; Hayasaka, T.; Ugajin, S. Asymmetric synthesis of optically active β -substituted ketones by highly enantioselective catalytic conjugate addition of dialkylzinc reagent to enones using a catalyst system of nickel(II)-chiral ligand-achiral ligand in acetonitrile/toluene. *J. Chem. Soc., Chem. Commun.* **1989**, 516-517. DOI: 10.1039/C39890000516.
- [122] Jansen, J.F.G.A.; Feringa, B.L. Enantioselective conjugate addition of Grignard reagents to enones catalyzed by chiral zinc(II) complexes. *J. Org. Chem.* **1990**, 55, 4168-4175.
- [123] Corey, E.J.; Naef, R.; Hannon, F.J. Enantioselective conjugate addition of rationally designed chiral cuprate reagents to 2-cycloalkenones. *J. Am. Chem. Soc.* **1986**, 108, 7114-7116.
- [124] Bertz, S.H.; Dabbagh, G.; Sundararajan, G. Asymmetric induction with amidocuprates. *J. Org. Chem.* **1986**, 51, 4953-4959.
- [125] Hayashi, T.; Kanehira, K.; Tsuchiya, H.; Kumada, M. New Chiral Ligands designed for Palladium-catalysed Asymmetric Allylic Alkylation. *J. Chem. Soc., Chem. Commun.* **1982**, 1162-1164. DOI: 10.1039/C39820001162.